

Project Data Page:





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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\110-C-AKChannel side= CS- 05-03-26Latest.pvdb

Date of Analysis : Sep 8, 2026 6:24pm

PV Elite 28, January 2026



PV Elite 28 Licensee: SL User
 FileName : 110-C-AK_Channel side= CS- 05-03-26_Latest Page 2 of 216
 Vessel Design Summary: Step: 33 6:24pm Sep 8,2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 2250.000 mm ID
 Vessel Design Length, Tangent to Tangent 11169.40 mm
 Specified Datum Line Distance 0.00 mm
 Shell Side Design Temperature 130 °C
 Channel Side Design Temperature 82 °C
 Shell Side Design Pressure 380.000 KPa.
 Channel Side Design Pressure 1000.000 KPa.
 Shop Shell Side Test Pressure 1457.651 KPa.
 Field Shell Side Test Pressure 494.000 KPa.
 Shop Channel Side Test Pressure 1589.649 KPa.
 Field Channel Side Test Pressure 1300.000 KPa.
 Wind Design Code ASCE/SEI 7-16
 Earthquake Design Code ASCE/SEI 7-16

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	
Shell	SA-240 304	...	...	S30400	
Flange	SA-266 2	...	...	K03506	
Nozzle	SA-106 B	...	...	K03006	
Nozzle	SA-312 TP304	...	...	S30400	
Re-Pad	SA-312 TP304	...	...	S30400	
Re-Pad	SA-516 70	...	...	K02700	
Nozzle Flg	SA-105	...	...	K03504	
Nozzle Flg	SA-182 F304	...	> 130	S30400	
Tubes	SA-213 TP316L	...	...	S31603	
Tubesheet	SA-266 2	...	...	K03506	
Flg Bolting	SA-193 B7	...	<= 64	G41400	
Hrz Bolting	SA-307 A	...	6 <= t <= 100		
Saddles	SA-516 70	...	...	K02700	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (KPa. & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
LHS Blind Flg	1025.044	103.42	1081.000	3.2000	N/A	No
LHS CH.Flbg	1022.545	103.42	1291.000	3.2000	N/A	No
LHS CH Shell	1022.086	103.42	1094.000	3.2000	N/A	No
main Shell 1	402.062	51.71	791.000	0.8000	N/A	No
main Shell 2	402.062	51.71	1815.000	0.8000	N/A	No
main Shell 2	402.062	51.71	791.000	0.8000	N/A	No
RHS Ch Shell	1022.086	103.42	1300.000	3.2000	N/A	No
RHS Ch Flg	1022.545	103.42	1291.000	3.2000	N/A	No
RHS Blind Flang	1025.005	103.42	1082.000	3.2000	N/A	No



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 FileName : 110-C-AK_Channel side= CS- 05-03-26_Latest Page 3 of 216
 Vessel Design Summary: Step: 33 6:24pm Sep 8,2026

Liquid Level: 2555.00 mm Dens.: 0.001 kg./cm³ Sp. Gr.: 1.000

Element Types and Properties:

Element Type	"To" Elev mm	Element Length mm	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
Body Flg	163.3	157.0	165.0	157.0	151.2	...	1.00	1.00
Body Flg	434.7	265.0	222.0	214.0	208.9	...	1.00	1.00
Cylinder	1434.7	1000.0	14.0	14.0	13.1	...	0.85	0.85
Cylinder	4724.7	3240.0	12.0	11.0	5.8	...	0.85	0.85
Cylinder	6474.7	1750.0	24.0	24.0	5.8	...	0.85	0.85
Cylinder	9684.7	3210.0	12.0	11.0	5.8	...	0.85	0.85
Cylinder	10734.7	1000.0	16.0	16.0	13.1	...	0.85	0.85
Body Flg	11006.0	265.0	222.0	214.0	208.9	...	1.00	1.00
Body Flg	11169.4	157.0	165.0	157.0	151.2	...	1.00	1.00

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure KPa.	External M.A.W.P. KPa.
10	20	157	115.291	103.419	No Calc
20	30	214	208.94	103.419	No Calc
30	Tsht	14	8.3849	103.419	548
Tsht	40	14	No Calc	103.419	No Calc
40	50	11	10.5149	51.7095	58.4
50	60	24	10.5806	51.7095	436
60	Tsht	11	10.5149	51.7095	58.4
Tsht	70	11	No Calc	51.7095	No Calc
70	80	16	8.3903	103.419	706
80	90	214	208.94	103.419	No Calc
90	100	157	115.291	103.419	No Calc

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required cm^4	Ring Inertia Available cm^4
10	20	No Calc	No Calc	No Calc	No Calc
20	30	No Calc	No Calc	No Calc	No Calc
30	Tsht	999.97	5816.62	No Calc	No Calc
Tsht	40	No Calc	No Calc	No Calc	No Calc
40	50	8200	9210.43	No Calc	No Calc
50	60	8200	186576	No Calc	No Calc
60	Tsht	8200	9210.43	No Calc	No Calc
Tsht	70	No Calc	No Calc	No Calc	No Calc
70	80	1000.03	8770.39	No Calc	No Calc
80	90	No Calc	No Calc	No Calc	No Calc
90	100	No Calc	No Calc	No Calc	No Calc

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
N5-Tube Inlet	WRC-297	0.971	Shell	Passed
N1-Shell inlet	WRC-297	0.920	Shell	Passed
N4-Shell outle	WRC-297	0.967	Shell	Passed
N6-Tube Outlet	WRC-297	0.974	Shell	Passed



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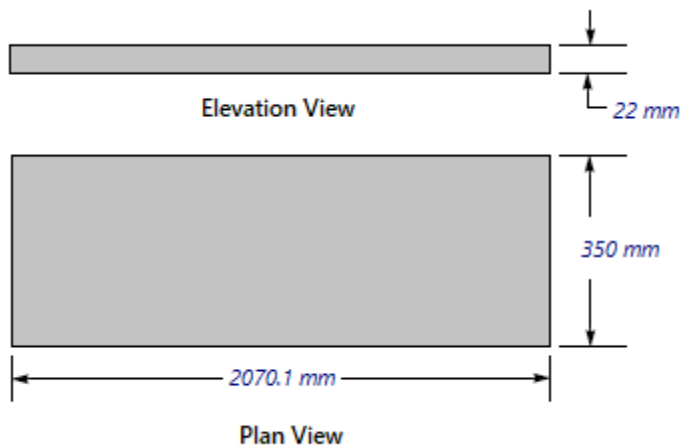
Foundation/Support Summary

Saddle Parameters:

Saddle Width	325.000	mm
Saddle Bearing Angle	167.000	deg.
Centerline Dimension	1760.000	mm
Wear Pad Width	560.000	mm
Wear Pad Thickness	20.000	mm
Wear Pad Bearing Angle	180.000	deg.
Distance from Saddle to Tangent	900.000	mm

Baseplate Length	2070.100	mm
Baseplate Thickness	22.000	mm
Baseplate Width	350.000	mm
Number of Ribs (including outside ribs)	5	
Rib Thickness	14.000	mm
Web Thickness	14.000	mm
Height of Center Web	730.000	mm
Number of Bolts in Baseplate	8	

Baseplate Sketch



Summary of Maximum Saddle Loads, Operating Case, Un-Factored:

Maximum Vertical Saddle Load	60925.73	Kgf
Maximum Transverse Saddle Shear Load	11228.62	Kgf
Maximum Longitudinal Saddle Shear Load	5963.35	Kgf

Summary of Maximum Saddle Loads, Pressure Test Case:

Maximum Vertical Saddle Load	52978.78	Kgf
Maximum Transverse Saddle Shear Load	312.62	Kgf
Maximum Longitudinal Saddle Shear Load	250.84	Kgf

Weights:

Fabricated - Bare w/o removable internals	63867.9	kg.
Shop Test - Fabricated + water (full)	104247.3	kg.
Shipping - Fab. + rem. intls.+ shipping app.	63867.9	kg.
Erected - Fab. + rem. intls.+ insul. (etc)	63867.9	kg.
Empty - Fab. + intls. + details + wghts.	63867.9	kg.
Operating - Empty + operating liquid (no ca)	104247.3	kg.
Field Test - Empty weight + test medium (full)	104247.3	kg.



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 Nozzle Summary: Step: 31 6:24pm Sep 8,2026

Nozzle Calculation Summary (KPa. & mm):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
N5-Tube Inlet	1094	OK	...	OK	9.33	OK	Passed
N1-Shell inlet	1815	OK	...	OK	9.13	OK	Passed
N4-Shell outlet	1815	OK	...	OK	9.13	OK	Passed
N6-Tube Outlet	1300	OK	...	OK	9.33	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 1815 Nozzle : N1-Shell inlet [Shellside]
 Minimum MAWP Nozzles : 1094 Nozzle : N5-Tube Inlet [Tubeside]

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	X Coordinate mm	Layout Angle deg	Dia. Limit mm
30	N5-Tube Inlet	934.700	0.000	861.301
50	N1-Shell inlet	5598.701	0.000	1736.000
50	N4-Shell outlet	5598.701	180.000	1744.000
70	N6-Tube Outlet	10234.701	180.000	861.301

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.
 lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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 FileName : 110-C-AK_Channel side= CS- 05-03-26_Latest Page 6 of 216
 Nozzle Schedule: Step: 30 6:24pm Sep 8,2026

Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
N5-Tube Inlet	18.000 in	40	WNF	18.000	14.275	657.20	18.00	243.47	150
N6-Tube Outlet	18.000 in	40	WNF	18.000	14.275	657.20	18.00	243.47	150
N1-Shell inlet	36.000 in Actual		WNF	36.000	24.000	1614.40	24.00	421.09	150
N4-Shell outlet	36.000 in Actual		WNF	36.000	22.000	1514.40	22.00	421.09	150

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
N5-Tube Inl	SA-106 B	...	10.000	14.000	18.000	...
N6-Tube Out	SA-106 B	...	10.000	14.000	18.000	...
N1-Shell in	SA-312 TP304	22.000	20.000	20.000	24.000	...
N4-Shell ou	SA-312 TP304	22.000	20.000	20.000	22.000	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
N5-Tube Inlet	934.700	0.0	200.00	0.00	LHS CH Shell
N6-Tube Outlet	10234.700	180.0	200.00	0.00	RHS Ch Shell
N1-Shell inlet	5598.701	0.0	300.00	0.00	main Shell 2
N4-Shell outlet	5598.701	180.0	300.00	0.00	main Shell 2

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Bill of Material: Step: 29 6:24pm Sep 8,2026

Bill of Materials:

QTY	DESCRIPTION	MATERIAL
2	BLIND FLANGE: 165.0mm THK X 2505.0mm OD	SA-266 2
2	WELD NECK FLANGE: 222.0mm THK X 2505.0mm OD	SA-266 2
1	CYLINDER: 14.0mm THK X 2250.0mm ID X 1000.0mm	SA-516 70
1	CYLINDER: 12.0mm THK X 2250.0mm ID X 3210.0mm	SA-240 304
1	CYLINDER: 16.0mm THK X 2250.0mm ID X 1000.0mm	SA-516 70
10	BAFFLES: 2224.0mm DIA X 8.0mm THK	SA-516 70
2	TUBESHEET: 2278.0mm X 50.0mm THK	SA-266 2
4500	TUBES: 8300.0mm X 19.0mm DIA X 1.6mm THK	SA-213 TP316L
2	SADDLE: 325.0mm X 167 DEG	SA-516 70
2	GASKET: 2326.0mm OD X 2282.0mm ID	...
160	BODY FLANGE BOLTS: 34.9mm DIA	SA-193 B7
320	NUTS FOR BODY FLANGE BOLTS: 34.9mm DIA	...
2	CYLINDRICAL SEGMENT 12.0mm THK X 2981.7mm X 3240.0mm	SA-240 304
1	CYLINDRICAL SEGMENT 12.0mm THK X 1105.1mm X 3240.0mm	SA-240 304
2	CYLINDRICAL SEGMENT 24.0mm THK X 2981.7mm X 1750.0mm	SA-240 304
1	CYLINDRICAL SEGMENT 24.0mm THK X 1105.1mm X 1750.0mm	SA-240 304
1	NAMEPLATE	...



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 FileName : 110-C-AK_Channel side= CS- 05-03-26_Latest Page 8 of 216
 MDMT Summary: Step: 32 6:24pm Sep 8,2026

Minimum Design Metal Temperature Results Summary (°C):

Description	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm	E*	PWHT reqd
Notes								
main Shell 1 [16]		-196						
main Shell 2 [16]		-196						
main Shell 2 [16]		-196						
N1-Shell inlet[1]	B	-2	-104	-29	0.113	24.000	1.00	No
N4-Shell outle[1]	B	-5	-104	-29	0.127	22.000	1.00	No
Tubesheet: SS[13]	B	-22	-104	-29	0.200	12.500	1.00	No
Warmest MDMT:								
		-2	-104					
LHS Blind Flg[11]	B	11	4		0.871	39.250	1.00	No
LHS CH.Flgs [11]	B	-18	-39	-29	0.634	14.000	1.00	No
RHS Ch Flg [11]	B	-18	-39	-29	0.634	14.000	1.00	No
RHS Blind Fla[11]	B	11	4		0.871	39.250	1.00	No
LHS CH Shell [8]	B	-18	-28	-29	0.823	14.000	0.85	No
RHS Ch Shell [8]	B	-14	-31	-29	0.695	16.000	0.85	No
N5-Tube Inlet [1]	B	-1	-104	-29	0.182	12.490	1.00	No
Nozzle Flg [5]	A	-18	-44					
N6-Tube Outlet[1]	B	-1	-104	-29	0.185	12.490	1.00	No
Nozzle Flg [5]	A	-18	-43					
Tubesheet: CS[14]	B	-18	-34	-29	0.722	14.000	1.00	No
Bolting [21]		-48						
Bolting [21]		-48						
Warmest MDMT:								
		11	4					

Exchanger Side	Computed MDMT	Required MDMT	Pass/Fail
Shell	-104.0	-12.0	Pass
Channel/Tube	4.0	-12.0	Fail

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.
 UCS-66(g) was not considered.
 UCS-66(i) was not considered.



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MDMT Summary: Step: 32 6:24pm Sep 8, 2026

Notes:

Impact test temps were not entered in and not considered in the analysis.
UCS-66(i) applies to impact tested materials not by specification and
UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).
The Basic MDMT includes the (30F) PWHT credit if applicable.

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FileName : 110-C-AK_Channel side= CS- 05-03-26_Latest Page 10 of 216
Warnings and Errors: Step: 0 6:24pm Sep 8, 2026

Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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 FileName : 110-C-AK_Channel side= CS- 05-03-26_Latest Page 11 of 216
 Input Echo: Step: 1 6:24pm Sep 8,2026

Units used in this Analysis (MM):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm
Force	Pounds	0.4536	Kgf
Mass	Pounds	0.4536	kg.
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	Kg-m.
Stress	lbs./sq.in.	0.0069	N./mm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	6.8946	KPa.
Elast. Modulus	lbs./sq.in.	6.8946	KPa.
Pipe Density	lbs./cu.in.	0.0277	kg./cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Wind Speed	miles/hr	1.6093	Km/hr
Tray Weight	lbs./sq.ft.	0.0005	Kgs/cm ²
Inertia	in.4	41.6231	cm4
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8825	Kgs/m ²
Elevation	Feet	304.8000	mm
Volume	in.3	16.3900	cm ³
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Exchanger Design Pressures and Temperatures

Shell Side Design Pressure	380	KPa.
Channel Side Design Pressure	999.99	KPa.
Shell Side Design Temperature	130.0	°C
Channel Side Design Temperature	82.0	°C
Radiography, Shell Side	RT-1	
Radiography, Channel Side	RT-1	
Service Type, Shell Side	None	
Service Type, Channel Side	None	
MDMT (CET), Shell Side	-12.0	°C
MDMT (CET), Tube Side	-12.0	°C
User defined MAWP, Shell Side	0	KPa.
User defined MAWP, Channel Side	0	KPa.
User defined MAPnc, Shell Side	0	KPa.
User defined MAPnc, Channel Side	0	KPa.
User defined Test Pres., Shell Side	0	KPa.
User defined Test Pres., Channel Side	0	KPa.

Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Type of Construction	Welded	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	UG-99(c)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	No



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 FileName : 110-C-AK_Channel side= CS- 05-03-26_Latest Page 12 of 216
 Input Echo: Step: 1 6:24pm Sep 8,2026

Field Pressure Test:

Type of Pressure Test UG-99(b) Note [35]
 Pressure Test Position Horizontal
 Test Performed in Corroded Condition Yes

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	ASCE/SEI 7-16	
Wind Load Reduction Scale Factor	1.100	
Basic Wind Speed	92.7	Km/hr
Surface Roughness Category	C: Open Terrain	
Type of Surface	Moderately Smooth	
Base Elevation	27500	mm
Percent Wind for Hydrotest	50.0	
Using User defined Wind Press. Vs Elev.	N	
Height of Hill or Escarpment H or Hh	0	mm
Distance Upwind of Crest Lh	0	mm
Distance from Crest to the Vessel x	0	mm
Type of Terrain (Hill, Escarpment)	Flat	
Damping Factor (Beta) for Wind (Ope)	0.0100	
Damping Factor (Beta) for Wind (Empty)	0.0000	
Damping Factor (Beta) for Wind (Filled)	0.0000	
Site Elevation above Sea Level	0	mm

Seismic Design Code	ASCE/SEI 7-16	
Seismic Load Reduction Scale Factor	1.000	
Importance Factor	1.250	
Table Value Fa	1.600	
Table Value Fv	2.400	
Max. Mapped Res. Acceleration [Ss]	0.147	
Max. Eff. Ground Acceleration [S]	0.065	
Force Modification Factor R	3.000	
Site Class	D	
Component Elevation Ratio z/h	0.000	
Amplification Factor Ap	1.000	
Force Factor	0.000	
Consider Vertical Acceleration	Yes	
Minimum Acceleration Multiplier	0.000	
User Value of Sds (used if > 0)	0.000	
User Value of Sd1 (used if > 0)	0.000	
Moment Reduction Factor Tau	1.000	

M.A.W.P + Static Head to Nozzle	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y



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Use ASME VIII-1 Appendix 1-9 N
 Perform Blast Load Analysis No
 Material Database Year Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk) Yes
 Do not use Nozzle MDMT Interpretation VIII-1 01-37 No
 Use Table G instead of exact equation for "A" Yes
 Shell Head Joints are Tapered Yes
 Compute "K" in corroded condition Yes
 Use Code Case 2286 No
 Use the MAWP to compute the MDMT Yes
 For thickness ratios <= 0.35, MDMT will be -155F (-104C) Yes
 For PWHT & P1 Materials the MDMT can be < -55F (-48C) No
 Using Metric Material Databases, ASME II D Yes
 Calculate B31.3 type stress for Nozzles with Loads Yes
 Reduce the MDMT due to lower membrane stress Yes
 Consider Longitudinal Stress in MDMT Calculations Yes

Complete Listing of Vessel Elements and Details:

Element#	1/9	
Element From Node	10	
Element To Node	20	
Element Type	Flange	
Description	LHS Blind Flg	
Distance "FROM" to "TO"	157	mm
Flange Outside Diameter	2505	mm
Element Thickness	157	mm
Internal Corrosion Allowance	3.2	mm
Nominal Thickness	165	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	1000	KPa.
Design Temperature Internal Pressure	82	°C
Design External Pressure	103.42	KPa.
Design Temperature External Pressure	82	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-266 2	
Allowable Stress, Ambient	138	N./mm ²
Allowable Stress, Operating	138	N./mm ²
Allowable Stress, Hydrotest	225	N./mm ²
Material Density	0.00775	kg./cm ³
P Number Thickness	32.004	mm
Yield Stress, Operating	230.08	N./mm ²
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K03506	
Product Form	Forgings	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	
Element From Node	10	
Detail Type	Liquid	
Detail ID	Liquid: 10	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	2555	mm
Liquid Density	0.0009984	kg./cm ³

Element# 2/9
 Element From Node 20



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Element To Node	30
Element Type	Flange
Description	LHS CH.Fl
Distance "FROM" to "TO"	265 mm
Flange Inside Diameter	2250 mm
Element Thickness	214 mm
Internal Corrosion Allowance	3.2 mm
Nominal Thickness	222 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1000 KPa.
Design Temperature Internal Pressure	82 °C
Design External Pressure	103.42 KPa.
Design Temperature External Pressure	82 °C
Effective Diameter Multiplier	1.2
Material Name	SA-266 2
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No

Element From Node	20
Detail Type	Liquid
Detail ID	Liquid: 20
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	2300 mm
Liquid Density	0.0009984 kg./cm ³

Element#	3/9
Element From Node	30
Element To Node	40
Element Type	Cylinder
Description	LHS CH Shell
Distance "FROM" to "TO"	1000 mm
Inside Diameter	2250 mm
Element Thickness	14 mm
Internal Corrosion Allowance	3.2 mm
Nominal Thickness	14 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1000 KPa.
Design Temperature Internal Pressure	82 °C
Design External Pressure	103.42 KPa.
Design Temperature External Pressure	82 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Allowable Stress, Ambient	138 N./mm ²
Allowable Stress, Operating	138 N./mm ²
Allowable Stress, Hydrotest	234 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	32.004 mm
Yield Stress, Operating	242.6 N./mm ²
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K02700
Product Form	Plate
Efficiency, Longitudinal Seam	0.85
Efficiency, Circumferential Seam	0.85
Weld is pre-Heated	No

Element From Node	30
Detail Type	Liquid
Detail ID	Liquid: 30
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	2250 mm
Liquid Density	0.0009984 kg./cm ³



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Element From Node	30
Detail Type	Nozzle
Detail ID	N5-Tube Inlet
Dist. from "FROM" Node / Offset dist	500 mm
Nozzle Diameter	18 in.
Nozzle Schedule	40
Nozzle Class	150
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	110.78 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-106 B

Element From Node	30
Detail Type	Weight
Detail ID	Partition weigh
Dist. from "FROM" Node / Offset dist	500 mm
Miscellaneous Weight	200 Kg
Offset from Element Centerline	0 mm

Element#	4/9
Element From Node	40
Element To Node	50
Element Type	Cylinder
Description	main Shell 1
Distance "FROM" to "TO"	3240 mm
Inside Diameter	2250 mm
Element Thickness	11 mm
Internal Corrosion Allowance	0.8 mm
Nominal Thickness	12 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	380 KPa.
Design Temperature Internal Pressure	130 °C
Design External Pressure	51.709 KPa.
Design Temperature External Pressure	130 °C
Effective Diameter Multiplier	1.2
Material Name	SA-240 304
Allowable Stress, Ambient	138 N./mm ²
Allowable Stress, Operating	106.2 N./mm ²
Allowable Stress, Hydrotest	184.5 N./mm ²
Material Density	0.008027 kg./cm ³
P Number Thickness	0 mm
Yield Stress, Operating	159.6 N./mm ²
External Pressure Chart Name	HA-1
UNS Number	S30400
Product Form	Plate
Efficiency, Longitudinal Seam	0.85
Efficiency, Circumferential Seam	0.85
Weld is pre-Heated	No

Element From Node	40
Detail Type	Saddle
Detail ID	Left Saddle flo
Dist. from "FROM" Node / Offset dist	900 mm
Width of Saddle	325 mm
Height of Saddle at Bottom	1760 mm
Saddle Contact Angle	167.0
Height of Composite Ring Stiffener	0 mm
Width of Wear Plate	560 mm
Thickness of Wear Plate	20 mm
Contact Angle, Wear Plate (degrees)	180.0
Friction coefficient	0.3000001
Moment Factor	3.0



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Dimension E at base (optional)	0	mm
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	mm
Element From Node	40	
Detail Type	Liquid	
Detail ID	Liquid: 40	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	2250	mm
Liquid Density	0.0009984	kg./cm ³
Element From Node	40	
Detail Type	Weight	
Detail ID	Fire prrofig	
Dist. from "FROM" Node / Offset dist	900.01	mm
Miscellaneous Weight	450	Kgf
Offset from Element Centerline	1760	mm

Element#	5/9
Element From Node	50
Element To Node	60
Element Type	Cylinder
Description	main Shell 2
Distance "FROM" to "TO"	1750 mm
Inside Diameter	2250 mm
Element Thickness	24 mm
Internal Corrosion Allowance	0.8 mm
Nominal Thickness	24 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	380 KPa.
Design Temperature Internal Pressure	130 °C
Design External Pressure	51.709 KPa.
Design Temperature External Pressure	130 °C
Effective Diameter Multiplier	1.2
Material Name	SA-240 304
Efficiency, Longitudinal Seam	0.85
Efficiency, Circumferential Seam	0.85
Weld is pre-Heated	No

Element From Node	50
Detail Type	Liquid
Detail ID	Liquid: 50
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	2250 mm
Liquid Density	0.0009984 kg./cm ³

Element From Node	50
Detail Type	Nozzle
Detail ID	N1-Shell inlet
Dist. from "FROM" Node / Offset dist	874 mm
Nozzle Diameter	36 in.
Nozzle Schedule	STD
Nozzle Class	150
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	537.98 Kgf
Grade of Attached Flange	GR 2.1
Nozzle Matl	SA-312 TP304

Element From Node	50
Detail Type	Nozzle
Detail ID	N4-Shell outlet



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Dist. from "FROM" Node / Offset dist	874	mm
Nozzle Diameter	36	in.
Nozzle Schedule	STD	
Nozzle Class	150	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	536.9	Kgf
Grade of Attached Flange	GR 2.1	
Nozzle Matl	SA-312 TP304	

Element#	6/9
Element From Node	60
Element To Node	70
Element Type	Cylinder
Description	main Shell 2
Distance "FROM" to "TO"	3210 mm
Inside Diameter	2250 mm
Element Thickness	11 mm
Internal Corrosion Allowance	0.8 mm
Nominal Thickness	12 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	380 KPa.
Design Temperature Internal Pressure	130 °C
Design External Pressure	51.709 KPa.
Design Temperature External Pressure	130 °C
Effective Diameter Multiplier	1.2
Material Name	SA-240 304
Efficiency, Longitudinal Seam	0.85
Efficiency, Circumferential Seam	0.85
Weld is pre-Heated	No

Element From Node	60
Detail Type	Saddle
Detail ID	Right saddle fi
Dist. from "FROM" Node / Offset dist	2290 mm
Width of Saddle	325 mm
Height of Saddle at Bottom	1760 mm
Saddle Contact Angle	167.0
Height of Composite Ring Stiffener	0 mm
Width of Wear Plate	560 mm
Thickness of Wear Plate	20 mm
Contact Angle, Wear Plate (degrees)	180.0
Friction coefficient	0.0
Moment Factor	3.0
Dimension E at base (optional)	0 mm
Circumferential Eff. over Saddle	1.0
Circumferential Eff. at Midspan	1.0
Tangent to Tangent dist. (optional)	0 mm

Element From Node	60
Detail Type	Liquid
Detail ID	Liquid: 40
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	2250 mm
Liquid Density	0.0009984 kg./cm³

Element From Node	60
Detail Type	Weight
Detail ID	Fireproofing we
Dist. from "FROM" Node / Offset dist	2290 mm
Miscellaneous Weight	450 Kgf
Offset from Element Centerline	1760 mm



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Element#	7/9
Element From Node	70
Element To Node	80
Element Type	Cylinder
Description	RHS Ch Shell
Distance "FROM" to "TO"	1000 mm
Inside Diameter	2250 mm
Element Thickness	16 mm
Internal Corrosion Allowance	3.2 mm
Nominal Thickness	16 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1000 KPa.
Design Temperature Internal Pressure	82 °C
Design External Pressure	103.42 KPa.
Design Temperature External Pressure	82 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Allowable Stress, Ambient	138 N./mm ²
Allowable Stress, Operating	138 N./mm ²
Allowable Stress, Hydrotest	234 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	32.004 mm
Yield Stress, Operating	242.6 N./mm ²
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K02700
Product Form	Plate
Efficiency, Longitudinal Seam	0.85
Efficiency, Circumferential Seam	0.85
Weld is pre-Heated	No
Element From Node	70
Detail Type	Liquid
Detail ID	Liquid: 50
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	2250 mm
Liquid Density	0.0009984 kg./cm ³
Element From Node	70
Detail Type	Nozzle
Detail ID	N6-Tube Outlet
Dist. from "FROM" Node / Offset dist	500 mm
Nozzle Diameter	18 in.
Nozzle Schedule	40
Nozzle Class	150
Layout Angle	180.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	110.78 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-106 B

Element#	8/9
Element From Node	80
Element To Node	90
Element Type	Flange
Description	RHS Ch Flg
Distance "FROM" to "TO"	265 mm
Flange Inside Diameter	2250 mm
Element Thickness	214 mm
Internal Corrosion Allowance	3.2 mm
Nominal Thickness	222 mm



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External Corrosion Allowance	0	mm
Design Internal Pressure	1000	KPa.
Design Temperature Internal Pressure	82	°C
Design External Pressure	103.42	KPa.
Design Temperature External Pressure	82	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-266 2	
Allowable Stress, Ambient	138	N./mm ²
Allowable Stress, Operating	138	N./mm ²
Allowable Stress, Hydrotest	225	N./mm ²
Material Density	0.00775	kg./cm ³
P Number Thickness	32.004	mm
Yield Stress, Operating	230.08	N./mm ²
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K03506	
Product Form	Forgings	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	
Element From Node	80	
Detail Type	Liquid	
Detail ID	Liquid: 60	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	2300	mm
Liquid Density	0.0009984	kg./cm ³

Element#	9/9
Element From Node	90
Element To Node	100
Element Type	Flange
Description	RHS Blind Flang
Distance "FROM" to "TO"	157 mm
Flange Outside Diameter	2505 mm
Element Thickness	157 mm
Internal Corrosion Allowance	3.2 mm
Nominal Thickness	165 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1000 KPa.
Design Temperature Internal Pressure	82 °C
Design External Pressure	103.42 KPa.
Design Temperature External Pressure	82 °C
Effective Diameter Multiplier	1.2
Material Name	SA-266 2
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No
Element From Node	90
Detail Type	Liquid
Detail ID	Liquid: 70
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	2551 mm
Liquid Density	0.0009984 kg./cm ³



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XY Coordinate Calculations: Step: 2 6:24pm Sep 8,2026

XY Coordinate Calculations:

From	To	X (Horiz.) mm	Y (Vert.) mm	DX (Horiz.) mm	DY (Vert.) mm
LHS Blind Flg		163.35	...	157	...
LHS CH.Fl g		434.7	...	265	...
LHS CH Shell		1434.7	...	1000	...
main Shell 1		4724.7	...	3240	...
main Shell 2		6474.7	...	1750	...
main Shell 2		9684.7	...	3210	...
RHS Ch Shell		10734.7	...	1000	...
RHS Ch Flg		11006	...	265	...
RHS Blind Flang		11169.4	...	157	...

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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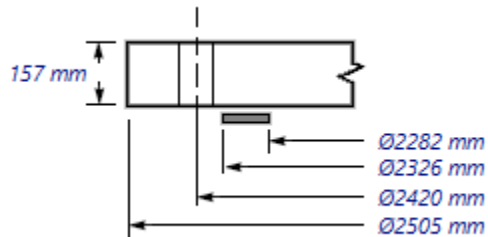
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 Flg Calc [Int P]: New Flange Flng: 9 6:24pm Sep 8,2026

Flange Input Data Values

Description: New Flange :

LHS Blind Flg

Flange Type	TEMA Channel Cover		
Design Pressure	P	1025.04	KPa.
Design Temperature		82	°C
Internal Corrosion Allowance	ci	3.2000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Outside Diameter	A	2505.000	mm
Flange Thickness	t	157.0000	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	138.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Perimeter along the Center of the Bolts	L	7602.654	mm
Youngs Modulus for Blind Flange Matl.	Sy	197846.66	N./mm ²
Allowed Channel Cover Deflection	dfAllow	2.8125	mm
Diameter of Bolt Circle	C	2420.000	mm
Nominal Bolt Diameter	a	34.9250	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		80	
Flange Face Outside Diameter	Fod	2337.000	mm
Flange Face Inside Diameter	Fid	2250.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	2326.000	mm
Gasket Inside Diameter	Gi	2282.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	6.3500	mm
Raised face depth	Fd	0.0000	mm
Length of Partition Gasket	lp	2300.0000	mm
Width of Partition Gasket	tp	10.0000	mm
Partition Gasket Factor	mPart	3.0000	
Partition Gasket Design Seating Stress	yPart	68.95	N./mm ²





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 Flg Calc [Int P]: New Flange Flng: 9 6:24pm Sep 8,2026

Gasket Contact Width,	$N = (G_o - G_i)/2$	22.000 mm
Basic Gasket Width,	$b_o = N/2$	11.000 mm
Effective Gasket Width,	$b = C_b \cdot \sqrt{b_o}$	8.358 mm
Gasket Reaction Diameter,	$G = G_o - 2 \cdot b$	2309.285 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 2309^2 \cdot 1025$$

$$= 437808.312 \text{ Kg}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P + 2 \cdot l_p \cdot b_{Part} \cdot m_{Part} \cdot P$$

$$= 2 \cdot 8.36 \cdot \pi \cdot 2309 \cdot 3 \cdot 1025$$

$$+ 2 \cdot 2300 \cdot 5 \cdot 3 \cdot 1025$$

$$= 45240.332 \text{ Kg}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(W_{o,1} + W_{o,2} + H'p, 0)$$

$$= \max(437808 + 45240 + 0, 0)$$

$$= 483048.594 \text{ Kg}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.9 \cdot 8.36 \cdot \pi \cdot 2309 + 68.9 \cdot 5 \cdot 2300$$

$$= 507155.281 \text{ Kg}$$

Required Bolt Area [Am]:

$$= \max((W_o + F_a + 4M_e/G) / S_b, W_g / S_a)$$

$$= \max((483049 + 0 + 4 \cdot 0 / 2309) / 172, 507155 / 172)$$

$$= 289.167 \text{ cm}^2$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	289.167	596.128	
Radial Distance between Bolts and Edge:	34.925	42.500	
Circ. Spacing between the Bolts:	77.800	95.009	...

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= A_b \cdot S_a / (y \cdot \pi (G_o + G_i))$$

$$= 596 \cdot 172 / (68.9 \cdot \pi (2326 + 2282))$$

$$= 10.272 \text{ mm}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= S_a \cdot A_b$$

$$= 172 \cdot 596$$

$$= 1045517.50 \text{ Kg}$$

Gasket Load for the Operating Condition [HG]:

$$= W_o$$

$$= 483048.59 \text{ Kg}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (2420 - 2309) / 2$$

$$= 55.3576 \text{ mm}$$

Flange Stresses



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 Flg Calc [Int P]: New Flange Flng: 9 6:24pm Sep 8,2026

Tangential Flange Stress, Flat Head (4.6.2.3), Operating [STo]:
 $= 1.9 \cdot Wm1 \cdot hG \cdot Bsc / (t^2 \cdot G) + C \cdot Z \cdot Peq \cdot G^2 / t^2$
 $= 1.9 \cdot 483049 \cdot 55.4 \cdot 1 / (154^2 \cdot 2309) + 0.3 \cdot 1 \cdot 1025 \cdot 2309^2 / 154^2$
 $= 78.45 \text{ N./mm}^2$

Tangential Flange Stress, Flat Head (4.6.2.4), Seating [STa]:
 $= 1.9 \cdot W \cdot hG \cdot Bsc / (t^2 \cdot G)$
 $= 1.9 \cdot 1045517 \cdot 55.4 \cdot 1 / (154^2 \cdot 2309)$
 $= 19.74 \text{ N./mm}^2$

Bolt Stress, Operating [BSO]:
 $= Wm1 / Ab$
 $= 483049 / 596$
 $= 79.47 \text{ N./mm}^2$

Bolt Stress, Seating [BSa]:
 $= (Wm2 / Ab)$
 $= (507155 / 596)$
 $= 83.43 \text{ N./mm}^2$

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Actual	Gasket Seating Allowed
Tangential Flange	78.45	138.00	19.74	138.00
Bolting	79.47	172.00	83.43	172.00

TEMA Channel Cover Deflection Calculations:

Actual deflection at center per TEMA formula 2.6724 mm
 Allowed deflection at center of Cover 2.8125 mm

Reqd. Blind Flange Thickness at Center 119.165 mm
 Reqd. Blind Flange Thickness at Gasket 61.373 mm
 Min. Required Flange Thickness per TEMA 151.202 mm

Estimated M.A.W.P. (Operating) 1107 KPa.
 Estimated Finished Weight of Flange at given Thk. 5997.0 kg.
 Estimated Unfinished Weight of Forging at given Thk 5997.0 kg.

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.87, Temperature Reduction per Fig. UCS 66.1 = 7 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 11 °C
 Min Metal Temp. at Required thickness (UCS 66.1) 4 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Actual	Gasket Seating Allowed
Tangential Flange	75.29	138.00	18.95	138.00
Bolting	79.47	172.00	83.43	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 150.343 mm

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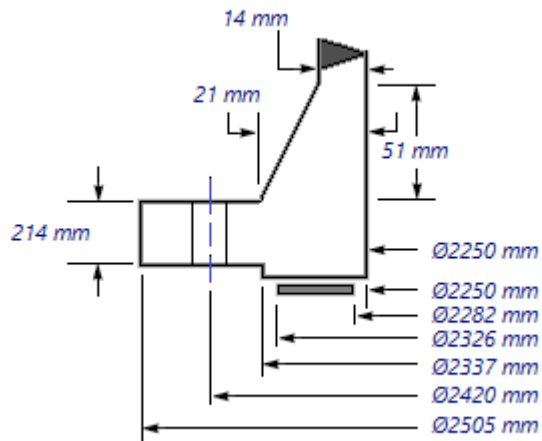
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Flange Input Data Values

Description: New Flange :

LHS CH.Flq

Flange Type	Integral Weld Neck, Integral type		
Design Pressure	P	1022.54	KPa.
Design Temperature		82	°C
Internal Corrosion Allowance	ci	3.2000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	2250.000	mm
Flange Outside Diameter	A	2505.000	mm
Flange Thickness	t	214.0000	mm
Thickness of Hub at Small End	go	14.0000	mm
Thickness of Hub at Large End	gl	21.0000	mm
Length of Hub	h	51.0000	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	138.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Diameter of Bolt Circle	C	2420.000	mm
Nominal Bolt Diameter	a	34.9250	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		80	
Flange Face Outside Diameter	Fod	2337.000	mm
Flange Face Inside Diameter	Fid	2250.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	2326.000	mm
Gasket Inside Diameter	Gi	2282.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	6.3500	mm
Raised face depth	Fd	0.0000	mm
Length of Partition Gasket	lp	2300.0000	mm
Width of Partition Gasket	tp	10.0000	mm
Partition Gasket Factor	mPart	3.0000	
Partition Gasket Design Seating Stress	yPart	68.95	N./mm ²



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Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + C_a)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (1023 \cdot (2250/2 + 3.2)) / (138 \cdot 1 - 0.6 \cdot 1023) + C_a$$

$$= 11.5976 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 10.8) / (1128 + 0.6 \cdot 10.8)$$

$$= 1313.398 \text{ KPa.}$$

Hub Small End Hub MAPnc:

$$= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 14) / (1125 + 0.6 \cdot 14)$$

$$= 1704.476 \text{ KPa.}$$

Corroded Flange Thickness, $t_c = T - c_i$	210.800 mm
Corroded Flange ID, $B_{cor} = B + 2 \cdot c_i$	2256.400 mm
Corroded Large Hub, $g1_{cor} = g1 - c_i$	17.800 mm
Corroded Small Hub, $g0_{cor} = g0 - c_i$	10.800 mm
Code R Dimension, $R = ((C - B_{cor})/2) - g1_{cor}$	64.000 mm

Gasket Contact Width, $N = (G_o - G_i)/2$	22.000 mm
Basic Gasket Width, $b_o = N/2$	11.000 mm
Effective Gasket Width, $b = C_b \cdot \sqrt{b_o}$	8.358 mm
Gasket Reaction Diameter, $G = G_o - 2 \cdot b$	2309.285 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 2309^2 \cdot 1023$$

$$= 436740.719 \text{ Kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P + 2 \cdot l_p \cdot b_{part} \cdot m_{part} \cdot P$$

$$= 2 \cdot 8.36 \cdot \pi \cdot 2309 \cdot 3 \cdot 1023$$

$$+ 2 \cdot 2300 \cdot 5 \cdot 3 \cdot 1023$$

$$= 45130.016 \text{ Kgf}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot B_{cor}^2 \cdot P$$

$$= \pi/4 \cdot 2256^2 \cdot 1023$$

$$= 416966.219 \text{ Kgf}$$



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Pressure Force on Flange Face [Ht]:

$$= H - H_d$$

$$= 436741 - 416966$$

$$= 19774.492 \text{ Kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(W_{o,1} + W_{o,2} + H'p, 0)$$

$$= \max(436741 + 45130 + 0, 0)$$

$$= 481870.719 \text{ Kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.9 \cdot 8.36 \cdot \pi \cdot 2309 + 68.9 \cdot 5 \cdot 2300$$

$$= 507155.281 \text{ Kgf}$$

Required Bolt Area [Am]:

$$= \max((W_o + F_a + 4Me/G) / S_b, W_g / S_a)$$

$$= \max((481871 + 0 + 4 \cdot 0 / 2309) / 172, 507155 / 172)$$

$$= 289.167 \text{ cm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmx]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 \cdot 34.9 + 6 \cdot 211 / (3 + 0.5)$$

$$= 431.221 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 2420 \cdot \sin(\pi / 80)$$

$$= 95.009 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{ Bs / (2a + t) }, 1)$$

$$= \max(\sqrt{ 95 / (2 \cdot 34.9 + 211) }, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	289.167	596.128	
Radial Distance between Hub and Bolts:	47.625	64.000	
Radial Distance between Bolts and Edge:	34.925	42.500	
Circ. Spacing between the Bolts:	77.800	95.009	431.221

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= A_b \cdot S_a / (y \cdot \pi (G_o + G_i))$$

$$= 596 \cdot 172 / (68.9 \cdot \pi (2326 + 2282))$$

$$= 10.272 \text{ mm}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 19 mm
 Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= S_a \cdot A_b$$

$$= 172 \cdot 596$$

$$= 1045517.50 \text{ Kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= W_o - H$$

$$= 481871 - 436741$$

$$= 45130.00 \text{ Kgf}$$



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Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:
 $= (C - G) / 2$
 $= (2420 - 2309) / 2$
 $= 55.3576 \text{ mm}$

Distance to Face Pressure Reaction [ht]:
 $= 0.5 ((C - B) / 2 + hg)$
 $= 0.5 ((2420 - 2256) / 2 + 55.4)$
 $= 68.5788 \text{ mm}$

Distance to End Pressure Reaction [hd]:
 $= (C - B - g1) / 2$
 $= (2420 - 2256 - 17.8) / 2$
 $= 72.8999 \text{ mm}$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	416966.	72.8999	1.0000	30397.
Face Pressure, Mt	19774.	68.5788	1.0000	1356.
Gasket Load, Mg	45130.	55.3576	1.0000	2498.
Gasket Seating, Matm	1045518.	55.3576	1.0000	57878.

Total Moment for Operation, Mo 34252. Kg-m.
 Total Moment for Gasket seating, Mg 57878. Kg-m.

Effective Hub Length, $h_o = \sqrt{B_{cor} \cdot g_{oCor}}$ 156.106 mm
 Hub Ratio, $h/h_o = H_L / H_o$ 0.327
 Thickness Ratio, $g1/g_o = (g1_{cor}/g_{oCor})$ 1.648

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.871
 Factor V per 4.16.5 0.335
 Factor f per 4.16.5 1.320

Stress factor from Table 4.16.4 [K]:

$= A / B$
 $= 2505/2256$
 $= 1.110$

Stress factor from Table 4.16.4 [Y]:

$= (0.66845 + (5.7169 (K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$
 $= (0.66845 + (5.7169 (1.11^2 \cdot \log_{10}(1.11))) / (1.11^2 - 1)) / (1.11 - 1)$
 $= 18.553$

Stress factor from Table 4.16.4 [T]:

$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$
 $= (1.11^2 (1 + 8.55246 \cdot \log_{10}(1.11)) - 1) / ((1.0472 + 1.9448 \cdot 1.11^2) \cdot (1.11 - 1))$
 $= 1.874$

Stress factor from Table 4.16.4 [U]:

$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136 (K^2 - 1) \cdot (K - 1))$
 $= (1.11^2 (1 + 8.55246 \cdot \log_{10}(1.11)) - 1) / (1.36136 (1.11^2 - 1) \cdot (1.11 - 1))$
 $= 20.388$

Stress factor from Table 4.16.4 [Z]:

$= (K^2 + 1) / (K^2 - 1)$
 $= (1.11^2 + 1) / (1.11^2 - 1)$
 $= 9.603$

Stress factor from Table 4.16.4 for integral flanges [d]:

$= U \cdot g_o^2 \cdot h_o / V$



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$$= 20.4 \cdot 10.8^2 \cdot 156/0.33$$

$$= 1109739.375 \text{ mm}^3$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$= F / h_0$$

$$= 0.87/156$$

$$= 0.006 \text{ 1/mm}$$

Stress factor from Table 4.16.4 [L]:

$$= (t \cdot e + 1) / T + t^3/d$$

$$= (211 \cdot 0.0056 + 1) / 1.87 + 211^3/1109739$$

$$= 9.603$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot Mop / Bcor) / (L \cdot g^{12})$$

$$= (1.32 \cdot 34252/2256) / (9.6 \cdot 17.8^2)$$

$$= 64.59 \text{ N./mm}^2$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot Matm / Bcor) / (L \cdot g^{12})$$

$$= (1.32 \cdot 57878/2256) / (9.6 \cdot 17.8^2)$$

$$= 109.14 \text{ N./mm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$$

$$= (2.57 \cdot 34252/2256) / (9.6 \cdot 211^2)$$

$$= 0.90 \text{ N./mm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm/Bcor) / (L \cdot t^2)$$

$$= (2.57 \cdot 57878/2256) / (9.6 \cdot 211^2)$$

$$= 1.51 \text{ N./mm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRO$$

$$= (18.6 \cdot 34252 / (211^2 \cdot 2256)) - 9.6 \cdot 0.9$$

$$= 53.55 \text{ N./mm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRA$$

$$= (18.6 \cdot 57878 / (211^2 \cdot 2256)) - 9.6 \cdot 1.51$$

$$= 90.49 \text{ N./mm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (SHo + \max(SRo, STo)) / 2$$

$$= (64.6 + \max(0.9, 53.5)) / 2$$

$$= 59.07 \text{ N./mm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (SHa + \max(SRa, STa)) / 2$$

$$= (109 + \max(1.51, 90.5)) / 2$$

$$= 99.81 \text{ N./mm}^2$$

Bolt Stress, Operating [BSO]:

$$= Wm1 / Ab$$

$$= 481871/596$$

$$= 79.27 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (Wm2 / Ab)$$

$$= (507155/596)$$

$$= 83.43 \text{ N./mm}^2$$



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Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	64.59	207.00	109.14	207.00
Radial Flange	0.90	138.00	1.51	138.00
Tangential Flange	53.55	138.00	90.49	138.00
Maximum Average	59.07	138.00	99.81	138.00
Bolting	79.27	172.00	83.43	172.00

Minimum Required Flange Thickness [Rigidity] 208.940 mm

Estimated M.A.W.P. (Operating) 1313 KPa.
 Estimated Finished Weight of Flange at given Thk. 1628.8 kg.
 Estimated Unfinished Weight of Forging at given Thk 1955.9 kg.

Flange Stress results per WRC-538 at computed S_{fmax} (N./mm²):

		Actual	Operating Allowed	stress ratio	Gasket Actual	Seating Allowed	stress ratio
Longitudinal Hub	Sh	452.03	499.99	0.904	452.03	499.99	0.904
Tangential Flange	St	374.99	374.99	1.000	374.99	374.99	1.000
Tangential Flange	Sto	135.43	250.00	0.542	135.43	250.00	0.542
Combined	Sh/f + St	717.63	749.99	0.957	717.63	749.99	0.957
Combined	Sr + St	381.27	749.99	0.508	381.27	749.99	0.508
Combined	Sh + Sto	587.46	749.99	0.783	587.46	749.99	0.783
Radial Flange	Sr	6.28	499.99	0.013	6.28	499.99	0.013
Maximum Stress Ratio				1.000			1.000



Coefficients for f and $f2$

	f	$f2$	$f3$
a	-0.71375912	-0.01447638	-0.06456312
b	-0.12846279	-0.27814745	-0.08232239
c	1.08037907	0.01035395	0.13691677
d	0.99766848	1.37984158	-0.77879888
e	1.21823466	-0.34346764	0.14909019
f	0.01024612	0.02536192	0.00660062
g	0.4262313	-0.0014508	0.0616252
h	1.4204976	2.25543162	0.74639834
i	-0.70278181	-0.60889597	-0.079595
j	-0.02483937	-0.15735462	0.09851673
k	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g1}{g0} \right) + e \left(\frac{h}{h0} \right) + g \left(\frac{g1}{g0} \right)^2 + i \left(\frac{h}{h0} \right)^2 + k \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}{\left[1 + b \left(\frac{g1}{g0} \right) + d \left(\frac{h}{h0} \right) + f \left(\frac{g1}{g0} \right)^2 + h \left(\frac{h}{h0} \right)^2 + j \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}$$

Hub Stress Correction Factor [f]:

= max(1; F1num / F1den)
 = max(1; 1.69/1.28)
 = 1.3193

Hub Stress Correction Factor [f2]:

= max(0.25; fn2) if h/ho < 0.35, else max (0.25; fn3)
 = max(0.25; 0.48/1.22) if h/ho < 0.35, else max (0.25; 0.31/0.76)
 = 0.3953

In this case, the hub ratio h/ho, 0.3267 was < 0.35.

PV Elite iterates on the assembly bolt load stress, Sfmax, until it exceeds the limits provided by WRC 538.

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress	Sbmax	507.5 N./mm ²
Minimum permissible bolt stress	Sbmin	290.0 N./mm ²
Max. permissible bolt stress prior to flange damage	Sfmax	712.8 N./mm ²
Target assembly gasket stress	SgT	207.0 N./mm ²
Maximum permissible gasket stress	Sgmax	276.0 N./mm ²
Minimum gasket operating stress	Sgmin-O	97.0 N./mm ²
Minimum gasket seating stress	Sgmin-S	140.0 N./mm ²
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	Phig	0.700



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Root Area of a Single Bolt	Abs	7.4516 cm ²
Max. permissible single flange rotation for gasket	Thetagmax	1.000 deg

Calculated Results:

Flange Material Ambient Yield Stress	Sya	250.0 N./mm ²
Flange Material Operating Yield Stress	Syo	230.1 N./mm ²
Single flange rotation at Sfmax	Thetafmax	0.610 deg

Gasket Contact Area [Ag]:
 $= \pi/4 (GO.D.^2 - GI.D.^2) + (PartFactor \cdot Ip \cdot bPart)$
 $= \pi/4 (2326^2 - 2282^2) + (0.5 \cdot 2300 \cdot 5)$
 $= 1649.9 \text{ cm}^2$

Target Bolt Stress [Sbsel]: (Eq. O-1)
 $= SgT \cdot Ag / (nb \cdot Ab)$
 $= 207 \cdot 1650 / (80 \cdot 7.45)$
 $= 572.9 \text{ N./mm}^2$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)
 $= \min(Sbsel; Sbmax)$
 $= \min(573; 508)$
 $= 507.5 \text{ N./mm}^2$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)
 $= \max(Sbsel; Sbmin)$
 $= \max(508; 290)$
 $= 507.5 \text{ N./mm}^2$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)
 $= \min(Sbsel; Sfmax)$
 $= \min(508; 713)$
 $= 507.5 \text{ N./mm}^2$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)
 $:: Sbsel \geq Sgmin \cdot S (Ag / (nb \cdot Abs))$
 $:: 508 \geq 140 (1650 / (80 \cdot 7.45))$
 $:: 507.5 \geq 387.5 \text{ N./mm}^2$ Check Passes

Check if Gasket Operating Stress is maintained. (Eq. O-8)
 $:: Sbsel \geq (Sgmin \cdot O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$
 $:: 508 \geq (97 \cdot 1650 + \pi/4 \cdot 1023 \cdot 2282^2) / (0.7 \cdot 7.45 \cdot 80)$
 $:: 507.5 \geq 483.8 \text{ N./mm}^2$ Check Passes

Check if Gasket Maximum Stress is achieved. (Eq. O-9)
 $:: Sbsel \leq Sgmax (Ag / (nb \cdot Abs))$
 $:: 508 \leq 276 (1650 / (80 \cdot 7.45))$
 $:: 507.5 \leq 763.9 \text{ N./mm}^2$ Check Passes

Check if Flange Rotation Limit is exceeded. (Eq. O-10)
 $:: Sbsel \leq Sfmax (thetagmax / thetafmax)$
 $:: 508 \leq 713 (1/0.61)$
 $:: 507.5 \leq 1168.4 \text{ N./mm}^2$ Check Passes

Assembly Bolt Torque [Tb]: (Eq. O-2)
 $= Sbsel \cdot K \cdot Abs \cdot \phi b$
 $= 508 \cdot 0.2 \cdot 7.45 \cdot 34.9$
 $= 269.4 \text{ Kg-m.}$

Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 118 N./mm²
 Bolt Allowable Stress for: SA-193 B7 Sb 171 N./mm²

Note: Since Sb >= SbB8, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:



Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 57878 / 1 \cdot 9807 \cdot 0.33 / (8.99 \cdot 201334 \cdot 10.8^2 \cdot 156 \cdot 0.3)$
 $= 1.000$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 34252 / 1 \cdot 9807 \cdot 0.33 / (8.99 \cdot 197847 \cdot 10.8^2 \cdot 156 \cdot 0.3)$
 $= 0.602$ (should be ≤ 1)

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 57878 / 1 \cdot 9807 \cdot 0.33 / (9.6 \cdot 201334 \cdot 10.8^2 \cdot 156 \cdot 0.3)$
 $= 0.937$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 34252 / 1 \cdot 9807 \cdot 0.33 / (9.6 \cdot 197847 \cdot 10.8^2 \cdot 156 \cdot 0.3)$
 $= 0.564$ (should be ≤ 1)

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.63, Temperature Reduction per Fig. UCS 66.1 = 20 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-18 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-39 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	65.12	207.00	107.86	207.00
Radial Flange	1.25	138.00	2.08	138.00
Tangential Flange	48.37	138.00	80.13	138.00
Maximum Average	56.74	138.00	94.00	138.00
Bolting	79.27	172.00	83.43	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 198.145 mm

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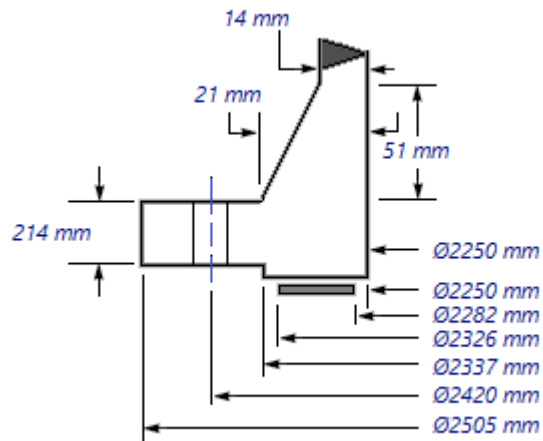
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Flange Input Data Values

Description: New Flange :

RHS Ch Flg

Flange Type	Integral Weld Neck		
Design Pressure	P	1022.54	KPa.
Design Temperature		82	°C
Internal Corrosion Allowance	ci	3.2000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	2250.000	mm
Flange Outside Diameter	A	2505.000	mm
Flange Thickness	t	214.0000	mm
Thickness of Hub at Small End	go	14.0000	mm
Thickness of Hub at Large End	gl	21.0000	mm
Length of Hub	h	51.0000	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	138.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Diameter of Bolt Circle	C	2420.000	mm
Nominal Bolt Diameter	a	34.9250	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		80	
Flange Face Outside Diameter	Fod	2337.000	mm
Flange Face Inside Diameter	Fid	2250.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	2326.000	mm
Gasket Inside Diameter	Gi	2282.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	6.3500	mm
Raised face depth	Fd	0.0000	mm
Length of Partition Gasket	lp	2300.0000	mm
Width of Partition Gasket	tp	10.0000	mm
Partition Gasket Factor	mPart	3.0000	
Partition Gasket Design Seating Stress	yPart	68.95	N./mm ²



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Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + C_a)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (1023 \cdot (2250/2 + 3.2)) / (138 \cdot 1 - 0.6 \cdot 1023) + C_a$$

$$= 11.5976 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 10.8) / (1128 + 0.6 \cdot 10.8)$$

$$= 1313.398 \text{ KPa.}$$

Hub Small End Hub MAPnc:

$$= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 14) / (1125 + 0.6 \cdot 14)$$

$$= 1704.476 \text{ KPa.}$$

Corroded Flange Thickness, $t_c = T - c_i$	210.800 mm
Corroded Flange ID, $B_{cor} = B + 2 \cdot c_i$	2256.400 mm
Corroded Large Hub, $g1_{cor} = g1 - c_i$	17.800 mm
Corroded Small Hub, $g0_{cor} = g0 - c_i$	10.800 mm
Code R Dimension, $R = ((C - B_{cor})/2) - g1_{cor}$	64.000 mm

Gasket Contact Width, $N = (G_o - G_i)/2$	22.000 mm
Basic Gasket Width, $b_o = N/2$	11.000 mm
Effective Gasket Width, $b = C_b \cdot \sqrt{b_o}$	8.358 mm
Gasket Reaction Diameter, $G = G_o - 2 \cdot b$	2309.285 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 2309^2 \cdot 1023$$

$$= 436740.719 \text{ Kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P + 2 \cdot l_p \cdot b_{Part} \cdot m_{Part} \cdot P$$

$$= 2 \cdot 8.36 \cdot \pi \cdot 2309 \cdot 3 \cdot 1023$$

$$+ 2 \cdot 2300 \cdot 5 \cdot 3 \cdot 1023$$

$$= 45130.016 \text{ Kgf}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot B_{cor}^2 \cdot P$$

$$= \pi/4 \cdot 2256^2 \cdot 1023$$

$$= 416966.219 \text{ Kgf}$$



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Pressure Force on Flange Face [Ht]:

$$= H - H_d$$

$$= 436741 - 416966$$

$$= 19774.492 \text{ Kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(W_{o,1} + W_{o,2} + H'p, 0)$$

$$= \max(436741 + 45130 + 0, 0)$$

$$= 481870.719 \text{ Kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.9 \cdot 8.36 \cdot \pi \cdot 2309 + 68.9 \cdot 5 \cdot 2300$$

$$= 507155.281 \text{ Kgf}$$

Required Bolt Area [Am]:

$$= \max((W_o + F_a + 4Me/G) / S_b, W_g / S_a)$$

$$= \max((481871 + 0 + 4 \cdot 0 / 2309) / 172, 507155 / 172)$$

$$= 289.167 \text{ cm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 \cdot 34.9 + 6 \cdot 211 / (3 + 0.5)$$

$$= 431.221 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 2420 \cdot \sin(\pi / 80)$$

$$= 95.009 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{ Bs / (2a + t) }, 1)$$

$$= \max(\sqrt{ 95 / (2 \cdot 34.9 + 211) }, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	289.167	596.128	
Radial Distance between Hub and Bolts:	47.625	64.000	
Radial Distance between Bolts and Edge:	34.925	42.500	
Circ. Spacing between the Bolts:	77.800	95.009	431.221

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= A_b \cdot S_a / (y \cdot \pi (G_o + G_i))$$

$$= 596 \cdot 172 / (68.9 \cdot \pi (2326 + 2282))$$

$$= 10.272 \text{ mm}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 19 mm
 Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= S_a \cdot A_b$$

$$= 172 \cdot 596$$

$$= 1045517.50 \text{ Kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= W_o - H$$

$$= 481871 - 436741$$

$$= 45130.00 \text{ Kgf}$$



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Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:
 $= (C - G) / 2$
 $= (2420 - 2309) / 2$
 $= 55.3576 \text{ mm}$

Distance to Face Pressure Reaction [ht]:
 $= 0.5 ((C - B) / 2 + hg)$
 $= 0.5 ((2420 - 2256) / 2 + 55.4)$
 $= 68.5788 \text{ mm}$

Distance to End Pressure Reaction [hd]:
 $= (C - B - g1) / 2$
 $= (2420 - 2256 - 17.8) / 2$
 $= 72.8999 \text{ mm}$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	416966.	72.8999	1.0000	30397.
Face Pressure, Mt	19774.	68.5788	1.0000	1356.
Gasket Load, Mg	45130.	55.3576	1.0000	2498.
Gasket Seating, Matm	1045518.	55.3576	1.0000	57878.

Total Moment for Operation, Mo 34252. Kg-m.
 Total Moment for Gasket seating, Mg 57878. Kg-m.

Effective Hub Length, $h_o = \sqrt{B_{cor} \cdot g_{oCor}}$ 156.106 mm
 Hub Ratio, $h/h_o = H_L / H_o$ 0.327
 Thickness Ratio, $g1/g_o = (g1_{cor}/g_{oCor})$ 1.648

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.871
 Factor V per 4.16.5 0.335
 Factor f per 4.16.5 1.320

Stress factor from Table 4.16.4 [K]:

$= A / B$
 $= 2505/2256$
 $= 1.110$

Stress factor from Table 4.16.4 [Y]:

$= (0.66845 + (5.7169 (K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$
 $= (0.66845 + (5.7169 (1.11^2 \cdot \log_{10}(1.11))) / (1.11^2 - 1)) / (1.11 - 1)$
 $= 18.553$

Stress factor from Table 4.16.4 [T]:

$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$
 $= (1.11^2 (1 + 8.55246 \cdot \log_{10}(1.11)) - 1) / ((1.0472 + 1.9448 \cdot 1.11^2) \cdot (1.11 - 1))$
 $= 1.874$

Stress factor from Table 4.16.4 [U]:

$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136 (K^2 - 1) \cdot (K - 1))$
 $= (1.11^2 (1 + 8.55246 \cdot \log_{10}(1.11)) - 1) / (1.36136 (1.11^2 - 1) \cdot (1.11 - 1))$
 $= 20.388$

Stress factor from Table 4.16.4 [Z]:

$= (K^2 + 1) / (K^2 - 1)$
 $= (1.11^2 + 1) / (1.11^2 - 1)$
 $= 9.603$

Stress factor from Table 4.16.4 for integral flanges [d]:

$= U \cdot g_o^2 \cdot h_o / V$



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$$= 20.4 \cdot 10.8^2 \cdot 156/0.33$$

$$= 1109739.375 \text{ mm}^3$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$= F / h_0$$

$$= 0.87/156$$

$$= 0.006 \text{ 1/mm}$$

Stress factor from Table 4.16.4 [L]:

$$= (t \cdot e + 1) / T + t^3/d$$

$$= (211 \cdot 0.0056 + 1) / 1.87 + 211^3/1109739$$

$$= 9.603$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot Mop / Bcor) / (L \cdot g^{12})$$

$$= (1.32 \cdot 34252/2256) / (9.6 \cdot 17.8^2)$$

$$= 64.59 \text{ N./mm}^2$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot Matm / Bcor) / (L \cdot g^{12})$$

$$= (1.32 \cdot 57878/2256) / (9.6 \cdot 17.8^2)$$

$$= 109.14 \text{ N./mm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$$

$$= (2.57 \cdot 34252/2256) / (9.6 \cdot 211^2)$$

$$= 0.90 \text{ N./mm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm/Bcor) / (L \cdot t^2)$$

$$= (2.57 \cdot 57878/2256) / (9.6 \cdot 211^2)$$

$$= 1.51 \text{ N./mm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$$

$$= (18.6 \cdot 34252 / (211^2 \cdot 2256)) - 9.6 \cdot 0.9$$

$$= 53.55 \text{ N./mm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa$$

$$= (18.6 \cdot 57878 / (211^2 \cdot 2256)) - 9.6 \cdot 1.51$$

$$= 90.49 \text{ N./mm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (SHo + \max(SRo, STo)) / 2$$

$$= (64.6 + \max(0.9, 53.5)) / 2$$

$$= 59.07 \text{ N./mm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (SHa + \max(SRa, STa)) / 2$$

$$= (109 + \max(1.51, 90.5)) / 2$$

$$= 99.81 \text{ N./mm}^2$$

Bolt Stress, Operating [BSO]:

$$= Wm1 / Ab$$

$$= 481871/596$$

$$= 79.27 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (Wm2 / Ab)$$

$$= (507155/596)$$

$$= 83.43 \text{ N./mm}^2$$



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Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	64.59	207.00	109.14	207.00
Radial Flange	0.90	138.00	1.51	138.00
Tangential Flange	53.55	138.00	90.49	138.00
Maximum Average	59.07	138.00	99.81	138.00
Bolting	79.27	172.00	83.43	172.00

Minimum Required Flange Thickness [Rigidity] 208.940 mm

Estimated M.A.W.P. (Operating) 1313 KPa.
 Estimated Finished Weight of Flange at given Thk. 1628.8 kg.
 Estimated Unfinished Weight of Forging at given Thk 1955.9 kg.

Flange Stress results per WRC-538 at computed Sfmax (N./mm²):

		Actual	Operating Allowed	stress ratio	Gasket Actual	Seating Allowed	stress ratio
Longitudinal Hub	Sh	452.03	499.99	0.904	452.03	499.99	0.904
Tangential Flange	St	374.99	374.99	1.000	374.99	374.99	1.000
Tangential Flange	Sto	135.43	250.00	0.542	135.43	250.00	0.542
Combined	Sh/f + St	717.63	749.99	0.957	717.63	749.99	0.957
Combined	Sr + St	381.27	749.99	0.508	381.27	749.99	0.508
Combined	Sh + Sto	587.46	749.99	0.783	587.46	749.99	0.783
Radial Flange	Sr	6.28	499.99	0.013	6.28	499.99	0.013
Maximum Stress Ratio				1.000			1.000



Coefficients for f and $f2$

	f	$f2$	$f3$
a	-0.71375912	-0.01447638	-0.06456312
b	-0.12846279	-0.27814745	-0.08232239
c	1.08037907	0.01035395	0.13691677
d	0.99766848	1.37984158	-0.77879888
e	1.21823466	-0.34346764	0.14909019
f	0.01024612	0.02536192	0.00660062
g	0.4262313	-0.0014508	0.0616252
h	1.4204976	2.25543162	0.74639834
i	-0.70278181	-0.60889597	-0.079595
j	-0.02483937	-0.15735462	0.09851673
k	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g1}{g0} \right) + e \left(\frac{h}{h0} \right) + g \left(\frac{g1}{g0} \right)^2 + i \left(\frac{h}{h0} \right)^2 + k \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}{\left[1 + b \left(\frac{g1}{g0} \right) + d \left(\frac{h}{h0} \right) + f \left(\frac{g1}{g0} \right)^2 + h \left(\frac{h}{h0} \right)^2 + j \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}$$

Hub Stress Correction Factor [f]:

= max(1; F1num / F1den)
 = max(1; 1.69/1.28)
 = 1.3193

Hub Stress Correction Factor [f2]:

= max(0.25; fn2) if h/ho < 0.35, else max (0.25; fn3)
 = max(0.25; 0.48/1.22) if h/ho < 0.35, else max (0.25; 0.31/0.76)
 = 0.3953

In this case, the hub ratio h/ho, 0.3267 was < 0.35.

PV Elite iterates on the assembly bolt load stress, Sfmax, until it exceeds the limits provided by WRC 538.

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress	Sbmax	507.6 N./mm ²
Minimum permissible bolt stress	Sbmin	290.0 N./mm ²
Max. permissible bolt stress prior to flange damage	Sfmax	712.8 N./mm ²
Target assembly gasket stress	SgT	207.0 N./mm ²
Maximum permissible gasket stress	Sgmax	276.0 N./mm ²
Minimum gasket operating stress	Sgmin-O	97.0 N./mm ²
Minimum gasket seating stress	Sgmin-S	140.0 N./mm ²
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	Phig	0.700



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Root Area of a Single Bolt	Abs	7.4516 cm ²
Max. permissible single flange rotation for gasket	Thetagmax	1.000 deg

Calculated Results:

Flange Material Ambient Yield Stress	Sya	250.0 N./mm ²
Flange Material Operating Yield Stress	Syo	230.1 N./mm ²
Single flange rotation at Sfmax	Thetafmax	0.610 deg

Gasket Contact Area [Ag]:
 $= \pi/4 (GO.D.^2 - GI.D.^2) + (PartFactor \cdot Ip \cdot bPart)$
 $= \pi/4 (2326^2 - 2282^2) + (0.5 \cdot 2300 \cdot 5)$
 $= 1649.9 \text{ cm}^2$

Target Bolt Stress [Sbsel]: (Eq. O-1)
 $= SgT \cdot Ag / (nb \cdot Ab)$
 $= 207 \cdot 1650 / (80 \cdot 7.45)$
 $= 572.9 \text{ N./mm}^2$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)
 $= \min(Sbsel; Sbmax)$
 $= \min(573; 508)$
 $= 507.6 \text{ N./mm}^2$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)
 $= \max(Sbsel; Sbmin)$
 $= \max(508; 290)$
 $= 507.6 \text{ N./mm}^2$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)
 $= \min(Sbsel; Sfmax)$
 $= \min(508; 713)$
 $= 507.6 \text{ N./mm}^2$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)
 $:: Sbsel \geq Sgmin \cdot S(Ag / (nb \cdot Abs))$
 $:: 508 \geq 140 (1650 / (80 \cdot 7.45))$
 $:: 507.6 \geq 387.5 \text{ N./mm}^2$ Check Passes

Check if Gasket Operating Stress is maintained. (Eq. O-8)
 $:: Sbsel \geq (Sgmin \cdot O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$
 $:: 508 \geq (97 \cdot 1650 + \pi/4 \cdot 1023 \cdot 2282^2) / (0.7 \cdot 7.45 \cdot 80)$
 $:: 507.6 \geq 483.8 \text{ N./mm}^2$ Check Passes

Check if Gasket Maximum Stress is achieved. (Eq. O-9)
 $:: Sbsel \leq Sgmax(Ag / (nb \cdot Abs))$
 $:: 508 \leq 276 (1650 / (80 \cdot 7.45))$
 $:: 507.6 \leq 763.9 \text{ N./mm}^2$ Check Passes

Check if Flange Rotation Limit is exceeded. (Eq. O-10)
 $:: Sbsel \leq Sfmax(thetagmax / thetafmax)$
 $:: 508 \leq 713 (1/0.61)$
 $:: 507.6 \leq 1168.4 \text{ N./mm}^2$ Check Passes

Assembly Bolt Torque [Tb]: (Eq. O-2)
 $= Sbsel \cdot K \cdot Abs \cdot \phi b$
 $= 508 \cdot 0.2 \cdot 7.45 \cdot 34.9$
 $= 269.4 \text{ Kg-m.}$

Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 118 N./mm²
 Bolt Allowable Stress for: SA-193 B7 Sb 171 N./mm²

Note: Since Sb >= SbB8, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:



Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 57878 / 1 \cdot 9807 \cdot 0.33 / (8.99 \cdot 201334 \cdot 10.8^2 \cdot 156 \cdot 0.3)$
 $= 1.000$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 34252 / 1 \cdot 9807 \cdot 0.33 / (8.99 \cdot 197847 \cdot 10.8^2 \cdot 156 \cdot 0.3)$
 $= 0.602$ (should be ≤ 1)

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 57878 / 1 \cdot 9807 \cdot 0.33 / (9.6 \cdot 201334 \cdot 10.8^2 \cdot 156 \cdot 0.3)$
 $= 0.937$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 34252 / 1 \cdot 9807 \cdot 0.33 / (9.6 \cdot 197847 \cdot 10.8^2 \cdot 156 \cdot 0.3)$
 $= 0.564$ (should be ≤ 1)

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.63, Temperature Reduction per Fig. UCS 66.1 = 20 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-18 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-39 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	65.12	207.00	107.86	207.00
Radial Flange	1.25	138.00	2.08	138.00
Tangential Flange	48.37	138.00	80.13	138.00
Maximum Average	56.74	138.00	94.00	138.00
Bolting	79.27	172.00	83.43	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 198.145 mm

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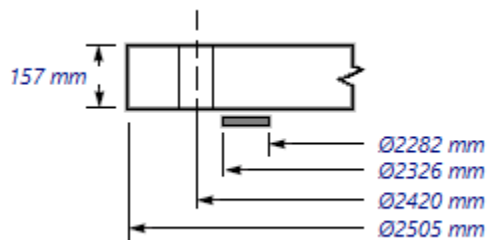
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Flange Input Data Values

Description: New Flange :

RHS Blind Flang

Flange Type	TEMA Channel Cover		
Design Pressure	P	1025.01	KPa.
Design Temperature		82	°C
Internal Corrosion Allowance	ci	3.2000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Outside Diameter	A	2505.000	mm
Flange Thickness	t	157.0000	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	138.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Perimeter along the Center of the Bolts	L	7602.654	mm
Youngs Modulus for Blind Flange Matl.	Sy	197846.66	N./mm ²
Allowed Channel Cover Deflection	dfAllow	2.8125	mm
Diameter of Bolt Circle	C	2420.000	mm
Nominal Bolt Diameter	a	34.9250	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		80	
Flange Face Outside Diameter	Fod	2337.000	mm
Flange Face Inside Diameter	Fid	2250.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	2326.000	mm
Gasket Inside Diameter	Gi	2282.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	6.3500	mm
Raised face depth	Fd	0.0000	mm
Length of Partition Gasket	lp	2300.0000	mm
Width of Partition Gasket	tp	10.0000	mm
Partition Gasket Factor	mPart	3.0000	
Partition Gasket Design Seating Stress	yPart	68.95	N./mm ²





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 FileName : 110-C-AK_Channel side= CS- 05-03-26_Latest Page 43 of 216
 Flg Calc [Int P]: New Flange Flng: 12 6:24pm Sep 8,2026

Gasket Contact Width,	$N = (G_o - G_i)/2$	22.000 mm
Basic Gasket Width,	$b_o = N/2$	11.000 mm
Effective Gasket Width,	$b = C_b \cdot \sqrt{b_o}$	8.358 mm
Gasket Reaction Diameter,	$G = G_o - 2 \cdot b$	2309.285 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 2309^2 \cdot 1025$$

$$= 437791.594 \text{ Kg}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P + 2 \cdot l_p \cdot b_{Part} \cdot m_{Part} \cdot P$$

$$= 2 \cdot 8.36 \cdot \pi \cdot 2309 \cdot 3 \cdot 1025$$

$$+ 2 \cdot 2300 \cdot 5 \cdot 3 \cdot 1025$$

$$= 45238.609 \text{ Kg}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(W_o,1 + W_o,2 + H'p, 0)$$

$$= \max(437792 + 45239 + 0, 0)$$

$$= 483030.250 \text{ Kg}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.9 \cdot 8.36 \cdot \pi \cdot 2309 + 68.9 \cdot 5 \cdot 2300$$

$$= 507155.281 \text{ Kg}$$

Required Bolt Area [Am]:

$$= \max((W_o + F_a + 4M_e/G) / S_b, W_g / S_a)$$

$$= \max((483030 + 0 + 4 \cdot 0 / 2309) / 172, 507155 / 172)$$

$$= 289.167 \text{ cm}^2$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	289.167	596.128	
Radial Distance between Bolts and Edge:	34.925	42.500	
Circ. Spacing between the Bolts:	77.800	95.009	...

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= A_b \cdot S_a / (y \cdot \pi (G_o + G_i))$$

$$= 596 \cdot 172 / (68.9 \cdot \pi (2326 + 2282))$$

$$= 10.272 \text{ mm}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= S_a \cdot A_b$$

$$= 172 \cdot 596$$

$$= 1045517.50 \text{ Kg}$$

Gasket Load for the Operating Condition [HG]:

$$= W_o$$

$$= 483030.25 \text{ Kg}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (2420 - 2309) / 2$$

$$= 55.3576 \text{ mm}$$

Flange Stresses



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 FileName : 110-C-AK_Channel side= CS- 05-03-26_Latest Page 44 of 216
 Flg Calc [Int P]: New Flange Flng: 12 6:24pm Sep 8,2026

Tangential Flange Stress, Flat Head (4.6.2.3), Operating [STo]:
 $= 1.9 \cdot Wm1 \cdot hG \cdot Bsc / (t^2 \cdot G) + C \cdot Z \cdot Peq \cdot G^2 / t^2$
 $= 1.9 \cdot 483030 \cdot 55.4 \cdot 1 / (154^2 \cdot 2309) + 0.3 \cdot 1 \cdot 1025 \cdot 2309^2 / 154^2$
 $= 78.45 \text{ N./mm}^2$

Tangential Flange Stress, Flat Head (4.6.2.4), Seating [STa]:
 $= 1.9 \cdot W \cdot hG \cdot Bsc / (t^2 \cdot G)$
 $= 1.9 \cdot 1045517 \cdot 55.4 \cdot 1 / (154^2 \cdot 2309)$
 $= 19.74 \text{ N./mm}^2$

Bolt Stress, Operating [BSO]:
 $= Wm1 / Ab$
 $= 483030 / 596$
 $= 79.46 \text{ N./mm}^2$

Bolt Stress, Seating [BSa]:
 $= (Wm2 / Ab)$
 $= (507155 / 596)$
 $= 83.43 \text{ N./mm}^2$

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Actual	Gasket Seating Allowed
Tangential Flange	78.45	138.00	19.74	138.00
Bolting	79.46	172.00	83.43	172.00

TEMA Channel Cover Deflection Calculations:

Actual deflection at center per TEMA formula 2.6723 mm
 Allowed deflection at center of Cover 2.8125 mm

Reqd. Blind Flange Thickness at Center 119.162 mm
 Reqd. Blind Flange Thickness at Gasket 61.373 mm
 Min. Required Flange Thickness per TEMA 151.200 mm

Estimated M.A.W.P. (Operating) 1107 KPa.
 Estimated Finished Weight of Flange at given Thk. 5997.0 kg.
 Estimated Unfinished Weight of Forging at given Thk 5997.0 kg.

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.87, Temperature Reduction per Fig. UCS 66.1 = 7 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 11 °C
 Min Metal Temp. at Required thickness (UCS 66.1) 4 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Actual	Gasket Seating Allowed
Tangential Flange	75.28	138.00	18.95	138.00
Bolting	79.46	172.00	83.43	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 150.343 mm

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 FileName : 110-C-AK_Channel side= CS- 05-03-26_Latest Page 45 of 216
 Internal Pressure Calculations: Step: 7 6:24pm Sep 8,2026

Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd KPa.	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) N./mm ²	Element#
LHS Blind Flg		1025	165	3.2	2505	138	1
LHS CH.Fl g		1022.5	222	3.2	2250	138	2
LHS CH Shell		1022.1	14	3.2	2250	117.3	3
main Shell 1		402.06	12	0.8	2250	90.27	4
main Shell 2		402.06	24	0.8	2250	90.27	5
main Shell 2		402.06	12	0.8	2250	90.27	6
RHS Ch Shell		1022.1	16	3.2	2250	117.3	7
RHS Ch Flg		1022.5	222	3.2	2250	138	8
RHS Blind Flang		1025	165	3.2	2505	138	9

Element Required Thickness and MAWP :

From	To	Design Pressure KPa.	M.A.W.P. Corroded KPa.	M.A.P. New & Cold KPa.	Minimum Thickness mm	Required Thickness mm	Element#
LHS Blind Flg		1000	1081	1242	157	151.202	1
LHS CH.Fl g		1000	1291	1704	214	208.94	2
LHS CH Shell		1000	1094	1449	14	13.0829	3
main Shell 1		380	791	1140	11	5.82804	4
main Shell 2		380	1815	2471	24	5.82804	5
main Shell 2		380	791	1140	11	5.82804	6
RHS Ch Shell		1000	1300	1654	16	13.0829	7
RHS Ch Flg		1000	1291	1704	214	208.94	8
RHS Blind Flang		1000	1082	1242	157	151.2	9

Summary of Heat Exchanger Maximum Allowable Working Pressures :

Note:
 For Exchanger designs, the following values include MAWPs that consider the tubesheet, tubes, tube/tubesheet joint etc. These values were determined by iteration. Review the tubesheet analysis report for more information.

Shell Side MAWP = 791 KPa.
 Shell Side MAPnc = 1140 KPa.
 Channel Side MAWP = 1081 KPa.
 Channel Side MAPnc = 1242 KPa.

Elements Suitable for Design Internal Pressure.

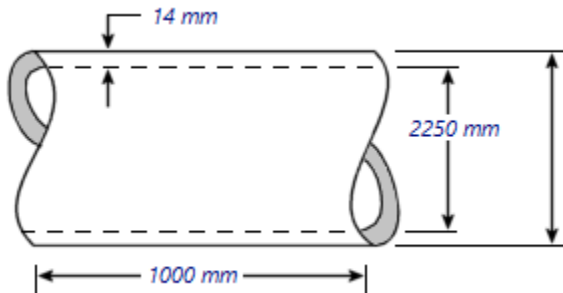
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 FileName : 110-C-AK_Channel side= CS- 05-03-26_Latest Page 46 of 216
 Element: LHS CH Shell Step: 8 6:24pm Sep 8,2026

Internal Pressure Calculation Results:

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Internal Pressure Results for: LHS CH Shell

Cylindrical Shell Design Information:			
Design Pressure P	1022.086 KPa.	Int. Design Temperature	82 °C
Material	SA-516 70	Ext. Design Temperature	82 °C
Length	1000.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.2000 mm	Allowable Stress (ope) S	138.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	2250.000 mm	Actual Stress	126.3 N./mm ²
Spec. Min. Thk t,e, or sa	14.0000 mm	Spec. Nominal Thk.	14.000 mm
Surface Area	71565.5 cm ²	SG of Contents	1.000
Element Volume	3976790.8 cm ³	Weight of Contents	3974.4 Kg
Empty Weight	771.8 Kg	Operating Weight	4746.1 Kg
		Comp. All. Ope. Stress	108.4 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (1022 \cdot 1128) / (138 \cdot 0.85 - 0.6 \cdot 1022) \\
 &= 9.8829 + 3.2000 = 13.0829 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 22.086 KPa.

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 0.85 \cdot 10.8) / (1128 + 0.6 \cdot 10.8) \\
 &= 1116 - 22.1 = 1094 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 0.85 \cdot 14) / (1125 + 0.6 \cdot 14) \\
 &= 1449 \text{ KPa.}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (1022 \cdot (1128 + 0.6 \cdot 10.8)) / (0.85 \cdot 10.8) \\
 &= 126.341 \text{ N./mm}^2
 \end{aligned}$$



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 FileName : 110-C-AK_Channel side= CS- 05-03-26_Latest Page 47 of 216
 Element: LHS CH Shell Step: 8 6:24pm Sep 8,2026

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 0.618 %

Minimum Design Metal Temperature Results:

Govrn. thk, $t_g = 14$, $t_r = 10.5$, $c = 3.2$ mm, $E^* = 0.85$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.82$, Temp. Reduction = 10 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-18 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-28 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	1081.470 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	14.000 mm	Basic MDMT	-18 °C
Stress/Thk. Ratio	0.823	Temperature Difference	10 °C
UCS-66(c) red.	No	Computed Min. Temp.	-28 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	85 %

EXTERNAL PRESSURE RESULTS for: LHS CH Shell

Cylindrical Shell From 30 to Tsht Ext. Chart: CS-2 at 82 °C

Elastic Modulus from Chart: CS-2 at 82 °C: 199943392 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
10.800	2278.00	999.97	210.93	0.4390	0.0010457	86.66

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 86.7) / (3 \cdot 211) = 548 \text{ KPa.}$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
5.185	2278.00	999.97	439.35	0.4390	0.0003409	34.08

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 34.1) / (3 \cdot 439) = 103 \text{ KPa.}$$

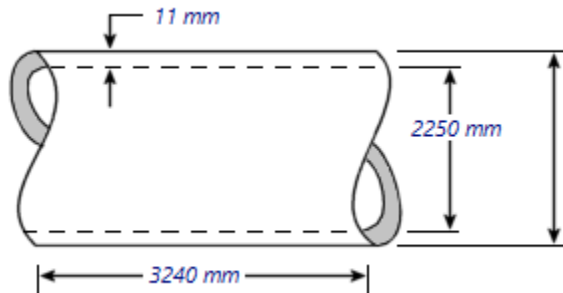
Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
10.800	2278.00	5816.62	210.93	2.5534	0.0001637	16.37

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 16.4) / (3 \cdot 211) = 103 \text{ KPa.}$$



Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: main Shell 1

Cylindrical Shell Design Information:					
Design Pressure	P	402.062 KPa.	Int. Design Temperature		130 °C
Material		SA-240 304	Ext. Design Temperature		130 °C
Length		3240.000 mm	External Pressure Chart		HA-1
			UNS Number		S30400
Int. Corr. All.	c	0.8000 mm	Allowable Stress (ope)	S	106.2 N./mm ²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	2250.000 mm	Actual Stress		52.5 N./mm ²
Spec. Min. Thk t,e, or sa		11.0000 mm	Spec. Nominal Thk.		12.000 mm
Surface Area		231465.1 cm ²	SG of Contents		1.000
Element Volume		8728423.0 cm ³	Weight of Contents		8723.1 Kg
Empty Weight		2217.9 Kg	Operating Weight		10941.0 Kg
			Comp. All. Ope. Stress		69.8 N./mm ²
Radiography Information:					
User Defined			User Defined		
Circ. Joint Efficiency		85 %	Long. Joint Eff.	E or v	85 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (402 \cdot 1126) / (106 \cdot 0.85 - 0.6 \cdot 402) \\
 &= 5.0280 + 0.8000 = 5.8280 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 22.062 KPa.

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (106 \cdot 0.85 \cdot 10.2) / (1126 + 0.6 \cdot 10.2) \\
 &= 813 - 22.1 = 791 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 0.85 \cdot 11) / (1125 + 0.6 \cdot 11) \\
 &= 1140 \text{ KPa.}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (402 \cdot (1126 + 0.6 \cdot 10.2)) / (0.85 \cdot 10.2) \\
 &= 52.495 \text{ N./mm}^2
 \end{aligned}$$



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 Element: main Shell 1 Step: 9 6:24pm Sep 8,2026

% Elongation per Table UG-79-1 (50*tnom/Rf)*(1-Rf/Ro) 0.531 %
 Note: Please Check Requirements of Table UHA-44 for Elongation limits.

SA-240 304, Min Metal Temp without impact per UHA-51: -196 °C

EXTERNAL PRESSURE RESULTS for: main Shell 1

Cylindrical Shell From 40 to 50 Ext. Chart: HA-1 at 130 °C

Elastic Modulus from Chart: HA-1 at 130 °C: 185038288 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
10.200	2272.00	8200.00	222.75	3.6092	0.0001055	9.76

$$MAEP = (4*B)/(3*(Do/t)) = (4*9.76)/(3*223) = 58.4 \text{ KPa.}$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
9.715	2272.00	8200.00	233.87	3.6092	0.0000980	9.07

$$MAEP = (4*B)/(3*(Do/t)) = (4*9.07)/(3*234) = 51.7 \text{ KPa.}$$

Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
10.200	2272.00	9210.43	222.75	4.0539	0.0000934	8.64

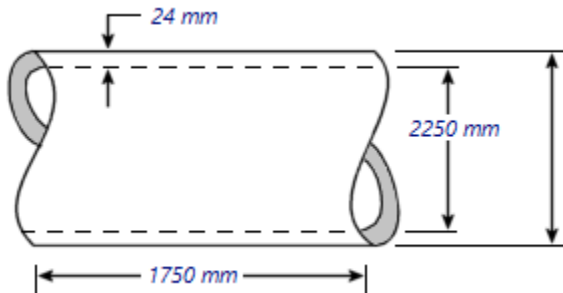
$$MAEP = (4*B)/(3*(Do/t)) = (4*8.64)/(3*223) = 51.7 \text{ KPa.}$$

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Internal Pressure Calculation Results:

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Internal Pressure Results for: main Shell 2

Cylindrical Shell Design Information:					
Design Pressure	P	402.062 KPa.	Int. Design Temperature		130 °C
Material		SA-240 304	Ext. Design Temperature		130 °C
Length		1750.000 mm	External Pressure Chart		HA-1
			UNS Number		S30400
Int. Corr. All.	c	0.8000 mm	Allowable Stress (ope)	S	106.2 N./mm ²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	2250.000 mm	Actual Stress		23.2 N./mm ²
Spec. Min. Thk t,e, or sa		24.0000 mm	Spec. Nominal Thk.		24.000 mm
Surface Area		126339.1 cm ²	SG of Contents		1.000
Element Volume		4714425.5 cm ³	Weight of Contents		4711.5 Kgf
Empty Weight		2408.6 Kgf	Operating Weight		7120.1 Kgf
			Comp. All. Ope. Stress		84.8 N./mm ²
Radiography Information:					
User Defined			User Defined		
Circ. Joint Efficiency		85 %	Long. Joint Eff.	E or v	85 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (402 \cdot 1126) / (106 \cdot 0.85 - 0.6 \cdot 402) \\
 &= 5.0280 + 0.8000 = 5.8280 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 22.062 KPa.

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (106 \cdot 0.85 \cdot 23.2) / (1126 + 0.6 \cdot 23.2) \\
 &= 1837 - 22.1 = 1815 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 0.85 \cdot 24) / (1125 + 0.6 \cdot 24) \\
 &= 2471 \text{ KPa.}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (402 \cdot (1126 + 0.6 \cdot 23.2)) / (0.85 \cdot 23.2) \\
 &= 23.239 \text{ N./mm}^2
 \end{aligned}$$



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 FileName : 110-C-AK_Channel side= CS- 05-03-26_Latest Page 51 of 216
 Element: main Shell 2 Step: 10 6:24pm Sep 8,2026

% Elongation per Table UG-79-1 (50*tnom/Rf)*(1-Rf/Ro) 1.055 %
 Note: Please Check Requirements of Table UHA-44 for Elongation limits.

SA-240 304, Min Metal Temp without impact per UHA-51: -196 °C

EXTERNAL PRESSURE RESULTS for: main Shell 2

Cylindrical Shell From 50 to 60 Ext. Chart: HA-1 at 130 °C

Elastic Modulus from Chart: HA-1 at 130 °C: 185038288 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
23.200	2298.00	8200.00	99.05	3.5683	0.0003503	32.42

$$MAEP = (4*B)/(3*(Do/t)) = (4*32.4)/(3*99.1) = 436 \text{ KPa.}$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
9.781	2298.00	8200.00	234.95	3.5683	0.0000985	9.11

$$MAEP = (4*B)/(3*(Do/t)) = (4*9.11)/(3*235) = 51.7 \text{ KPa.}$$

Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
23.200	2298.00	186576.27	99.05	50.0000	0.0001125	10.41

$$MAEP = (4*B)/(3*(Do/t)) = (4*10.4)/(3*99.1) = 140 \text{ KPa.}$$

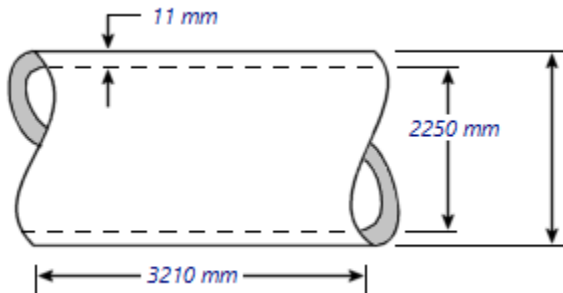
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 Element: main Shell_2 Step: 11 6:24pm Sep 8,2026

Internal Pressure Calculation Results:

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Internal Pressure Results for: main Shell 2

Cylindrical Shell Design Information:			
Design Pressure P	402.062 KPa.	Int. Design Temperature	130 °C
Material	SA-240 304	Ext. Design Temperature	130 °C
Length	3210.000 mm	External Pressure Chart	HA-1
		UNS Number	S30400
Int. Corr. All. c	0.8000 mm	Allowable Stress (ope) S	106.2 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	2250.000 mm	Actual Stress	52.5 N./mm ²
Spec. Min. Thk t,e, or sa	11.0000 mm	Spec. Nominal Thk.	12.000 mm
Surface Area	229321.8 cm ²	SG of Contents	1.000
Element Volume	8647604.0 cm ³	Weight of Contents	8642.3 Kg
Empty Weight	2197.4 Kg	Operating Weight	10839.7 Kg
		Comp. All. Ope. Stress	69.8 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (402 \cdot 1126) / (106 \cdot 0.85 - 0.6 \cdot 402)$$

$$= 5.0280 + 0.8000 = 5.8280 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 22.062 KPa.

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (106 \cdot 0.85 \cdot 10.2) / (1126 + 0.6 \cdot 10.2)$$

$$= 813 - 22.1 = 791 \text{ KPa.}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 0.85 \cdot 11) / (1125 + 0.6 \cdot 11)$$

$$= 1140 \text{ KPa.}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (402 \cdot (1126 + 0.6 \cdot 10.2)) / (0.85 \cdot 10.2)$$

$$= 52.495 \text{ N./mm}^2$$



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 FileName : 110-C-AK_Channel side= CS- 05-03-26_Latest Page 53 of 216
 Element: main Shell_2 Step: 11 6:24pm Sep 8,2026

% Elongation per Table UG-79-1 (50*tnom/Rf)*(1-Rf/Ro) 0.531 %
 Note: Please Check Requirements of Table UHA-44 for Elongation limits.

SA-240 304, Min Metal Temp without impact per UHA-51: -196 °C

EXTERNAL PRESSURE RESULTS for: main Shell 2

Cylindrical Shell From 60 to Tsht Ext. Chart: HA-1 at 130 °C

Elastic Modulus from Chart: HA-1 at 130 °C: 185038288 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
10.200	2272.00	8200.00	222.75	3.6092	0.0001055	9.76

$$MAEP = (4*B)/(3*(Do/t)) = (4*9.76)/(3*223) = 58.4 \text{ KPa.}$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
9.715	2272.00	8200.00	233.87	3.6092	0.0000980	9.07

$$MAEP = (4*B)/(3*(Do/t)) = (4*9.07)/(3*234) = 51.7 \text{ KPa.}$$

Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
10.200	2272.00	9210.43	222.75	4.0539	0.0000934	8.64

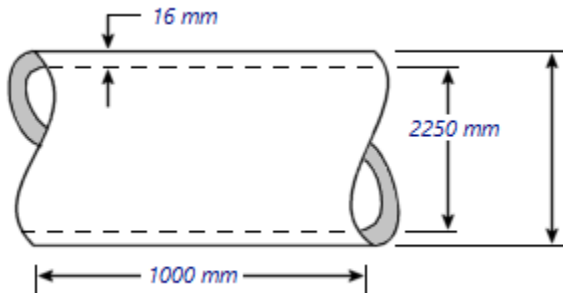
$$MAEP = (4*B)/(3*(Do/t)) = (4*8.64)/(3*223) = 51.7 \text{ KPa.}$$



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 FileName : 110-C-AK_Channel side= CS- 05-03-26_Latest Page 54 of 216
 Element: RHS Ch Shell Step: 12 6:24pm Sep 8,2026

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: RHS Ch Shell

Cylindrical Shell Design Information:			
Design Pressure P	1022.086 KPa.	Int. Design Temperature	82 °C
Material	SA-516 70	Ext. Design Temperature	82 °C
Length	1000.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.2000 mm	Allowable Stress (ope) S	138.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	2250.000 mm	Actual Stress	106.7 N./mm ²
Spec. Min. Thk t,e, or sa	16.0000 mm	Spec. Nominal Thk.	16.000 mm
Surface Area	71691.1 cm ²	SG of Contents	1.000
Element Volume	3976790.8 cm ³	Weight of Contents	3974.4 Kg
Empty Weight	882.8 Kg	Operating Weight	4857.2 Kg
		Comp. All. Ope. Stress	69.8 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (1022 \cdot 1128) / (138 \cdot 0.85 - 0.6 \cdot 1022) \\
 &= 9.8829 + 3.2000 = 13.0829 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 22.086 KPa.

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 0.85 \cdot 12.8) / (1128 + 0.6 \cdot 12.8) \\
 &= 1322 - 22.1 = 1300 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 0.85 \cdot 16) / (1125 + 0.6 \cdot 16) \\
 &= 1654 \text{ KPa.}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (1022 \cdot (1128 + 0.6 \cdot 12.8)) / (0.85 \cdot 12.8) \\
 &= 106.713 \text{ N./mm}^2
 \end{aligned}$$



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 FileName : 110-C-AK_Channel side= CS- 05-03-26_Latest Page 55 of 216
 Element: RHS Ch Shell Step: 12 6:24pm Sep 8,2026

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 0.706 %

Minimum Design Metal Temperature Results:

Govrn. thk, $t_g = 16$, $t_r = 10.5$, $c = 3.2$ mm, $E^* = 0.85$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.69$, Temp. Reduction = 17 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-31 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	1081.470 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	16.000 mm	Basic MDMT	-14 °C
Stress/Thk. Ratio	0.695	Temperature Difference	17 °C
UCS-66(c) red.	No	Computed Min. Temp.	-31 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	85 %

EXTERNAL PRESSURE RESULTS for: RHS Ch Shell

Cylindrical Shell From 70 to 80 Ext. Chart: CS-2 at 82 °C

Elastic Modulus from Chart: CS-2 at 82 °C: 199943392 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
12.800	2282.00	1000.03	178.28	0.4382	0.0013740	94.47

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 94.5) / (3 \cdot 178) = 706 \text{ KPa.}$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
5.190	2282.00	1000.03	439.67	0.4382	0.0003411	34.11

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 34.1) / (3 \cdot 440) = 103 \text{ KPa.}$$

Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
12.800	2282.00	8770.39	178.28	3.8433	0.0001384	13.84

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 13.8) / (3 \cdot 178) = 103 \text{ KPa.}$$



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 FileName : 110-C-AK_Channel side= CS- 05-03-26_Latest Page 56 of 216
 Shop Pressure Test Results: Step: 13 6:24pm Sep 8,2026

Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	1028.735 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	494.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	1457.651 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	870.468 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	543.400 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	543.400 KPa.
App 27-4	= M.A.W.P.	791.335 KPa.

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	1405.911 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	1300.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	1589.649 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	1189.617 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	1430.000 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	1430.000 KPa.
App 27-4	= M.A.W.P.	1081.470 KPa.

Horizontal Test performed per: UG-99c

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & KPa.):

From To	Stress	Allowable	Ratio	Pressure
LHS CH Shell	153.5	234.0	0.656	1611.70
main Shell 1	179.1	184.5	0.971	1479.71
main Shell 2	82.7	184.5	0.448	1479.71
main Shell 2	179.1	184.5	0.971	1479.71
RHS Ch Shell	134.5	234.0	0.575	1611.70

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N5-Tube Inlet	Nozzle	118.00	118.00	1.000
N5-Tube Inlet	Pad	138.00	137.51	1.004
N1-Shell inlet	Nozzle	138.00	133.20	1.036
N1-Shell inlet	Pad	138.00	138.00	1.000
N4-Shell outlet	Nozzle	138.00	133.20	1.036
N4-Shell outlet	Pad	138.00	138.00	1.000
N6-Tube Outlet	Nozzle	118.00	118.00	1.000
N6-Tube Outlet	Pad	138.00	137.51	1.004

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
LHS Blind Flg	138.00	138.00	1.000
LHS CH.Fl g	138.00	138.00	1.000
LHS CH Shell	138.00	138.00	1.000
main Shell 1	138.00	106.20	1.299
main Shell 2	138.00	106.20	1.299
main Shell 2	138.00	106.20	1.299



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FileName : 110-C-AK_Channel side= CS- 05-03-26_Latest Page 57 of 216
Shop Pressure Test Results: Step: 13 6:24pm Sep 8,2026

RHS Ch Shell	138.00	138.00	1.000
RHS Ch Flg	138.00	138.00	1.000
RHS Blind Flang	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Exchanger Materials (N./mm²):

Description	Ambient	Operating	Ratio
Tube Material	115.00	115.00	1.000
Tubesheet Material	138.00	138.00	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
N5-Tube Inlet	25.17	216.00	0.117
N1-Shell inlet	27.60	184.50	0.150
N4-Shell outlet	30.16	184.50	0.163
N6-Tube Outlet	25.17	216.00	0.117

Elements Suitable for Test Pressure.

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Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	1028.735 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	494.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	1457.651 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	870.468 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	543.400 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	543.400 KPa.
App 27-4	= M.A.W.P.	791.335 KPa.

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	1405.911 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	1300.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	1589.649 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	1189.617 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	1430.000 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	1430.000 KPa.
App 27-4	= M.A.W.P.	1081.470 KPa.

UG-99(b)(35), Test Pressure Calculation [Shell Side]:

$$= \text{Test Factor} \cdot \text{Design Pressure} \cdot \text{Stress Ratio}$$

$$= 1.3 \cdot 380 \cdot 1$$

$$= 494.000 \text{ KPa.}$$

UG-99(b) Note 35, Test Pressure Calculation [Channel Side]:

$$= \text{Test Factor} \cdot \text{Design Pressure} \cdot \text{Stress Ratio}$$

$$= 1.3 \cdot 1000 \cdot 1$$

$$= 1300.000 \text{ KPa.}$$

Horizontal Test performed per: UG-99b (Note 35)

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & KPa.):

From To	Stress	Allowable	Ratio	Pressure
LHS CH Shell	163.4	234.0	0.698	1322.05
main Shell 1	67.4	184.5	0.365	516.05
main Shell 2	29.8	184.5	0.162	516.05
main Shell 2	67.4	184.5	0.365	516.05
RHS Ch Shell	138.0	234.0	0.590	1322.05

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N5-Tube Inlet	Nozzle	118.00	118.00	1.000
N5-Tube Inlet	Pad	138.00	137.51	1.004
N1-Shell inlet	Nozzle	138.00	133.20	1.036
N1-Shell inlet	Pad	138.00	138.00	1.000
N4-Shell outlet	Nozzle	138.00	133.20	1.036
N4-Shell outlet	Pad	138.00	138.00	1.000
N6-Tube Outlet	Nozzle	118.00	118.00	1.000
N6-Tube Outlet	Pad	138.00	137.51	1.004

Minimum

1.000



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Field Pressure Test Results: Step: 14 6:24pm Sep 8,2026

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
LHS Blind Flg	138.00	138.00	1.000
LHS CH.Flг	138.00	138.00	1.000
LHS CH Shell	138.00	138.00	1.000
main Shell 1	138.00	106.20	1.299
main Shell 2	138.00	106.20	1.299
main Shell 2	138.00	106.20	1.299
RHS Ch Shell	138.00	138.00	1.000
RHS Ch Flg	138.00	138.00	1.000
RHS Blind Flang	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Exchanger Materials (N./mm²):

Description	Ambient	Operating	Ratio
Tube Material	115.00	115.00	1.000
Tubesheet Material	138.00	138.00	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
N5-Tube Inlet	20.64	216.00	0.096
N1-Shell inlet	9.62	184.50	0.052
N4-Shell outlet	10.52	184.50	0.057
N6-Tube Outlet	20.64	216.00	0.096

Elements Suitable for Test Pressure.

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 ASME TS Calc: Pressure Test Case: 1 6:24p Sep 8,2026

Input Echo, Tubesheet Number 1, Description: Fixed Tubesheet

Shell Data:

Main Shell Description: main Shell 1

Shell Test Pressure		494.00 KPa.
Shell Thickness	ts	12.0000 mm
Shell Internal Corrosion Allowance	cas	0.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	2250.000 mm
Shell Circumferential Joint Efficiency	Esw	0.850
Shell Temperature for Internal Pressure	Ts	21.11 °C
Shell Material		SA-240 304

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		S30400
Shell Allowable Stress at Temperature	Ss	138.00 N./mm ²
Shell Allowable Stress at Ambient		138.00 N./mm ²

Channel Data:

Channel Description: LHS CH Shell

Channel Type:		Cylinder
Channel Test Pressure		1300.00 KPa.
Channel Thickness	tc	14.0000 mm
Channel Internal Corrosion Allowance	cac	0.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	2250.000 mm
Channel Design Temperature	TEMPC	21.11 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	138.00 N./mm ²
Channel Allowable Stress at Ambient		138.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	4500
Tube Wall Thickness	et	1.6500 mm
Tube Outside Diameter	D	19.0500 mm
Total Straight Tube Length	Lt	8300.00 mm
Straight Tube Length (bet. inner tubsht faces)	L	8200.00 mm
Design Temperature of the Tubes		21.11 °C
Tube Material		SA-213 TP316L
Tube Material UNS Number		S31603
Is this a Welded Tube		No
Tube Material Specification used		Smls. tube
Tube Allowable Stress at Temperature		115.00 N./mm ²
Tube Allowable Stress At Ambient		115.00 N./mm ²
Tube Yield Stress At design Temperature	Syt	170.00 N./mm ²
Tube Pitch (Center to Center Spacing)	P	23.2100 mm
Tube Layout Pattern		Triangular

Fillet Weld Leg	af	1.6500 mm
Groove Weld Leg	ag	1.6500 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No

Radius to Outermost Tube Hole Center	ro	1107.050 mm
Largest Center-to-Center Tube Distance	U1	159.7280 mm
Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	0.0000 mm
Area of Tube Layout	Ap	20443.4 cm ²



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Perimeter of Tube Layout Cp 6127.4 mm

Tubesheet Data:

Tubesheet TYPE: Fixed Tubesheet Exchanger, Conf A

Tubesheet Design Metal Temperature T 21.11 °C
 Tubesheet Material SA-266 2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number		K03506
Tubesheet Allowable Stress at Temperature	S	138.00 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	138.00 N./mm ²
Thickness of Tubesheet	h	50.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	0.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	0.0000 mm
Tubesheet Outside Diameter	A	2278.000 mm

Area of the Untubed Lanes AL 0.0 cm²

Additional Data for Fixed/Floating Tubesheet Exchangers:

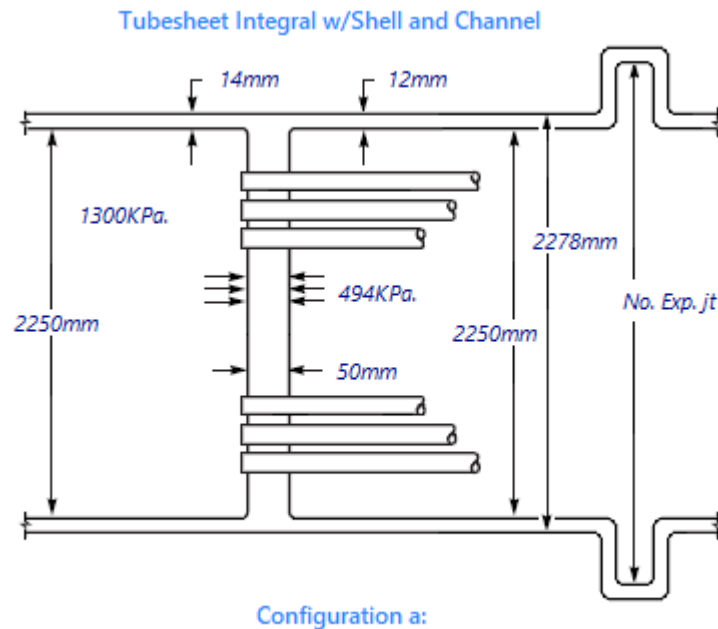
Unsupported Tube Span under consideration l 700.000 mm
 Tube End condition corresponding to Span (l) k 1.00

Ignore Radial Thermal Exp. effects (4.18.8/4.18.9) Yes
 Perform Differential Pressure Design No
 Run Multiple Load Cases Yes
 Shell Side Min. Design Pressure Psd,min -51.7095 KPa.
 Channel Side Min. Design Pressure Ptd,min -103.4190 KPa.

Mean Shell Metal Temp. along Shell len. Tsm 21.11 °C
 Mean Tube Metal Temp. along Tube length Ttm 21.11 °C
 Junction Stress Reduction option Perform Plastic Calculation

Tubesheet Integral with Both
 Tubesheet Extended as Flange No

Is Exchanger in Creep range (skip EP, Use 3S for Sps) No



ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties:

Shell - TE-1 Austenitic Group 3 Steels Group 1
 Shell - TM-1 Material Group G

Th. Exp. Coeff. Metal Temp. along Len	21.1 °C	0.0000152988 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.19512E+09 KPa.
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000152988 /°C
Elastic Mod. at Metal Temp. along Len	21.1 °C	0.19512E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.19512E+09 KPa.

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
 Channel - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tubes - TE-1 Austenitic Group 3 Steels Group 1
 Tubes - TM-1 Material Group G

Th. Exp. Coeff. Metal Temp. along Len	21.1 °C	0.0000152988 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.19512E+09 KPa.
Elastic Mod. at Metal Temp. along Len	21.1 °C	0.19512E+09 KPa.
Elastic Mod. at Tubsht. Design Temp.	21.1 °C	0.19512E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.19512E+09 KPa.

TubeSheet - TE-1 Carbon & Low Alloy Steels, Group 1
 TubeSheet - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Note:



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 FileName : 110-C-AK_Channel side= CS- 05-03-26_Latest Page 63 of 216
 ASME TS Calc: Pressure Test Case: 1 6:24p Sep 8, 2026

The Elasticity and Alpha values are taken from Tables in ASME II D.
 Please insure these properties are consistent with the
 type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:
 $= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= (1352 \cdot (19/2 - 0)) / (115 \cdot 1 + 0.4 \cdot 1352)$
 $= 0.1114 + 0.0000 = 0.1114 \text{ mm}$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart HA-4 at 21.11 °C
 Elastic Modulus for Material 193048800.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
1.6500	19.05	8200.00	11.55	50.0000	0.0082522	93.31

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 93.3) / (3 \cdot 11.5) = 10775 \text{ KPa.}$

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.3078	19.05	8200.00	61.88	50.0000	0.0002873	27.73

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 27.7) / (3 \cdot 61.9) = 597 \text{ KPa.}$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.3078 mm
Allowable Internal Pressure at Corroded thickness	21402.77 KPa.
Required Internal Design Pressure	1351.71 KPa.
Allowable External Pressure at Corroded thickness	10774.75 KPa.
Required External Design Pressure	597.42 KPa.
Required Thickness due to Shell Side pressure	0.3078 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Results for ASME Fixed Tubesheet Calculations for Configuration a:

Results for Tubesheet Calculations Original Thickness:

ASME Section VIII-2 Part 4.18.8.4 Step 1:

Tube expansion depth ratio $[\rho]$:
 $= l_{tx} / h$ (modified for corrosion if present)
 $= 0/50$
 $= 0.0000$ (must be $0 \leq \rho \leq 1$)

Effective tube hole diameter $[d^*]$:
 $= \max(dt - 2 \cdot tt(Et/E) \cdot (St/T/S) \cdot (\rho); dt - 2 \cdot tt)$
 $= \max(19 - 2 \cdot 1.65 (195117184/202701232) \cdot (115/138) \cdot (0); 19 - 2 \cdot 1.65)$
 $= 19.0500 \text{ mm}$

Equivalent outer tube limit circle diameter $[Do]$:

$= 2 \cdot ro + dt$
 $= 2 \cdot 1107 + 19$
 $= 2233.150 \text{ mm}$

Determine the basic ligament efficiency for shear $[\mu]$:

$= (p - dt) / p$
 $= (23.2 - 19) / 23.2$
 $= 0.179$



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Equivalent outer tube limit radius [ao]:

$$= D_o / 2$$

$$= 2233/2$$

$$= 1116.5751 \text{ mm}$$

Effective tube pitch [p*]:

$$= p / \sqrt{1 - 4 \cdot \min(AL \cdot CNVfactor; 4 \cdot D_o \cdot p) / (\pi \cdot D_o^2)}$$

$$= 23.2 / \sqrt{1 - 4 \cdot \min(0 \cdot 100 ; 4 \cdot 2233 \cdot 23.2) / (\pi \cdot 2233^2)}$$

$$= 23.2100 \text{ mm}$$

Effective ligament efficiency for bending [μ^*]:

$$= (p^* - d^*) / p^*$$

$$= (23.2 - 19) / 23.2$$

$$= 0.1792$$

Ratio [ps]:

$$= a_s / a_o$$

$$= 1125/1117$$

$$= 1.0075$$

Ratio [pc]:

$$= a_c / a_o$$

$$= 1125/1117$$

$$= 1.0075$$

Compute parameter [xt]:

$$= 1 - Nt / ((dt - 2 \cdot tt) / (2 \cdot a_o))^2$$

$$= 1 - 4500 / ((19 - 2 \cdot 1.65) / (2 \cdot 1117))^2$$

$$= 0.7762$$

Determine parameter [xs]:

$$= 1 - Nt / (dt / (2 \cdot a_o))^2$$

$$= 1 - 4500 / (19 / (2 \cdot 1117))^2$$

$$= 0.6725$$

Determine the value [h'g]:

$$= \max((h_g - CATC); 0) \text{ (For pressure only cases)}$$

$$= \max((0 - 0); 0)$$

$$= 0.000 \text{ mm}$$

ASME Section VIII-2 Part 4.18.8.4 Step 2:

Axial Shell Stiffness [Ks]:

$$= \pi \cdot t_s (D_s + t_s) E_s / L$$

$$= \pi \cdot 12 (2250 + 12) 195117184 / 8200$$

$$= 206919.7188 \text{ Kg/mm}$$

Axial Tube Stiffness [Kt]:

$$= \pi \cdot t_t (D_t - t_t) E_t / L$$

$$= \pi \cdot 1.65 (19 - 1.65) 195117184 / 8200$$

$$= 218.8574 \text{ Kg/mm}$$

Stiffness Factor [Ks,t]:

$$= K_s / (Nt \cdot K_t)$$

$$= 206920 / (4500 \cdot 219)$$

$$= 0.21010$$

Rigidity Ratio [J]:

$$= 1 / (1 + K_s / K_t)$$

$$= 1 / (1 + 206920 / 0)$$

$$= 1.0000 (= 1 \text{ if No Exp. Jt.})$$

Shell coefficient [β_s]:

$$= ((12(1 - \nu_s^2))^{1/4}) / ((D_s + t_s) t_s)$$

$$= ((12(1 - 0.31^2))^{1/4}) / ((2250 + 12) 12)$$



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= 0.0110 1/mm

Shell coefficient [ks]:

$$= \beta_s \cdot E_s \cdot t_s^3 / (6(1 - \nu_s^2))$$

$$= 0.011 \cdot 195117184 \cdot 12^3 / (6(1 - 0.31^2))$$

$$= 684787968.0000 \text{ KPa} \cdot \text{mm}^2$$

Shell coefficient [λs]:

$$= 6(D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$

$$= 6(2250 + 12) \cdot 684787968 / 50^3 \cdot (1 + 50 \cdot 0.011 + (50 \cdot 0.011)^2 / 2)$$

$$= 126577584.0000 \text{ KPa}$$

Shell coefficient [δs]:

$$= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot \nu_s / (2(D_s + t_s)))$$

$$= 2250^2 / (4 \cdot 195117184 \cdot 12) \cdot (1 - 2250 \cdot 0.31 / (2(2250 + 12)))$$

$$= 0.45717\text{E}+00 \text{ mm/N} \cdot \text{mm}^2$$

Channel coefficient [βc]:

$$= ((12(1 - \nu_c^2))^{1/4}) / \sqrt{ (D_c + t_c) t_c }$$

$$= ((12(1 - 0.3^2))^{1/4}) / \sqrt{ (2250 + 14) 14 }$$

$$= 0.0102 \text{ 1/mm}$$

Channel coefficient [kc]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - \nu_c^2))$$

$$= 0.01 \cdot 202701232 \cdot 14^3 / (6(1 - 0.3^2))$$

$$= 1040162944.0000 \text{ KPa} \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λc]:

$$= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (CmDelta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot CqDelta)$$

$$= 6(2250 + 14 / \cos(0)) \cdot 1040162944 / 50^3 \cdot (1 + 50 \cdot 0.01 \cdot (1 + 1) / 2 + (50 \cdot 0.01)^2 / 2 \cdot 1)$$

$$= 185476576.0 \text{ KPa}$$

Channel coefficient [δc]:

$$= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot \nu_c / (2(D_c + t_c)))$$

$$= 2250^2 / (4 \cdot 202701232 \cdot 14) \cdot (1 - 2250 \cdot 0.3 / (2(2250 + 14)))$$

$$= 0.37948\text{E}+00 \text{ mm/N} \cdot \text{mm}^2$$

Channel coefficient [zetac]:

$$zetac = 0$$

ASME Section VIII-2 Part 4.18.8.4 Step 3:

$h/p = 2.154244$; $\mu^* = 0.179233$; $E^* = 26634668 \text{ KPa}$
 $E^*/E = 0.131399$; $\nu^* = 0.491147$

Note: As h/p (2.154) is > 2, data values for $h/p = 2$ were used.

Tube Bundle Stiffness Factor [Xa]:

$$= ((24(1 - \nu^*) Nt \cdot Et \cdot t(dt - tt) a_o^2) / (E^* \cdot L \cdot H^3))^{1/4}$$

$$= ((24(1 - 0.49^2) 4500 \cdot 195117184 \cdot 1.65 (19 - 1.65) 1117^2) / (26634668 \cdot 8200 \cdot 50^3))^{1/4}$$

$$= 12.0328$$

Values from Table 4.18.3

$$Z_d = 0.000834 ; Z_v = 0.006876 ; Z_m = 0.117351$$

$$Z_a = 0.220245\text{E}+06 ; Z_w = 0.006876$$

ASME Section VIII-2 Part 4.18.8.4 Step 4:

Diameter ratio [K]:



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$$= A / D_o$$

$$= 2278/2233$$

$$= 1.0201$$

Compute coefficient [F]:

$$= (1 - \nu^*) / E^* \cdot (\lambda s + \lambda c + E \cdot \ln(K))$$

$$= (1 - 0.49) / 26634668 \cdot (126577584 + 185476576 + 202701232 \cdot \ln(1.02))$$

$$= 6.0388$$

Compute parameter [ϕ]:

$$= (1 + \nu^*) F$$

$$= (1 + 0.49) 6.04$$

$$= 9.0047$$

Compute parameter [Q1]:

$$= (\rho s - 1 - \phi \cdot Z_v) / (1 + \phi \cdot Z_m)$$

$$= (1.01 - 1 - 9 \cdot 0.0069) / (1 + 9 \cdot 0.12)$$

$$= -0.026436$$

Compute Parameter [Qz1]:

$$= (Z_d + Q1 \cdot Z_w) / 2 \cdot X_a^4$$

$$= (0.00083 + -0.026 \cdot 0.0069) / 2 \cdot 12^4$$

$$= 6.8328$$

Compute parameter [Qz2]:

$$= (Z_v + Q1 \cdot Z_m) / 2 \cdot X_a^4$$

$$= (0.0069 + -0.026 \cdot 0.12) / 2 \cdot 12^4$$

$$= 39.5558$$

Compute parameter [U]:

$$= (Z_w + (\rho s - 1) Z_m) X_a^4 / (1 + \phi \cdot Z_m)$$

$$= (0.0069 + (1.01 - 1) 0.12) 12^4 / (1 + 9 \cdot 0.12)$$

$$= 79.1116$$

ASME Section VIII-2 Part 4.18.8.4 Step 5:

Determine factor [γ_b]:

$$= 0 \text{ (config a)}$$

Compute Parameter [γ]:

$$= 0.000 \text{ mm (For Pressure only cases)}$$

Calculate parameter [ωs]:

$$= \rho s \cdot k_s \cdot \beta s \cdot \delta S (1 + h \cdot \beta s)$$

$$= 1.01 \cdot 684787968 \cdot 0.011 \cdot 0.45717472 (1 + 50 \cdot 0.011)$$

$$= 5388.3994 \text{ mm}^2$$

Calculate parameter [$\omega^* s$]:

$$= a o^2 (\rho s^2 - 1) \cdot (\rho s - 1) / 4 - \omega s$$

$$= 1117^2 (1.01^2 - 1) \cdot (1.01 - 1) / 4 - 5388$$

$$= -5352.7759 \text{ mm}^2$$

Parameter [ωc]:

$$= \rho c \cdot k_c [\delta C (C_m \theta + \beta C + C_q \Delta \cdot h \cdot \beta C^2) - \zeta c (C_m \theta + h \cdot C_q \theta \cdot h \cdot \beta C / 2)]$$

$$= 1.01 \cdot 1040162944 [0.37947923 (1 + 0.01 \cdot 1 \cdot 50) - 0.000 (1/50 + 1 \cdot 0.01/2)]$$

$$= 6134.2681 \text{ mm}^2$$

Parameter [$\omega^* c$]:

$$= a o^2 [(\rho c^2 + 1) \cdot (\rho c - 1) / 4 - (\rho s - 1) / 2] - \omega c$$

$$= 1117^2 [(1.01^2 + 1) \cdot (1.01 - 1) / 4 - (1.01 - 1) / 2] - 6134$$

$$= -6098.6445 \text{ mm}^2$$



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Pressure [P*S]:
 = 0 For Pressure only cases or Configurations d,e,f,A,B,C,D

Pressure [P*C]:
 = 0 For Pressure only cases or Configurations b,c,d,B,C,D

ASME Section VIII-2 Part 4.18.8.4 Step 6:

Pressure [P's]:

$$= P_s \{ x_s + 2(1 - x_s) v_t + [2/K_{st} \cdot (D_s/D_o)^2] v_s - [(p_s^2 - 1)/(J \cdot K_{st})] - [(1 - J)/(2J \cdot K_{st})][(D_j^2 - (D_s)^2)/D_o^2] \}$$

$$= 494 \{ 0.67 + 2(1 - 0.67)0.31 + [2/0.21 \cdot (2250/2233)^2]0.31 - [(1.01^2 - 1)/(1 \cdot 0.21)] - [(1 - 1)/(2 \cdot 1 \cdot 0.21)][(0^2 - (2250)^2)/2233^2] \}$$

 = 1876.7689 KPa.

Pressure [P't]:

$$= [x_t + 2(1 - x_t) v_t + 1/(J \cdot K_{st})] P_t$$

$$= [0.78 + 2(1 - 0.78)0.31 + 1/(1 \cdot 0.21)] 1300$$

 = 7376.9224 KPa.

Pressure [P_γ]:

$$= N_t \cdot K_t \cdot \gamma / (\pi \cdot a_o^2)$$

$$= 4500 \cdot 219 \cdot 0 / (3.14 \cdot 1117^2)$$

 = 0.0000 KPa.

Pressure [P_w]:

$$= -\gamma_b \cdot U \cdot W^* / (2 \cdot \pi \cdot a_o^2)$$

$$= -0 \cdot 79.1 \cdot 0 / (2 \cdot \pi \cdot 1117^2)$$

 = -0.0000 KPa.

Calculate the pressure [Prim]:

$$= -(U/a_o^2) \cdot (\omega \cdot s \cdot P_s - \omega \cdot c \cdot P_t)$$

$$= -(79.1/1117^2) \cdot (-5353 \cdot 494 - 6099 \cdot 1300)$$

 = -335.2929 KPa.

Calculate the pressure [P_ω]:

$$= (U/a_o^2) \cdot (\omega \cdot s \cdot P_s - \omega \cdot c \cdot P_t)$$

$$= (79.1/1117^2) \cdot (5388 \cdot 0 - 6134 \cdot 0)$$

 = 0.0000 KPa.

Determine the effective pressure [Pe]:

$$= J \cdot K_{st} / (1 + J \cdot K_{st} (Q_{z1} + (p_s - 1) Q_{z2})) \cdot (P_s - P_t + P_\gamma + P_\omega + P_w + P_{prim})$$

$$= 0.1000E+01 \cdot 0.21 / (1 + 1 \cdot 0.21 (6.83 + (1.01 - 1)39.6)) \cdot (1877 - 7377 + 0 + 0 + -0 + -335)$$

 = -490.7478 KPa.

ASME Section VIII-2 Part 4.18.8.4 Step 7:

Determine factor [Q2]:

$$= [((\omega \cdot s \cdot P_s - \omega \cdot c \cdot P_t) - (\omega \cdot s \cdot P_s - \omega \cdot c \cdot P_t)) + W^* \cdot \gamma_b / (2 \cdot \pi)] / (1 + \phi \cdot Z_m)$$

$$= [((-5353 \cdot 494 - 6099 \cdot 1300) - (5388 \cdot 0 - 6134 \cdot 0)) \cdot 0.0001 + 0 \cdot 0 / (2 \cdot \pi)] / (1 + 9 \cdot 0.12)$$

 = 261.9889 Kgf

Calculate factor [Q3]:

$$= Q_1 + 2 \cdot Q_2 / (P_e \cdot a_o^2)$$

$$= -0.026 + 2 \cdot 262 / (-491 \cdot 1117^2)$$

 = -0.03483

Fm Value from Table 4.18.3 = 0.017417



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Tubesheet Bending Stress - original thickness [σ]:

$$= (1.5 \cdot F_m / \mu \cdot) \cdot (2 \cdot a_o / (h - h'g))^2 \cdot P_e$$

$$= (1.5 \cdot 0.017 / 0.18) \cdot (2 \cdot 1117 / (50 - 0))^2 \cdot -491$$

$$= -142.7007 \text{ N./mm}^2$$

Allowable Tubesheet Bending Stress [σ allowed]:

$$= 1.5 \cdot \beta T \cdot S_y = 1.5 \cdot 0.95 \cdot 250 = 356 \text{ N./mm}^2, 4.18.233$$

Tubesheet Bending Stress - final thickness [σ_f]:

$$= (1.5 \cdot F_m / \mu \cdot) \cdot (2 \cdot a_o / (h - h'g))^2 \cdot P_e$$

$$= (1.5 \cdot 0.016 / 0.18) \cdot (2 \cdot 1117 / (28 - 0))^2 \cdot -427$$

$$= -356.2325 \text{ N./mm}^2$$

Reqd Tubesheet Thickness, for Bending Stress (including CA) [HReqB]:

$$= h + C_{ats} + C_{atc}$$

$$= 28 + 0 + 0$$

$$= 28.0495 \text{ mm}$$

ASME Section VIII-2 Part 4.18.8.4 Step 8:

Shear Stress check [τ limit]:

$$= (2 \cdot \mu \cdot h / a_o) \cdot (0.8 \cdot \beta T \cdot S_y)$$

$$= (2 \cdot 0.18 \cdot 50 / 1117) \cdot 190$$

$$= 3.0498 \text{ N./mm}^2$$

Shear Stress is not required to be computed; [P_e] <= τ ulimit

Could not converge on a minimum required tubesheet thk. to satisfy the Shear Stress criteria.



Use the best Estimated Required Tubesheet Thickness value as design guidance & Compare the Actual Stresses versus Allowable Stresses.

Tubesheet thickness = 13.7620 mm
 Tubesheet Shear Stress = 45.5505 N./mm²

Required Tubesheet Thickness for Given Loadings (Including CA) [H reqd]:

$$= \max(H_{reqB}, H_{reqS})$$

$$= \max(28, 13.8)$$

$$= 28.0495 \text{ mm}$$

ASME Section VIII-2 Part 4.18.8.4 Step 9:

The F_{tmin} and F_{tmax} Coefficients from Table 4.18.4:

$$F_{tmin} = -0.38, F_{tmax} = 6.23$$

First Extreme Tube Axial Stress from among all the tubes [σ_{t1}]:

$$= ((P_s \cdot x_s - P_t \cdot x_t) - P_e \cdot F_{tmin}) / (X_t - X_s)$$

$$= ((494 \cdot 0.67 - 1300 \cdot 0.78) - (-491) \cdot -0.38) / (0.78 - 0.67)$$

$$= -8.3220 \text{ N./mm}^2$$

Second Extreme value of Tube Axial Stress from among all the tubes [σ_{t2}]:

$$= ((P_s \cdot x_s - P_t \cdot x_t) - P_e \cdot F_{tmax}) / (X_t - X_s)$$

$$= ((494 \cdot 0.67 - 1300 \cdot 0.78) - (-491) \cdot 6.23) / (0.78 - 0.67)$$

$$= 22.9626 \text{ N./mm}^2$$

Maximum Tube Axial Stress [$\sigma_{t,max}$]:

$$= \max(|\sigma_{t,1}|, |\sigma_{t,2}|) = 23 \text{ N./mm}^2$$



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Allowable Tube Stress [σ_t, A]:
 $= \beta T \cdot S_{y,t} = 161.4971 \text{ N./mm}^2$

Check for Buckling as some of the tubes are in compression.

Determine the Factor of Safety [Fs]:
 $= \max((3.25 - 0.25(Z_d + Q_3 \cdot Z_w) X_{a^4}); 1.25)$
 $= \max((3.25 - 0.25(0.00083 + -0.035 \cdot 0.0069) 12^4); 1.25)$
 $= 2.0000 \text{ (Should be } \leq 2 \text{)}$

Determine the factor [rt]:
 $= (\sqrt{ \frac{dt^2}{4} + (\frac{dt}{2} - 2 \cdot tt)^2 }) / 4$
 $= (\sqrt{ \frac{19^2}{4} + (\frac{19}{2} - 2 \cdot 1.65)^2 }) / 4$
 $= 6.1794 \text{ mm}$

Determine the factor [Ct]:
 $= \sqrt{ \frac{2 \cdot \pi^2 \cdot E_t / S_{y,t}}{L^2} }$
 $= \sqrt{ \frac{2 \cdot \pi^2 \cdot 195117184 / 170}{L^2} }$
 $= 157.7078$

Determine the factor [Ft]:
 $= k \cdot L / r$
 $= 1 \cdot 700 / 6.18$
 $= 113.2791$

Buckling Allowable Stress [Stb]:
 $= (S_{y,t} / F_s) \cdot (1 - F_t / (2 \cdot C_t))$
 $= (170 / 2) \cdot (1 - 113 / (2 \cdot 158))$
 $= 161.497 \text{ N./mm}^2 \text{ (Never greater than } S_{ot} \text{)}$

Note: The Axial Compressive stress in tubes is within limits.

Largest tube-to-tubesheet Joint Load [Wt]:
 $= \sigma_{t,max} \cdot \text{Tube Area}$
 $= 23 \cdot 0.9$
 $= 211.19 \text{ Kgf}$

Tube Metal Area [At]:
 $= \pi (d_o - t) t$
 $= \pi (19 - 1.65) 1.65$
 $= 0.902 \text{ cm}^2$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:
 $= \pi \cdot t (d_o - t) S$
 $= \pi \cdot 1.65 (19 - 1.65) 161$
 $= 1485.317 \text{ Kgf}$

Groove Weld Strength [Fg]:
 $= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t \text{)}$
 $= 0.85 \cdot \pi \cdot 1.65 (19 + 0.67 \cdot 1.65) 161$
 $= 1462.4543 \text{ Kgf}$

Weld Strength Factor [fw]:
 $= S / S_w \text{ where } S_w = \min(S \text{ and } S_t)$
 $= 161 / 161$
 $= 1.000$

Design Strength Ratio [fd]:
 $= 1 \text{ for full strength welds}$

Fillet Weld Strength [Ff]:
 $= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t \text{)}$



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$$= 0.55 \cdot \pi \cdot 1.65 (19 + 0.67 \cdot 1.65) 161$$

$$= 946.2939 \text{ Kg}$$

Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{((0.75 \cdot do)^2 + 1.07 \cdot t \cdot (do - t) \cdot fw \cdot fd)} - 0.75 \cdot do \right)$$

$$= 2 \left(\sqrt{((0.75 \cdot 19)^2 + 1.07 \cdot 1.65 \cdot (19 - 1.65) \cdot 1 \cdot 1)} - 0.75 \cdot 19 \right)$$

$$= 2.075 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.0374 mm
 Minimum Required Groove Weld Leg [agr]: 1.0374 mm

Tube-Tubesheet Jt allowable, 1485 is >= tube strength 1485 Kg

Note: This tube-tubesheet joint is a Full Strength joint

ASME Section VIII-2 Part 4.18.8.4 Step 10:

Shell Axial Membrane Allowable Stress:
 $= \beta T \cdot Sy,s = 0.95 \cdot 205 = 292 \text{ N./mm}^2$

Axial Membrane Stress in Shell [σs,m]:

$$= ao^2 / ((Ds + ts)ts) \cdot [Pe + (ps^2 - 1)(Ps - Pt)] + as^2 \cdot Pt / ((Ds + ts)ts)$$

$$= 1117^2 / ((2250 + 12)12) \cdot [-491 + (1.01^2 - 1)(494 - 1300)] + 1125^2 \cdot 1300 / ((2250 + 12)12)$$

$$= 37.5153 \text{ N./mm}^2$$

ASME Section VIII-2 Part 4.18.8.4 Step 11:

Note:
 For a given Shell thickness of 12 mm, the minimum Shell length adjacent to the tubesheet should be 295.770 mm

Stress/Force Summary for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet Bend. Stress	142.7 <=	356.2 N./mm ²	Pass
Tubesheet Shear Stress	18.3 <=	190.0 N./mm ²	Pass
Maximum Tube Stress	23.0 <=	161.5 N./mm ²	Pass
Minimum Tube Stress (Buckling)	-8.3 <=	-161.5 N./mm ²	Pass
Maximum Force on any one Tube	211.2 <=	1485.3 Kg	Pass
Axial Membrane Stress in Shell	37.5 <=	292.1 N./mm ²	Pass

Thickness Results for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	28.049	50.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.037	1.650	Ok
Tube-Tubesheet Groove Weld Leg :	1.037	1.650	Ok

Fixed Tubesheet results per ASME Div. 2 Part 4.18.8 2025

Results for 3 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Stresses Bend Allwd Shear Allwd	Case Type	Pass/Fail
D1uc	35.369 ...	204	356 25 190	Ps+Pt-Th	D1 Ok
D2uc	14.966 ...	71	356 8 190	Ps+Pt-Th	D2 Ok
D3uc	28.049 ...	143	356 18 190	Ps+Pt-Th	D3 Ok
Max:	35.3692 ... mm	0.572	0.133 (Str. Ratio)		



Load Case Definitions:

[Ps & Pt]:
 Shell-side and Tube-side Design or Operating Pressures
 derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
 Load Case Tables

[(+)-Th]:
 With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:
 With or Without Corrosion Allowance

[D1, D2, D3]:
 Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:
 Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:
 Operating Load Cases using the Maximum and Minimum Operating Pressures and
 Operating Temperatures

Shell Axial Membrane Stress Summary:

Case#	---- Shell Stresses ----				:	---- Shell Band Stress ----				:	Pass Fail
	Ten	Allwd	Cmp	Allwd		Ten	Allwd	Cmp	Allwd		
D1uc	28	292	...	...	:	...	...	...	...	:	Ok
D2uc	6	292	...	...	:	...	...	...	...	:	Ok
D3uc	38	292	...	...	:	...	...	...	...	:	Ok
Max RATIO	0.128				:	...				:	...

Tube, Shell and Channel Stress Summary:

Case#	---- Tube Stresses ----				Tube Loads		Shell Stress		Channel Stress		Pass Fail
	Ten	Allwd	Cmp	Allwd	Ld	Allwd	Stress	Allwd	Stress	Allwd	
D1uc	30	161	-13	-161	279	1485	...	292	...	370	Ok
D2uc	9	161	-9	-161	83	1485	...	292	...	370	Ok
D3uc	23	161	-8	-161	211	1485	...	292	...	370	Ok
Max RATIO	0.188				0.077		0.188		...		...

Summary of Thickness Comparisons for 3 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	35.369	50.000	Ok
Tube thickness :	0.308	1.650	Ok
Tube-tubesheet fillet weld leg :	1.037	1.650	Ok
Tube-tubesheet groove weld leg :	1.037	1.650	Ok

Min Shell length of thk, (12.000) adj. to tubesheet: 295.770 mm

Note: This is a full strength Tube to Tubesheet Joint.

Summary of Axial Differential Expansion between Shell and Tubes :

Due to Thermal Expansion Shell Extends by : +.000 mm
 Due to Pressure Shell Extends by : +.035 mm
 Due to Pressure + Thermal Shell Extends by : +.035 mm



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ASME TS Calc: Pressure Test Case: 1 6:24p Sep 8, 2026

Summary of Exchanger Pressure and Thermal Displacements: (mm)

Load Case#	Due to Pressure		Due to Thermal Exp.		Computed Delta Ms	
	UnCorroded	Corroded	UnCorroded	Corroded	UnCorroded	Corroded
1	0.0347	...	...	...	...	...
2	-0.0322	...	...	...	...	...
3	0.0019	...	...	...	...	...
4	...	...	...	...	...	...
5	...	...	...	...	...	...
6	...	...	...	...	...	...
7	...	...	...	...	...	...
8	...	...	...	...	...	...

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External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2025

External Pressure Calculations:

From	To	Section Length mm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B N./mm ²
10	20	No Calc	...	153.8	No Calc	No Calc
20	30	No Calc	...	210.8	No Calc	No Calc
30	Tsht	999.97	2278	10.8	0.0010457	86.6632
Tsht	40	No Calc	...	10.8	No Calc	No Calc
40	50	8200	2272	10.2	0.00010554	9.76462
50	60	8200	2298	23.2	0.00035034	32.4155
60	Tsht	8200	2272	10.2	0.00010554	9.76462
Tsht	70	No Calc	...	10.2	No Calc	No Calc
70	80	1000.03	2282	12.8	0.001374	94.4685
80	90	No Calc	...	210.8	No Calc	No Calc
90	100	No Calc	...	153.8	No Calc	No Calc

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure KPa.	External M.A.W.P. KPa.
10	20	157	115.291	103.419	No Calc
20	30	214	208.94	103.419	No Calc
30	Tsht	14	8.3849	103.419	548
Tsht	40	14	No Calc	103.419	No Calc
40	50	11	10.5149	51.7095	58.4
50	60	24	10.5806	51.7095	436
60	Tsht	11	10.5149	51.7095	58.4
Tsht	70	11	No Calc	51.7095	No Calc
70	80	16	8.3903	103.419	706
80	90	214	208.94	103.419	No Calc
90	100	157	115.291	103.419	No Calc

Minimum

58.4

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	30	No Calc	No Calc	No Calc	No Calc
30	Tsht	999.97	5816.62	No Calc	No Calc
Tsht	40	No Calc	No Calc	No Calc	No Calc
40	50	8200	9210.43	No Calc	No Calc
50	60	8200	186576	No Calc	No Calc
60	Tsht	8200	9210.43	No Calc	No Calc
Tsht	70	No Calc	No Calc	No Calc	No Calc
70	80	1000.03	8770.39	No Calc	No Calc
80	90	No Calc	No Calc	No Calc	No Calc
90	100	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.



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 Element and Detail Weights: Step: 16 6:24pm Sep 8,2026

Element and Detail Weights:

From	To	Element Metal Wgt. kg.	Element ID Volume cm ³	Corroded Metal Wgt. kg.	Corroded ID Volume cm ³	Extra due Misc % kg.
10	20	6302.59	...	6302.59	...	630.259
20	30	1687.84	1090693	1639.91	1091848	168.784
30	40	771.764	3976790	596.203	3999446	77.1764
40	50	2217.89	8728423	2070.76	8746756	221.789
50	60	2408.57	4714425	2329.1	4724327	240.857
60	70	2197.35	8647604	2051.58	8665767	219.735
70	80	882.795	3976790	707.234	3999446	88.2795
80	90	1687.84	1090693	1639.91	1091848	168.784
90	100	6302.59	...	6302.59	...	630.259
Total		24459	32225422.00	23639	32319440.00	2445



For elements specified as shell side elements, the volume(s) shown above for those elements, reflects the displacement of the tubes.

Weight of Details:

From	Type	Weight of Detail kg.	X Offset, Dtl. Cent. mm	Y Offset, Dtl. Cent. mm	Z Offset Dtl. Cent. mm	Description
10	Liqd	...	78.5	-25	...	Liquid: 10
20	Liqd	1090.03	132.5	-25	...	Liquid: 20
30	Liqd	3974.36	500	...	...	Liquid: 30
30	Nozl	121.858	500	1353.6	...	N5-Tube Inlet
30	Wght	200	500	...	...	Partition weigh
40	Sadl	964.263	900	1427	...	Left Saddle flo
40	Liqd	8723.1	1620	...	...	Liquid: 40
40	Wght	450	900.005	-1760	...	Fire prrofig
50	Liqd	4711.55	875	...	...	Liquid: 50
50	Nozl	591.777	874	1582.2	...	N1-Shell inlet
50	Nozl	590.591	874	-1582.2	...	N4-Shell outlet
60	Sadl	964.263	2290	1427	...	Right saddle fi
60	Liqd	8642.33	1605	...	...	Liquid: 40
60	Wght	450	2290	-1760	...	Fireproofing we
70	Liqd	3974.36	500	...	...	Liquid: 50
70	Nozl	121.858	500	-1353.6	...	N6-Tube Outlet
80	Liqd	1090.03	132.5	-25	...	Liquid: 60
90	Liqd	...	78.5	-23	...	Liquid: 70
30	FTsh	1190.62	1025	...	...	Fixed Tubesheet
30	Tube	29746.5	5150	...	...	Tubes
30	RTsh	1190.62	9275	...	...	Tubesheet
40	Bafl	128.036	700	...	...	Baffles
40	Bafl	128.036	1400	...	...	Baffles
40	Bafl	128.036	2100	...	...	Baffles
40	Bafl	128.036	2800	...	...	Baffles
40	Bafl	128.036	3500	...	...	Baffles
40	Bafl	128.036	4200	...	...	Baffles
40	Bafl	128.036	4900	...	...	Baffles
40	Bafl	128.036	5600	...	...	Baffles
40	Bafl	128.036	6300	...	...	Baffles
40	Bafl	128.036	7000	...	...	Baffles

Total Weight of Each Detail Type:



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Saddles	1928.5
Liquid	32205.8
Nozzles	1426.1
Weights	1100.0
Exchanger Components	33408.1
Liquid in Tubes	7273.7

Sum of the Detail Weights 77342.2 kg.

Weight Summation Results: (kg.)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	26905.2	26905.2	26905.2	26905.2	26905.2	26905.2
Saddles	1928.5	1928.5	1928.5	1928.5	1928.5	1928.5
Nozzles	1426.1	1426.1	1426.1	1426.1	1426.1	1426.1
Wld Weights	200.0	200.0	200.0	200.0	200.0	200.0
Exchanger	33408.1	33408.1	33408.1	33408.1	33408.1	33408.1
Test Weights	...	900.0	...	...	...	...
Ope Weights	...	...	...	...	...	900.0
Ope. Liquid	...	...	...	...	...	32205.8
Tube Ope Lqd	...	...	...	...	...	7273.7
Test Liquid	...	32205.8	...	...	...	...
Tube Tst Lqd	...	7273.7	...	...	...	...
Totals	63867.9	104247.3	63867.9	63867.9	63867.9	104247.3

Miscellaneous Weight Percent: 10.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	63867.9 kg.
Shop Test Wt.	- Fabricated weight + water (full)	104247.3 kg.
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	63867.9 kg.
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	63867.9 kg.
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	63867.9 kg.
Operating Wt.	- Empty weight + operating liq. uncorroded	104247.3 kg.
Oper. Wt. + CA	- Corr wt. + operating liquid	103346.0 kg.
Field Test Wt.	- Empty weight + test medium (full)	104247.3 kg.

Exchanger Tube Data

Volume of Exchanger tubes :	7278123.0 cm ³
Weight of Ope Liq in tubes :	7273.7 kg.
Weight of Water in tubes :	7273.7 kg.

The Corroded Weight and thickness are used in the Horizontal Vessel Analysis (Ope Case) and Earthquake Load Calculations.

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
10	20	62268.9	...
20	30	30666.1	19297.2
30	40	71565.5	70685.8
40	50	231465	229022
50	60	126339	123700
60	70	229322	226902
70	80	71691.1	70685.8



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80	90	30666.1	19297.2
90	100	62268.9	...

Total	916252.8	759590.0 cm ²
Total	986.2	817.6 ft ²

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Nozzle Flange MAWP Results: (KPa. & °C)

Nozzle Description	Flange Rating Ope.	Ambient	Des. Temp	Class	Grade/Group	Equiv. Press	- - - UG-44	Max Pressure 50%	DNV
N5-Tube Inlet	1820.4	1960.0	82	150	GR 1.1	1795.5	1820.4	1820.4	935.1
N1-Shell inlet	1476.0	1910.0	130	150	GR 2.1	1070.4	869.4	1476.0	1143.7
N4-Shell outle	1476.0	1910.0	130	150	GR 2.1	1070.4	869.4	1476.0	1143.7
N6-Tube Outlet	1820.4	1960.0	82	150	GR 1.1	1795.5	1820.4	1820.4	935.1

Shellside Flange Rating

Lowest Flange Pressure Rating was (Ope)[ShellSide]: 869.430 KPa.
 Lowest Flange Pressure Rating was (Amb)[ShellSide]: 1910.006 KPa.

Channelside Flange Rating

Lowest Flange Pressure Rating was (Ope)[TubeSide]: 1820.434 KPa.
 Lowest Flange Pressure Rating was (Amb)[TubeSide]: 1960.006 KPa.

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm	Fm Value	Resolved Moment ME Kg-m.	Axial Force FE Kgf	MAWP PD KPa.	Maximum Pressure Pmax KPa.	Pass Fail
N5-Tube Inlet	500.888	1.20	4087	3442	1081.5	1820.4	Pass
N1-Shell inlet	942.975	0.31	16348	6884	791.3	869.4	Pass
N4-Shell outlet	942.975	0.31	16348	6884	791.3	869.4	Pass
N6-Tube Outlet	500.888	1.20	4087	3442	1081.5	1820.4	Pass

Evaluating Flange on Nozzle [N5-Tube Inlet]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [((PR - PD) + FM \cdot PR)]$$

$$= 16(4087) + 4 \cdot \text{abs}(3442) \cdot 500.888 \leq \pi \cdot 500.888^3 [(1820.434 - 1081.470) + 1.200 \cdot 1820.434]$$

$$= 0.72289E+05 \leq 0.11770E+06 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N5-Tube Inlet [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(1820.434, 1.200 \cdot 1820.434 + 1820.434 - 72288.7 / (\pi \cdot 500.888^3))$$

$$= \min(1820.434, 2209.423)$$

$$= 1820.434 \text{ KPa.}$$

Evaluating Flange on Nozzle [N1-Shell inlet]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [((PR - PD) + FM \cdot PR)]$$

$$= 16(16348) + 4 \cdot \text{abs}(6884) \cdot 942.975 \leq \pi \cdot 942.975^3 [(1476.034 - 791.335) + 0.314 \cdot 1476.034]$$

$$= 0.28754E+06 \leq 0.30851E+06 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N1-Shell inlet [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(1476.034, 0.314 \cdot 1476.034 + 1476.034 - 287535.4 / (\pi \cdot 942.975^3))$$

$$= \min(1476.034, 869.430)$$

$$= 869.430 \text{ KPa.}$$

Evaluating Flange on Nozzle [N4-Shell outlet]:



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$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR]$$
$$= 16(16348) + 4 \cdot \text{abs}(6884) 942.975 \leq \pi \cdot 942.975^3 [(1476.034 - 791.335) + .314 \cdot 1476.034]$$
$$= 0.28754E+06 \leq 0.30851E+06 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N4-Shell outlet [Pmax]:
$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$
$$= \min(1476.034, 0.314 \cdot 1476.034 + 1476.034 - 287535.4 / (\pi \cdot 942.975^3))$$
$$= \min(1476.034, 869.430)$$
$$= 869.430 \text{ KPa.}$$

Evaluating Flange on Nozzle [N6-Tube Outlet]:
$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR]$$
$$= 16(4087) + 4 \cdot \text{abs}(3442) 500.888 \leq \pi \cdot 500.888^3 [(1820.434 - 1081.470) + 1.200 \cdot 1820.434]$$
$$= 0.72289E+05 \leq 0.11770E+06 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N6-Tube Outlet [Pmax]:
$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$
$$= \min(1820.434, 1.200 \cdot 1820.434 + 1820.434 - 72288.7 / (\pi \cdot 500.888^3))$$
$$= \min(1820.434, 2209.423)$$
$$= 1820.434 \text{ KPa.}$$

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

Large Diameter Flange Pressure Ratings are per ASME B16.47 2025 Edition

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 Wind Load Calculation: Step: 18 6:24pm Sep 8,2026

Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	92.7 Km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	27500 mm
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm
Distance Upwind of Crest Lh	0 mm
Distance from Crest to the Vessel x	0 mm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:
 $= \min(0.85, 0.925((1 + 1.7 \cdot gQ \cdot I_{zbar} \cdot Q) / (1 + 1.7 \cdot gV \cdot I_{zbar})))$
 $= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.18 \cdot 0.91) / (1 + 1.7 \cdot 3.4 \cdot 0.18)))$
 $= \min(0.85, 0.88)$
 $= 0.850$

Natural Frequency of Vessel (Operating)	33.000 Hz
Natural Frequency of Vessel (Empty)	33.000 Hz
Natural Frequency of Vessel (Test)	33.000 Hz

User Entered Importance Factor is	1.000
Force Coefficient [Cf]	0.566
Structure Height to Diameter ratio	4.948

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of $[\alpha]$ and $[Zg]$:
 Exposure Category: C from Table 26.11-1
 $\alpha = 9.5 : Zg = 274320 \text{ mm}$

Effective Height $[z]$:
 $= \text{Centroid Height} + \text{Vessel Base Elevation}$
 $= 1760 + 27500 = 29260 \text{ mm}$
 $= 96 \text{ ft. Imperial Units}$

Velocity Pressure coefficient evaluated at height z $[Kz]$:
 Because z (96 ft.) $> 15 \text{ ft.}$
 $= 2.01(z / Zg)^{2/\alpha}$
 $= 2.01(96/900)^{2/9.5}$
 $= 1.25$

Type of Hill: No Hill



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Wind Directionality Factor [Kd]:
 = 1, per Table 26.6-1

As there is No Hill Present: [Kzt]:
 K1 = 0, K2 = 0, K3 = 0

Topographical Factor [Kzt]:
 = $(1 + K1 \cdot K2 \cdot K3)^2$
 = $(1 + 0 \cdot 0 \cdot 0)^2$
 = 1

Velocity Pressure evaluated at height z, Imperial Units [qz]:
 = $\max(16, 0.00256 \cdot Kz \cdot Kzt \cdot Kd \cdot Ke \cdot V(\text{mph})^2)$
 = $\max(16, 0.00256 \cdot 1.25 \cdot 1 \cdot 1 \cdot 1 \cdot 57.6^2)$
 = 16 psf [78.1] Kgs/m²

Force on the first element [F]:
 = $qz \cdot G \cdot Cf \cdot \text{WindArea}$
 = $16 \cdot 0.85 \cdot 0.57 \cdot 5.08$
 = 39.1 lbs. [17.7] Kgf

Element	Hgt (z) mm	K1	K2	K3	Kz	Kzt	qz Kgs/m ²
LHS Blind Flg	29260.0	0.000	0.000	0.000	1.255	1.000	78.120
LHS CH.Fl g	29260.0	0.000	0.000	0.000	1.255	1.000	78.120
LHS CH Shell	29260.0	0.000	0.000	0.000	1.255	1.000	78.120
main Shell 1	29260.0	0.000	0.000	0.000	1.255	1.000	78.120
main Shell 2	29260.0	0.000	0.000	0.000	1.255	1.000	78.120
main Shell 2	29260.0	0.000	0.000	0.000	1.255	1.000	78.120
RHS Ch Shell	29260.0	0.000	0.000	0.000	1.255	1.000	78.120
RHS Ch Flg	29260.0	0.000	0.000	0.000	1.255	1.000	78.120
RHS Blind Flang	29260.0	0.000	0.000	0.000	1.255	1.000	78.120

Wind Loads on Masses/Equipment/Piping

ID	Wind Area cm ²	Elevation mm	Pressure Kgs/m ²	Force Kgf
Partition wei	0.00	29260.00	78.12	0.00
Fire prroofing	0.00	27500.00	78.12	0.00
Fireproofing	0.00	27500.00	78.12	0.00

Wind Load Calculation:

From	To	Wind Height mm	Wind Diameter mm	Wind Area cm ²	Wind Pressure Kgs/m ²	Element Wind Load Kgf
10	20	29260	3006	4719.42	78.12	19.5039
20	30	29260	2700	7155	78.12	29.5694
30	40	29260	2733.6	27336	78.12	112.971
40	50	29260	2726.4	88335.4	78.12	365.063
50	60	29260	2757.6	48258	78.12	199.436
60	70	29260	2726.4	87517.4	78.12	361.683
70	80	29260	2738.4	27384	78.12	113.17
80	90	29260	2700	7155	78.12	29.5694
90	100	29260	3006	4719.42	78.12	19.5039

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.



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Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	1.250
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	3.000
Site Class	D
Component Elevation Ratio z/h	0.000
Amplification Factor Ap	1.000
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

Seismic Analysis Results:

$$\begin{aligned} SMS &= Fa * Ss = 1.6 * 0.15 = 0.24 \\ SM1 &= Fv * S1 = 2.4 * 0.065 = 0.16 \\ SDS &= 2/3 * Sms = 2/3 * 0.24 = 0.16 \\ SD1 &= 2/3 * Sm1 = 2/3 * 0.16 = 0.1 \end{aligned}$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 = $C_t \cdot h_n^x$ where $C_t = 0.020$, $x = 0.75$ and h_n = Structural Height (ft.)
 = $0.020(9.880.75)$
 = 0.111 seconds

The Coefficient C_u from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 = (1/Natural Frequency) = (1/33)
 = 0.030

Check the Value of T which is the smaller of $C_u \cdot T_a$ and T:
 = Minimum Value of ($1.7 \cdot 0.11$, 0.03) per 12.8.2
 = 0.030
 As the time period is < 0.06 second, use section 15.4.2.

Base Shear per equation 15.4-5, [V]:
 = $0.3 \cdot Sds \cdot W \cdot I$
 = $0.3 \cdot 0.16 \cdot 101418 \cdot 1.25$
 = 5963.351 Kgf

Vertical load per 12.4-4, [Ev]:
 = $0.2 \cdot Sds \cdot W$
 = $0.2 \cdot 0.16 \cdot 101418$
 = 3180.5 Kgf

Earthquake Load Calculation:

From	To	Earthquake Height mm	Earthquake Weight Kgf	Element Ope Load Kgf
10	20	1252.5	9219.77	591.378



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Earthquake Load Calculation: Step: 19 6:24pm Sep 8, 2026

20	30	1125	9219.77	531.177
30	40	1125	9219.77	531.177
40	50	1125	9219.77	531.177
40	50	1125	9219.77	531.177
50	60	1125	9219.77	531.177
60	0	1125	9219.77	531.177
60	70	1125	9219.77	531.177
70	80	1125	9219.77	531.177
80	90	1125	9219.77	531.177
90	100	1252.5	9219.77	591.378

The Earthquake Weight does not include the weight of the saddles.

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Center of Gravity Calculation: Step: 20 6:24pm Sep 8,2026

Shop/Field Installation Options :

Note : The CG is computed from the first Element From Node

Center of Gravity of the Saddles	5574.700 mm
Center of Gravity of the Liquid	5584.915 mm
Center of Gravity of the Nozzles	5596.308 mm
Center of Gravity of the Added Weights (Operating)	4731.066 mm
Center of Gravity of the Added Weights (Empty)	934.700 mm
Center of Gravity of the Tubesheet(s)	5584.700 mm
Center of Gravity of the Tubes	5584.700 mm
Center of Gravity of the Baffles	5334.699 mm
Center of Gravity of Bare Shell New and Cold	5606.551 mm
Center of Gravity of Bare Shell Corroded	5607.308 mm
Vessel CG in the Operating Condition	5578.245 mm
Vessel CG in the Fabricated (Shop/Empty) Condition	5574.289 mm
Vessel CG in the Test Condition	5578.266 mm

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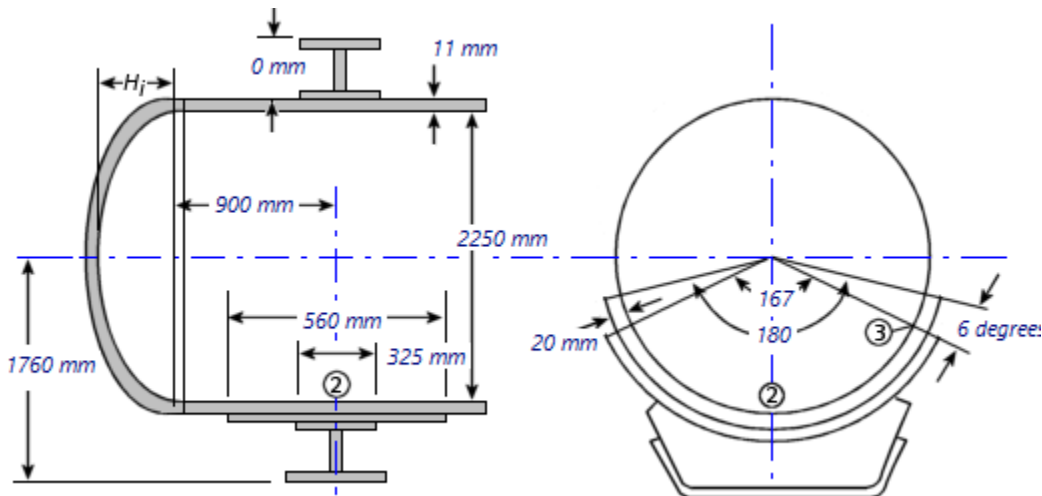
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 Saddle Calcs: Road Transport: Step: 21 6:24pm Sep 8,2026

ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Road Transport

Input and Calculated Values:

Vessel Mean Radius	Rm	1130.50	mm
Shell Thickness used in this Case	t	11.000	mm
Stiffened Vessel Length per 4.15.6	L	8200.00	mm
Distance from Saddle to Vessel tangent	a1 or a	900.00	mm
Saddle Width	b1 or b	325.00	mm
Saddle Bearing Angle	delta or theta	167.00	degrees
Wear Plate Width	b2 or b1	560.00	mm
Wear Plate Bearing Angle	delta2 or theta1	180.00	degrees
Wear Plate Thickness	e2 or tr	20.0	mm
Wear Plate Allowable Stress	fw or Sr	106.20	N./mm ²
Shell Allowable Stress used in Calculation		106.20	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1760.00	mm
Coefficient of Friction	mu	0.30	

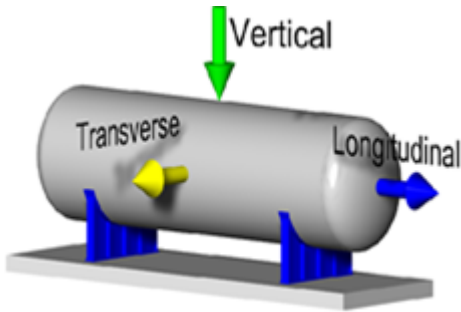


Results for Horizontal Vessel Analysis:

Saddle Force Q for this Transportation Case		41597.37	Kgf
Shipping Longitudinal Acceleration	gx	0.10	g
Shipping Transverse Acceleration	gz	0.10	g
Shipping Vertical Acceleration	gy	0.05	g
Vertical Acceleration Acting with Transverse	gw	1.00	g
Internal Pressure during Transport		0.00	KPa.
Transportation Wind Load Scalar Factor	wlsf	1.00	
Transportation Wind Velocity	wv	333.7	Km/hr



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Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	-11.34	-184.50
Long. Stress at Top of Midspan	11.34	106.20
Long. Stress at Bottom of Midspan	11.34	106.20
Long. Stress at Top of Saddles	0.98	106.20
Long. Stress at Bottom of Saddles	-0.58	-184.50
Long. Stress at Bottom of Saddles	0.58	106.20
Tangential Shear in Shell	16.92	84.96
Circ. Stress at Horn of Saddle	17.74	159.30
Circ. Compressive Stress in Shell	1.69	106.20

Intermediate Results: Saddle Reaction Q due to Road Transport Load:

Transverse Force due to Wind during Transport [F_{tw}]:
 = Transverse Area • 0.00256 • wv² • wlsf
 = 326 • 0.00256 • 207² • 1
 = 35853.9 lbf
 = 16263.309 Kgf

Longitudinal Force due to Wind during Transport [F_{lw}]:
 = π • D²/4 • 0.00256 • wv² • wlsf • Wind Dia. Mult
 = π • 7.45²/4 • 0.00256 • 207² • 1 • 1.2
 = 5764.8 lbf
 = 2614.9 Kgf

Transverse Force due to Lateral Acceleration [F_t]:
 = g_z • Shipping Weight • g_w + F_{tw}
 = 0.1 • 61939 • 1 + 16263
 = 22457.2 Kgf

Longitudinal Force due to Axial Acceleration [F_l]:
 = g_x • Shipping Weight • g_w + F_{lw}
 = 0.1 • 61939 • 1 + 2615
 = 8808.8 Kgf

Saddle Reaction Force due to Transverse Load [F_{wt}]:
 = F_t/Num of Saddles • B / E
 = 22457/2 • 1760/2070
 = 9546.6 Kgf

Saddle Reaction Force due to Longitudinal Load [F_{wl}]:
 = F_l • B / L_s
 = 8809 • 1760/6380
 = 2430.0 Kgf

Saddle Reaction Force due to Vertical Transportation Acceleration [F_{rv}]:
 = Saddle Load • g_y
 = 30525 • 0.05



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= 1526.228 Kgf

Load Combination Results due to Transport Loading [Q]:

= Saddle Load + Frv + max(Fwl, Fwt)
 = 30525 + 1526 + max(2430, 9547)
 = 41597.4 Kgf

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	42561.63	Kgf
Transverse Shear Load	11228.62	Kgf
Longitudinal Shear Load	8808.85	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

= Baseplate Length
 = 2070.100 mm

The Computed K values from Table 4.15.1:

K1 = 0.1974	K2 = 0.6609	K3 = 0.3428	K4 = 0.2477
K5 = 0.6419	K6 = 0.0229	K7 = 0.0159	K8 = 0.2738
K9 = 0.1786	K10 = 0.0228	K1* = 0.3348	K6p = 0.0174
K7p = 0.0121			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

= $-Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))/(1 + (4hm)/3L)]$
 = $-41597 \cdot 900 [1 - (1 - 900/8200 + (1130^2 - 0^2)/(2 \cdot 900 \cdot 8200))/(1 + (4 \cdot 0)/(3 \cdot 8200))]$
 = -867.4 Kg-m.

Moment per Equation 4.15.2 [M2]:

= $Q \cdot L/4 ((1 + 2(Rm^2 - hm^2)/(L^2))/(1 + (4hm^2)/(3L)) - 4a/L)$
 = $41597 \cdot 8200/4 ((1 + 2(1130^2 - 0^2)/(8200^2))/(1 + (4 \cdot 0)/(3 \cdot 8200)) - 4 \cdot 900/8200)$
 = 51079.6 Kg-m.

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

= $P \cdot Rm/(2t) - M2/(\pi \cdot Rm^2 t)$
 = $0 \cdot 1130/(2 \cdot 11) - 51080/(\pi \cdot 1130^2 \cdot 11)$
 = -11.34 N./mm²

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

= $P \cdot Rm/(2t) + M2/(\pi \cdot Rm^2 \cdot t)$
 = $0 \cdot 1130/(2 \cdot 11) + 51080/(\pi \cdot 1130^2 \cdot 11)$
 = 11.34 N./mm²

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_3]:

= $P \cdot Rm/(2t) - M1/(K1 \cdot \pi \cdot Rm^2 t)$
 = $0 \cdot 1130/(2 \cdot 11) - 867/(0.2 \cdot \pi \cdot 1130^2 \cdot 11)$
 = 0.98 N./mm²

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_4]:

= $P \cdot Rm/(2t) + M1/(K1 \cdot \pi \cdot Rm^2 \cdot t)$
 = $0 \cdot 1130/(2 \cdot 11) + 867/(0.33 \cdot \pi \cdot 1130^2 \cdot 11)$
 = -0.58 N./mm²

Maximum Shear Force in the Saddle (4.15.3) [T]:

= $Q(L - 2a)/(L + (4 \cdot hm/3))$
 = $41597 (8200 - 2 \cdot 900)/(8200 + (4 \cdot 0/3))$
 = 32466.2 Kgf



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Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 0.66 \cdot 32466 / (1130 \cdot 11)$$

$$= 16.92 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1130 \cdot 11)}$$

$$= 86.981 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(325 + 1.56 \cdot \sqrt{(1130 \cdot 11)}, 2 \cdot 900)$$

$$= 498.96 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= 1.0000 \text{ Materials are the same, transport case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr))$$

$$= -0.64 \cdot 41597 \cdot 0.1 / (499(11 + 1 \cdot 20))$$

$$= -1.69 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.29) [σ_7, r^*]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 12 \cdot K_7 \cdot Q \cdot R_m / (L(t + \eta \cdot tr)^2)$$

$$= -41597 / (4(11 + 1 \cdot 20)499) -$$

$$12 \cdot 0.016 \cdot 41597 \cdot 1130 / (8200(11 + 1 \cdot 20)^2)$$

$$= -17.74 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2070.1001	mm
Baseplate Thickness	Bpthk	22.0000	mm
Baseplate Width	Bpwid	350.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	14.0000	mm
Web Thickness	Webtk	14.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	730.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(1125 \cdot 11)} + 560)11^3) / 12$$

$$= 8.136 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 8.14 + 80.7(219 - 5.5)^2$$

$$= 36959.289 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	733.5	11.0	5.5	80.7	44379.1	0.370E+05
Wearplate	560.0	20.0	21.0	112.0	235200.0	0.442E+05
Web	14.0	582.0	322.0	81.5	2623656.2	0.316E+05
BasePlate	350.0	22.0	624.0	77.0	4804800.0	0.126E+06
Totals	...	...	...	351.2	7708035.5	0.239E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 7708035/351$$



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$$= 219.496 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 167/2$$

$$= 96.5$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(96.5) - 0.5 \cdot \sin(96.5)^2) / (\pi - 1.68 + \sin(96.5) \cos(96.5))$$

$$= 0.2924$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.29 \cdot 41597$$

$$= 12161.9258 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 4.4096 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1760 - 1125 \cdot \sin(167/2) / 1.46$$

$$= 993.013 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 12077.1885 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 10.8905 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(41597 + 964) \cdot 350) / (4 \cdot 2070 \cdot 173)}$$

$$= 17.475 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 167/2) (1125 + 11 + 20)$$

$$= 2297.138 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nrings} - 1)$$

$$= 2297 / (5 - 1)$$

$$= 574.285 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 574 \cdot 350 / 2$$

$$= 1004.998 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 42562 / (2070 \cdot 350)$$

$$= 5.874 \text{ Kgf/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 1005 \cdot 5.87$$

$$= 5903.698 \text{ Kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 43.5 + 40.2$$



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$$= 83.740 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P/A_r$$

$$= 5904/83.7$$

$$= 6.914 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	14.0	325.0	...	45.5	...	0.400E+04

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 325 - 14$$

$$= 311.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/83.7$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 325/2$$

$$= 162.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(4005/83.7)}$$

$$= 69.156 \text{ mm}$$

Length of Outer Rib, step 1, [Lv]:

$$= \text{Saddle Height} - \cos(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1760 - \cos(167/2) (1125 + 11 + 20) - 22$$

$$= 1607.137 \text{ mm}$$

Length of Outer Rib, step 2, [Lw]:

$$= 2 \cdot \sin(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk})$$

$$= 2 \cdot \sin(167/2) (1125 + 11 + 20)$$

$$= 2297.138 \text{ mm}$$

Length of Outer Rib, step 3, [Bd]:

$$= \text{Baseplate length} - \text{clearance}$$

$$= 2070 - 50.8$$

$$= 2019.300 \text{ mm}$$

Length of Outer Rib, step 4, [Dd]:

$$= (Lw - Bd) / 2$$

$$= (2297 - 2019) / 2$$

$$= 138.919 \text{ mm}$$

Length of Outer Rib [L]:

$$= \sqrt{(Lv^2 + Dd^2)}$$

$$= \sqrt{(1607^2 + 139^2)}$$

$$= 1613.130 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 199943008/260)}$$

$$= 123.211$$



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Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1613/69.2$$

$$= 23.326$$

Bending Moment [Rm]:

$$= Fl / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 8809 / (2 \cdot 2070) \cdot 574 \cdot 1613 / 2$$

$$= 985.537 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ (23.3 < 123) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (23.3)^2 / (2 \cdot 123^2)) 260 /$$

$$(5/3 + 3 \cdot (23.3) / (8 \cdot 123) - (23.3^3) / (8 \cdot 123^3))$$

$$= 147 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 6.91 / 147 + (986 \cdot 162 / 40049480) / 173$$

$$= 0.273$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	14.0	311.0	0.0	43.5	0.0	0.400E+04
Web	574.3	14.0	0.0	80.4	0.0	13.1
Totals	...	...	...	123.9	...	0.402E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 124$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 325 / 2$$

$$= 162.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - B_{pthk}$$

$$= 1760 - \cos(1156 / 2) (22 + 0 + 0) - 0$$

$$= 582.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (4018 / 124) }$$

$$= 56.936 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 582 / 56.9$$

$$= 10.222$$

Unit Force [Force,u]:

$$= Fl / (2 \cdot \text{Baseplate Length})$$

$$= 8809 / (2 \cdot 2070)$$

$$= 2.128 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 2.13 \cdot 574 \cdot 582$$

$$= 711.142 \text{ Kg-m.}$$



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Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:
 = Bending Moment / Section Modulus
 = 711/247247
 = 28.206 N./mm²

Compressive Allowable, $KL/r < C_c$ (10.2 < 123) per AISC E2-1 [Sca]:
 = (1-(KL/r)²/(2• C_c^2)) F_y /(5/3+3•(KL/r)/(8• C_c)-(KL/r)³)/(8• C_c^3)
 = (1-(10.2)²/(2 • 123²))260/
 (5/3+3•(10.2)/(8• 123)-(10.2³)/(8•123³)
 = 152.6 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)
 = $S_c/S_{ca} + (M_{base,c} \cdot C_{1/I})/S_{ba}$
 = 9.35/153 + (711 • 162/4018)/173
 = 0.224

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	1130.50	mm
Shell Thickness used in this Case	t	11.000	mm
Stiffened Vessel Length per 4.15.6	L	8200.00	mm
Distance from Saddle to Vessel tangent	a1 or a	900.00	mm
Saddle Width	b1 or b	325.00	mm
Saddle Bearing Angle	delta or theta	167.00	degrees
Wear Plate Width	b2 or b1	560.00	mm
Wear Plate Bearing Angle	delta2 or theta1	180.00	degrees
Wear Plate Thickness	e2 or tr	20.0	mm
Wear Plate Allowable Stress	fw or Sr	106.20	N./mm ²
Shell Allowable Stress used in Calculation		106.20	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1760.00	mm
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:

Saddle Force Q for this Transportation Case		41585.77	Kgf
Shipping Longitudinal Acceleration	gx	0.10	g
Shipping Transverse Acceleration	gz	0.10	g
Shipping Vertical Acceleration	gy	0.05	g
Vertical Acceleration Acting with Transverse	gw	1.00	g
Internal Pressure during Transport		0.00	KPa.
Transportation Wind Load Scalar Factor	wlsf	1.00	
Transportation Wind Velocity	wv	333.7	Km/hr

Horizontal Vessel Analysis Results:		Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top	of Midspan	-11.34	-184.50
Long. Stress at Top	of Midspan	11.34	106.20
Long. Stress at Bottom	of Midspan	11.34	106.20
Long. Stress at Top	of Saddles	0.98	106.20
Long. Stress at Bottom	of Saddles	-0.58	-184.50
Long. Stress at Bottom	of Saddles	0.58	106.20
Tangential Shear in Shell		16.92	84.96
Circ. Stress at Horn of Saddle		17.73	159.30
Circ. Compressive Stress in Shell		1.69	106.20



Intermediate Results: Saddle Reaction Q due to Road Transport Load:

Transverse Force due to Wind during Transport [Ftw]:
 = Transverse Area • 0.00256 • wv² • wlsf
 = 326 • 0.00256 • 207² • 1
 = 35853.9 lbf
 = 16263.309 Kgf

Longitudinal Force due to Wind during Transport [Flw]:
 = $\pi \cdot D^2/4 \cdot 0.00256 \cdot wv^2 \cdot wlsf \cdot \text{Wind Dia. Mult}$
 = $\pi \cdot 7.45^2/4 \cdot 0.00256 \cdot 207^2 \cdot 1 \cdot 1.2$
 = 5764.8 lbf
 = 2614.9 Kgf

Transverse Force due to Lateral Acceleration [Ft]:
 = gz • Shipping Weight • gw + Ftw
 = 0.1 • 61939 • 1 + 16263
 = 22457.2 Kgf

Longitudinal Force due to Axial Acceleration [Fl]:
 = gx • Shipping Weight • gw + Flw
 = 0.1 • 61939 • 1 + 2615
 = 8808.8 Kgf

Saddle Reaction Force due to Transverse Load [Fwt]:
 = Ft/Num of Saddles • B / E
 = 22457/2 • 1760/2070
 = 9546.6 Kgf

Saddle Reaction Force due to Longitudinal Load [Fwl]:
 = Fl • B / Ls
 = 8809 • 1760/6380
 = 2430.0 Kgf

Saddle Reaction Force due to Vertical Transportation Acceleration [Frv]:
 = Saddle Load • gy
 = 30514 • 0.05
 = 1525.676 Kgf

Load Combination Results due to Transport Loading [Q]:
 = Saddle Load + Frv + max(Fwl, Fwt)
 = 30514 + 1526 + max(2430, 9547)
 = 41585.8 Kgf

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	42550.03	Kgf
Transverse Shear Load	11228.62	Kgf
Longitudinal Shear Load	8808.85	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:
 = Baseplate Length
 = 2070.100 mm

The Computed K values from Table 4.15.1:

K1 = 0.1974	K2 = 0.6609	K3 = 0.3428	K4 = 0.2477
K5 = 0.6419	K6 = 0.0229	K7 = 0.0159	K8 = 0.2738
K9 = 0.1786	K10 = 0.0228	K1* = 0.3348	K6p = 0.0174
K7p = 0.0121			

The suffix 'p' denotes the values for a wear plate if it exists.



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Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))/(1 + (4hm)/3L)]$$

$$= -41586 \cdot 900 [1 - (1 - 900/8200 + (1130^2 - 0^2)/(2 \cdot 900 \cdot 8200))/(1 + (4 \cdot 0)/(3 \cdot 8200))]$$

$$= -867.2 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4 [(1 + 2(Rm^2 - hm^2)/(L^2))/(1 + (4hm^2)/(3L)) - 4a/L]$$

$$= 41586 \cdot 8200/4 [(1 + 2(1130^2 - 0^2)/(8200^2))/(1 + (4 \cdot 0)/(3 \cdot 8200)) - 4 \cdot 900/8200]$$

$$= 51065.4 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm/(2t) - M2/(\pi \cdot Rm^2 t)$$

$$= 0 \cdot 1130/(2 \cdot 11) - 51065/(\pi \cdot 1130^2 \cdot 11)$$

$$= -11.34 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm/(2t) + M2/(\pi \cdot Rm^2 \cdot t)$$

$$= 0 \cdot 1130/(2 \cdot 11) + 51065/(\pi \cdot 1130^2 \cdot 11)$$

$$= 11.34 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot Rm/(2t) - M1/(K1 \cdot \pi \cdot Rm^2 t)$$

$$= 0 \cdot 1130/(2 \cdot 11) - 867/(0.2 \cdot \pi \cdot 1130^2 \cdot 11)$$

$$= 0.98 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot Rm/(2t) + M1/(K1 \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 0 \cdot 1130/(2 \cdot 11) + 867/(0.33 \cdot \pi \cdot 1130^2 \cdot 11)$$

$$= -0.58 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a)/(L + (4 \cdot hm/3))$$

$$= 41586 (8200 - 2 \cdot 900)/(8200 + (4 \cdot 0/3))$$

$$= 32457.2 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (\pi \cdot Rm \cdot t)$$

$$= 0.66 \cdot 32457 / (\pi \cdot 1130 \cdot 11)$$

$$= 16.92 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(Rm \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1130 \cdot 11)}$$

$$= 86.981 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a)$$

$$= \min(325 + 1.56 \cdot \sqrt{(1130 \cdot 11)}, 2 \cdot 900)$$

$$= 498.96 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= 1.0000$$
 Materials are the same, transport case

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.64 \cdot 41586 \cdot 0.1 / (498(11 + 1 \cdot 20))$$

$$= -1.69 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, L < 8Rm (4.15.29) [σ_7, r_{*}]:

$$= -Q/(4(t + \eta \cdot tr)b_1) - 12 \cdot K7 \cdot Q \cdot Rm / (L(t + \eta \cdot tr)^2)$$

$$= -41586/(4(11 + 1 \cdot 20)498) -$$



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$$12 \cdot 0.016 \cdot 41586 \cdot 1130 / (8200(11+1 \cdot 20)^2) \\ = -17.73 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2070.1001	mm
Baseplate Thickness	Bpthk	22.0000	mm
Baseplate Width	Bpwid	350.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	14.0000	mm
Web Thickness	Webtk	14.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	730.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3)/12 \\ = ((1.56 \cdot \sqrt{(1125 \cdot 11)} + 560)11^3)/12 \\ = 8.136 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2 \\ = 8.14 + 80.7(219 - 5.5)^2 \\ = 36959.289 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	733.5	11.0	5.5	80.7	44379.1	0.370E+05
Wearplate	560.0	20.0	21.0	112.0	235200.0	0.442E+05
Web	14.0	582.0	322.0	81.5	2623656.2	0.316E+05
BasePlate	350.0	22.0	624.0	77.0	4804800.0	0.126E+06
Totals	...	...	...	351.2	7708035.5	0.239E+06

Distance to Centroid [C1]:

$$= AY / A \\ = 7708035/351 \\ = 219.496 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2 \\ = 180 - 167/2 \\ = 96.5$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta)\cos(\beta)) \\ = (1 + \cos(96.5) - 0.5 \cdot \sin(96.5)^2) / (\pi - 1.68 + \sin(96.5)\cos(96.5)) \\ = 0.2924$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q \\ = 0.29 \cdot 41586 \\ = 12158.5332 \text{ Kgf}$$

Tension Stress, St = (Fh/As) = 4.4083 N./mm²
 Allowed Stress, Sa = 0.6 * Yield Str = 155.9972 N./mm²

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians}) \\ = 1760 - 1125 \cdot \sin(167/2) / 1.46 \\ = 993.013 \text{ mm}$$



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Bending Moment, M = Fh * d = 12073.8193 Kg-m.

Bending Stress, Sb = (M * C1 / I) = 10.8875 N./mm²

Allowed Stress, Sa = 2/3 * Yield Str = 173.3302 N./mm²

Minimum Thickness of Baseplate per Moss:

= $\sqrt{\frac{3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}}{4 \cdot \text{BasePlateLength} \cdot \text{AllStress}}}$
 = $\sqrt{\frac{3(41586 + 964) \cdot 350}{4 \cdot 2070 \cdot 173}}$
 = 17.472 mm

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:
 = 2 * cos(90 - Saddle Angle/2) (Inside Radius + Shell Thk + Wear Plate Thk)
 = 2 * cos(90 - 167/2) (1125 + 11 + 20)
 = 2297.138 mm

Distance between Ribs [e]:
 = Web Length / (Nribs - 1)
 = 2297 / (5 - 1)
 = 574.285 mm

Baseplate Pressure Area [Ap]:
 = e * Bpwid / 2
 = 574 * 350/2
 = 1004.998 cm²

Bearing Pressure [Bp]:
 = (Q + SaddleWt) / (BasePlateLength * BasePlateWidth)
 = 42550 / (2070 * 350)
 = 5.873 Kg/cm²

Axial Load [P]:
 = Ap * Bp
 = 1005 * 5.87
 = 5902.088 Kg

Area of the Rib and Web [Ar]:
 = Rib Area + Web Area
 = 43.5 + 40.2
 = 83.740 cm²

Compressive Stress [Sc]:
 = P/Ar
 = 5902/83.7
 = 6.912 N./mm²

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	14.0	325.0	...	45.5	...	0.400E+04

Rib dimension [D]:
 = Saddle Width - Web Thickness
 = 325 - 14
 = 311.000 mm

Distance to Centroid from Datum [ytot]:
 = AY / A
 = 0/83.7
 = 0.000 mm



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Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 325 / 2$$

$$= 162.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{4005 / 83.7}$$

$$= 69.156 \text{ mm}$$

Length of Outer Rib, step 1, [Lv]:

$$= \text{Saddle Height} - \cos(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1760 - \cos(167/2) (1125 + 11 + 20) - 22$$

$$= 1607.137 \text{ mm}$$

Length of Outer Rib, step 2, [Lw]:

$$= 2 \cdot \sin(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk})$$

$$= 2 \cdot \sin(167/2) (1125 + 11 + 20)$$

$$= 2297.138 \text{ mm}$$

Length of Outer Rib, step 3, [Bd]:

$$= \text{Baseplate length} - \text{clearance}$$

$$= 2070 - 50.8$$

$$= 2019.300 \text{ mm}$$

Length of Outer Rib, step 4, [Dd]:

$$= (Lw - Bd) / 2$$

$$= (2297 - 2019) / 2$$

$$= 138.919 \text{ mm}$$

Length of Outer Rib [L]:

$$= \sqrt{Lv^2 + Dd^2}$$

$$= \sqrt{1607^2 + 139^2}$$

$$= 1613.130 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 199943008 / 260)}$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 1613 / 69.2$$

$$= 23.326$$

Bending Moment [Rm]:

$$= FI / (2 \cdot Bplen) \cdot e \cdot L / 2$$

$$= 8809 / (2 \cdot 2070) \cdot 574 \cdot 1613 / 2$$

$$= 985.537 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < Cc$ ($23.3 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (23.3)^2 / (2 \cdot 123^2)) 260 /$$

$$(5/3 + 3 \cdot (23.3) / (8 \cdot 123) - (23.3)^3 / (8 \cdot 123^3))$$

$$= 147 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc / Sca + (Rm \cdot C1 / I) / Sba$$

$$= 6.91 / 147 + (986 \cdot 162 / 40049480) / 173$$

$$= 0.273$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

B	D	Y	A	AY	Io
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Rib	14.0	311.0	0.0	43.5	0.0	0.400E+04
Web	574.3	14.0	0.0	80.4	0.0	13.1
Totals	...	...	...	123.9	...	0.402E+04

Distance to Centroid from Datum [ytot]:
 = AY / A
 = $0/124$
 = 0.000 mm

Distance to Centroid [C1]:
 = Saddle Width / 2
 = $325/2$
 = 162.500 mm

Length of Inner Rib [L]:
 = Saddle Height - Outside Radius - Bpthk
 = $1760 - \cos(1156/2)(22 + 0 + 0) - 0$
 = 582.000 mm

Radius of Gyration [r]:
 = $\sqrt{\text{Total Inertia} / \text{Total Area}}$
 = $\sqrt{4018/124}$
 = 56.936 mm

Slenderness ratio [KL/r]:
 = KL/r
 = $1 \cdot 582/56.9$
 = 10.222

Unit Force [Force,u]:
 = $FI / (2 \cdot \text{Baseplate Length})$
 = $8809 / (2 \cdot 2070)$
 = 2.128 Kg/mm

Moment at base of inner Rib [Mbase,c]:
 = Unit Force $\cdot e \cdot L$
 = $2.13 \cdot 574 \cdot 582$
 = 711.142 Kg-m.

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:
 = Bending Moment / Section Modulus
 = $711/247247$
 = 28.206 N./mm²

Compressive Allowable, $KL/r < C_c$ ($10.2 < 123$) per AISC E2-1 [Sca]:
 = $(1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$
 = $(1 - (10.2)^2 / (2 \cdot 123^2)) 260 /$
 = $(5/3 + 3 \cdot (10.2) / (8 \cdot 123) - (10.2^3) / (8 \cdot 123^3))$
 = 152.6 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)
 = $S_c / S_{ca} + (M_{base,c} \cdot C1/I) / S_{ba}$
 = $9.34/153 + (711 \cdot 162/4018) / 173$
 = 0.224

Review notes about nozzle loadings and supports in the Warnings report.

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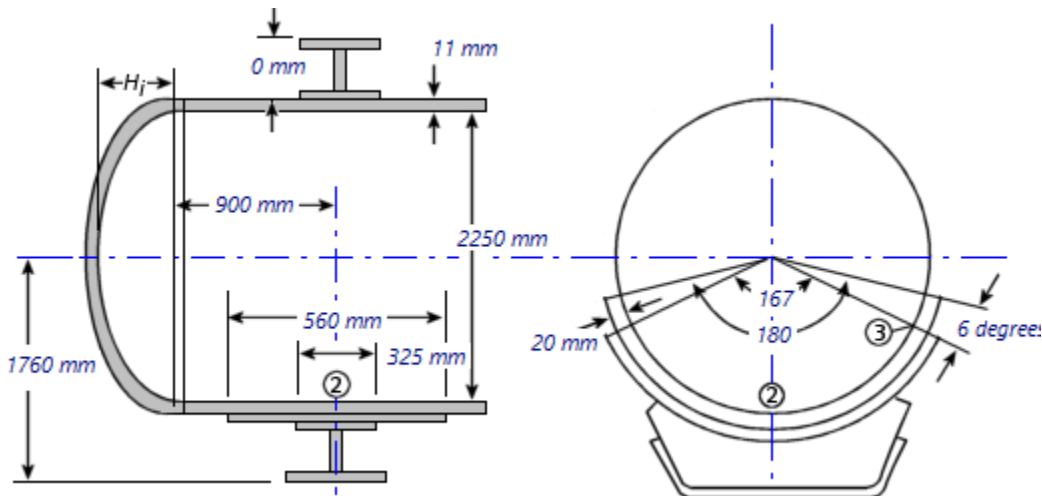
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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Operating Case

Input and Calculated Values:

Vessel Mean Radius	Rm	1130.90	mm
Shell Thickness used in this Case	t	11.000	mm
Stiffened Vessel Length per 4.15.6	L	8200.00	mm
Distance from Saddle to Vessel tangent	a1 or a	900.00	mm
Saddle Width	b1 or b	325.00	mm
Saddle Bearing Angle	delta or theta	167.00	degrees
Wear Plate Width	b2 or b1	560.00	mm
Wear Plate Bearing Angle	delta2 or theta1	180.00	degrees
Wear Plate Thickness	e2 or tr	20.0	mm
Wear Plate Allowable Stress	fw or Sr	106.20	N./mm ²
Shell Allowable Stress used in Calculation		106.20	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1760.00	mm
Coefficient of Friction	mu	0.30	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case		59846.61	Kgf
Pressure used in Analysis	P	391.027	KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	4.09	106.20
Long. Stress at Bottom of Midspan	39.26	106.20
Long. Stress at Top of Saddles	23.19	106.20
Long. Stress at Bottom of Saddles	20.79	106.20
Tangential Shear in Shell	26.25	84.96
Circ. Stress at Horn of Saddle	26.75	132.75
Circ. Compressive Stress in Shell	2.53	106.20



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Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (1250/2 + 0) \cdot 1760/2070$$

$$= 1594.7 \text{ Kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(502, 0) \cdot 1760/6380$$

$$= 138.4 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(\text{Fl,eq}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(5963, 0) \cdot 1760/6380$$

$$= 1645.1 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (5963/2 + 0) \cdot 1760/2070$$

$$= 7605.1 \text{ Kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 52242 + \max(138, 1595, 1645, 7605)$$

$$= 59846.6 \text{ Kgf}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766 (0 + \pi/4 (2.27 \cdot 1.2)^2)$$

$$= 4919.705 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	60810.88	Kgf
Transverse Shear Load	2981.68	Kgf
Longitudinal Shear Load	5963.35	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \text{Baseplate Length}$$

$$= 2070.100 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1974	K2 = 0.6609	K3 = 0.3428	K4 = 0.2477
K5 = 0.6419	K6 = 0.0229	K7 = 0.0159	K8 = 0.2738
K9 = 0.1786	K10 = 0.0228	K1* = 0.3348	K6p = 0.0174
K7p = 0.0121			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$$

$$= -59847 \cdot 900 [1 - (1 - 900/8200 + (1131^2 - 0^2)/(2 \cdot 900 \cdot 8200))/(1 + (4 \cdot 0)/(3 \cdot 8200))]$$

$$= -1244.6 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4 ((1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L)$$

$$= 59847 \cdot 8200/4 ((1 + 2(1131^2 - 0^2)/(8200^2))/(1 + (4 \cdot 0)/(3 \cdot 8200)) - 4 \cdot 900/8200)$$



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= 73492.1 Kg-m.

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 t)$$

$$= 391 \cdot 1131 / (2 \cdot 10.2) - 73492 / (\pi \cdot 1131^2 \cdot 10.2)$$

$$= 4.09 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t)$$

$$= 391 \cdot 1131 / (2 \cdot 10.2) + 73492 / (\pi \cdot 1131^2 \cdot 10.2)$$

$$= 39.26 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 t)$$

$$= 391 \cdot 1131 / (2 \cdot 10.2) - 1245 / (0.2 \cdot \pi \cdot 1131^2 \cdot 10.2)$$

$$= 23.19 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot Rm / (2t) + M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 391 \cdot 1131 / (2 \cdot 10.2) + 1245 / (0.33 \cdot \pi \cdot 1131^2 \cdot 10.2)$$

$$= 20.79 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot hm/3))$$

$$= 59847 (8200 - 2 \cdot 900) / (8200 + (4 \cdot 0/3))$$

$$= 46709.6 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (Rm \cdot t)$$

$$= 0.66 \cdot 46710 / (1131 \cdot 10.2)$$

$$= 26.25 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(Rm \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1131 \cdot 10.2)}$$

$$= 83.774 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a)$$

$$= \min(325 + 1.56 \cdot \sqrt{(1131 \cdot 10.2)}, 2 \cdot 900)$$

$$= 492.55 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [eta]:

$$= \min(Sr/S, 1)$$

$$= \min(106/106, 1)$$

$$= 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.64 \cdot 59847 \cdot 0.1 / (493(10.2 + 1 \cdot 20))$$

$$= -2.53 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8Rm$ (4.15.29) [σ_7, r_{*}]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 12 \cdot K7 \cdot Q \cdot Rm / (L(t + \eta \cdot tr)^2)$$

$$= -59847 / (4(10.2 + 1 \cdot 20)493) -$$

$$12 \cdot 0.016 \cdot 59847 \cdot 1131 / (8200(10.2 + 1 \cdot 20)^2)$$

$$= -26.75 \text{ N./mm}^2$$

Distance between Saddle Supports [Ls]:

$$= 6380.0 \text{ mm}$$

Free Un-Restrained Thermal Expansion between the Saddles [Exp]:

$$= \alpha \cdot Ls(\text{Design Temperature} - \text{Ambient Temperature})$$

$$= 0.16436E-04 \cdot 6380(130 - 21.1)$$

$$= 11.418 \text{ mm}$$



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Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2070.1001	mm
Baseplate Thickness	Bpthk	22.0000	mm
Baseplate Width	Bpwid	350.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	14.0000	mm
Web Thickness	Webtk	14.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	262.0	N./mm ²
Height of Web at Center	Hw,c	730.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t) + \text{WearPlateWidth})t^3)/12$$

$$= ((1.56 \cdot \sqrt{(1126 \cdot 10.2) + 560)10.2^3)/12$$

$$= 6.431 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{shell}}(C1 - Y)^2$$

$$= 6.43 + 74.2 (223 - 5.1)^2$$

$$= 35168.086 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² (2) cm ⁴
Shell	727.2	10.2	5.1	74.2	37827.3	0.352E+05
Wearplate	560.0	20.0	20.2	112.0	226240.0	0.460E+05
Web	14.0	582.0	321.2	81.5	2617137.0	0.309E+05
BasePlate	350.0	22.0	623.2	77.0	4798639.5	0.123E+06
Totals	...	...	...	344.7	7679843.5	0.236E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 7679843/345$$

$$= 222.829 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 167/2$$

$$= 96.5$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(96.5) - 0.5 \cdot \sin(96.5)^2) / (\pi - 1.68 + \sin(96.5) \cos(96.5))$$

$$= 0.2924$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.29 \cdot 59847$$

$$= 17497.5000 \text{ Kg}$$

Tension Stress, St = (Fh/As) = 6.3441 N./mm²
 Allowed Stress, Sa = 0.6 * Yield Str = 157.1972 N./mm²

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$$

$$= 1760 - 1126 \cdot \sin(167/2)/1.46$$

$$= 992.468 \text{ mm}$$

Bending Moment, M = Fh * d = 17366.0449 Kg-m.



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Bending Stress, $S_b = (M \cdot C_1 / I) = 16.1116 \text{ N./mm}^2$
 Allowed Stress, $S_a = 2/3 \cdot \text{Yield Str} = 174.6635 \text{ N./mm}^2$

Minimum Thickness of Baseplate per Moss:

$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress}))}$
 $= \sqrt{(3(59847 + 964) \cdot 350) / (4 \cdot 2070 \cdot 175)}$
 $= 20.808 \text{ mm}$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:
 $= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$
 $= 2 \cdot \cos(90 - 167/2) (1125 + 11 + 20)$
 $= 2297.138 \text{ mm}$

Distance between Ribs [e]:
 $= \text{Web Length} / (\text{Nr ribs} - 1)$
 $= 2297 / (5 - 1)$
 $= 574.285 \text{ mm}$

Baseplate Pressure Area [Ap]:
 $= e \cdot \text{Bpwid} / 2$
 $= 574 \cdot 350 / 2$
 $= 1004.998 \text{ cm}^2$

Bearing Pressure [Bp]:
 $= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$
 $= 60811 / (2070 \cdot 350)$
 $= 8.393 \text{ Kg/cm}^2$

Axial Load [P]:
 $= A_p \cdot B_p$
 $= 1005 \cdot 8.39$
 $= 8435.038 \text{ Kg}$

Area of the Rib and Web [Ar]:
 $= \text{Rib Area} + \text{Web Area}$
 $= 43.5 + 40.2$
 $= 83.740 \text{ cm}^2$

Compressive Stress [Sc]:
 $= P / A_r$
 $= 8435 / 83.7$
 $= 9.878 \text{ N./mm}^2$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	14.0	325.0	...	45.5	...	0.400E+04

Rib dimension [D]:
 $= \text{Saddle Width} - \text{Web Thickness}$
 $= 325 - 14$
 $= 311.000 \text{ mm}$

Distance to Centroid from Datum [ytot]:
 $= AY / A$
 $= 0 / 83.7$
 $= 0.000 \text{ mm}$

Distance to Centroid [C1]:
 $= \text{Saddle Width} / 2$



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$$= 325/2$$

$$= 162.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(4005/83.7)}$$

$$= 69.156 \text{ mm}$$

Length of Outer Rib, step 1, [Lv]:

$$= \text{Saddle Height} - \cos(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1760 - \cos(167/2) (1125 + 11 + 20) - 22$$

$$= 1607.137 \text{ mm}$$

Length of Outer Rib, step 2, [Lw]:

$$= 2 \cdot \sin(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk})$$

$$= 2 \cdot \sin(167/2) (1125 + 11 + 20)$$

$$= 2297.138 \text{ mm}$$

Length of Outer Rib, step 3, [Bd]:

$$= \text{Baseplate length} - \text{clearance}$$

$$= 2070 - 50.8$$

$$= 2019.300 \text{ mm}$$

Length of Outer Rib, step 4, [Dd]:

$$= (Lw - Bd) / 2$$

$$= (2297 - 2019) / 2$$

$$= 138.919 \text{ mm}$$

Length of Outer Rib [L]:

$$= \sqrt{Lv^2 + Dd^2}$$

$$= \sqrt{1607^2 + 139^2}$$

$$= 1613.130 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 199943008 / 262)}$$

$$= 122.739$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1613 / 69.2$$

$$= 23.326$$

Bending Moment [Rm]:

$$= FI / (2 \cdot Bplen) \cdot e \cdot L / 2$$

$$= 5963 / (2 \cdot 2070) \cdot 574 \cdot 1613 / 2$$

$$= 667.182 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < Cc$ ($23.3 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (23.3)^2 / (2 \cdot 123^2)) 262 / (5/3 + 3 \cdot (23.3) / (8 \cdot 123) - (23.3^3) / (8 \cdot 123^3))$$

$$= 148.1 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 9.88 / 148 + (667 \cdot 162 / 40049480) / 175$$

$$= 0.219$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	14.0	311.0	0.0	43.5	0.0	0.400E+04



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Web	574.3	14.0	0.0	80.4	0.0	13.1
Totals	...	...	...	123.9	...	0.402E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/124$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 325/2$$

$$= 162.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 1760 - \cos(1156/2)(22 + 0 + 0) - 0$$

$$= 582.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{4018/124}$$

$$= 56.936 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 582/56.9$$

$$= 10.222$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 5963 / (2 \cdot 2070)$$

$$= 1.440 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 1.44 \cdot 574 \cdot 582$$

$$= 481.424 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 481/247247$$

$$= 19.095 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($10.2 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (10.2)^2 / (2 \cdot 123^2)) 262 / (5/3 + 3 \cdot (10.2) / (8 \cdot 123) - (10.2)^3 / (8 \cdot 123^3))$$

$$= 153.8 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (Mbase,c \cdot C1/I) / Sba$$

$$= 13.4/154 + (481 \cdot 162/4018) / 175$$

$$= 0.196$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	8
Total Number of Bolts in Tension/Baseplate	Nbt	4
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-307 A
Bolt Allowable Stress	Stba	138.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	76.2 mm
User Defined Area in a Single Bolt	UserBoltArea	4.0400 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²



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Area Available in a Single Bolt	BltArea	3.0428 cm ²
Saddle Load QO (Weight)	QO	53205.8 Kgf
Saddle Load QL (Wind/Seismic contribution)	QL	1645.1 Kgf
Maximum Transverse Force	Ft	2981.7 Kgf
Maximum Longitudinal Force	Fl	5963.4 Kgf
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τb,l]:

$$= Fl / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 5963 / (3.04 \cdot 8)$$

$$= 24.0 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [τb,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 2982 / (3.04 \cdot 8)$$

$$= 12.0 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} QO > QL \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= Fl / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 5963 / (82.8 \cdot 8)$$

$$= 0.8829 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 1760 \cdot 2982 + 0$$

$$= 5247.85 \text{ Kg-m.}$$

Eccentricity (e):

$$= Rmom / QO$$

$$= 5248 / 53206$$

$$= 98.63 \text{ mm} < Bplen/6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$

$$= \max[0, 0.88, 0]$$

$$= 0.8829 \text{ cm}^2$$

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	1130.90	mm
Shell Thickness used in this Case	t	11.000	mm
Stiffened Vessel Length per 4.15.6	L	8200.00	mm
Distance from Saddle to Vessel tangent	a1 or a	900.00	mm
Saddle Width	b1 or b	325.00	mm
Saddle Bearing Angle	delta or theta	167.00	degrees
Wear Plate Width	b2 or b1	560.00	mm
Wear Plate Bearing Angle	delta2 or theta1	180.00	degrees
Wear Plate Thickness	e2 or tr	20.0	mm
Wear Plate Allowable Stress	fw or Sr	106.20	N./mm ²



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Shell Allowable Stress used in Calculation	106.20	N./mm ²
Head Allowable Stress used in Calculation	138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle	1.00	
Circumferential Efficiency at Mid-Span	1.00	
Distance from Saddle Base to Centerline	B	1760.00 mm
Coefficient of Friction	mu	0.00

Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case	59961.47	Kgf
Pressure used in Analysis	P	391.027 KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	4.06	106.20
Long. Stress at Bottom of Midspan	39.30	106.20
Long. Stress at Top of Saddles	23.19	106.20
Long. Stress at Bottom of Saddles	20.79	106.20
Tangential Shear in Shell	26.30	84.96
Circ. Stress at Horn of Saddle	26.80	132.75
Circ. Compressive Stress in Shell	2.54	106.20

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= Ftr(Ft/Num \text{ of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3 (1250/2 + 0) \cdot 1760/2070$
 $= 1594.7 \text{ Kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(Fl,wind; \text{Sum of X Forces}) \cdot B / Ls$
 $= \max(502, 0) \cdot 1760/6380$
 $= 138.4 \text{ Kgf}$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(Fl,eq; \text{Sum of X Forces}) \cdot B / Ls$
 $= \max(5963, 0) \cdot 1760/6380$
 $= 1645.1 \text{ Kgf}$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= Ftr(Ft/Num \text{ of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3 (5963/2 + 0) \cdot 1760/2070$
 $= 7605.1 \text{ Kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(Fwl, Fwt, Fsl, Fst)$
 $= 52356 + \max(138, 1595, 1645, 7605)$
 $= 59961.5 \text{ Kgf}$

Longitudinal Wind Force [Fl]:
 $= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(OD \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 766 (0 + \pi/4(2.27 \cdot 1.2)^2)$
 $= 4919.705 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	60925.73	Kgf
Transverse Shear Load	2981.68	Kgf
Longitudinal Shear Load	5963.35	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1



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Saddle Dimension [E]:

= Baseplate Length
 = 2070.100 mm

The Computed K values from Table 4.15.1:

K1 = 0.1974 K2 = 0.6609 K3 = 0.3428 K4 = 0.2477
 K5 = 0.6419 K6 = 0.0229 K7 = 0.0159 K8 = 0.2738
 K9 = 0.1786 K10 = 0.0228 K1* = 0.3348 K6p = 0.0174
 K7p = 0.0121

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

= $-Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))/(1 + (4hm)/3L)]$
 = $-59961 \cdot 900 [1 - (1 - 900/8200 + (1131^2 - 0^2)/(2 \cdot 900 \cdot 8200))/(1 + (4 \cdot 0)/(3 \cdot 8200))]$
 = -1247.0 Kg-m.

Moment per Equation 4.15.2 [M2]:

= $Q \cdot L/4 [(1 + 2(Rm^2 - hm^2)/(L^2))/(1 + (4hm^2)/(3L)) - 4a/L]$
 = $59961 \cdot 8200/4 [(1 + 2(1131^2 - 0^2)/(8200^2))/(1 + (4 \cdot 0)/(3 \cdot 8200)) - 4 \cdot 900/8200]$
 = 73633.2 Kg-m.

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

= $P \cdot Rm/(2t) - M2/(\pi \cdot Rm^2 t)$
 = $391 \cdot 1131/(2 \cdot 10.2) - 73633/(\pi \cdot 1131^2 \cdot 10.2)$
 = 4.06 N./mm²

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

= $P \cdot Rm/(2t) + M2/(\pi \cdot Rm^2 \cdot t)$
 = $391 \cdot 1131/(2 \cdot 10.2) + 73633/(\pi \cdot 1131^2 \cdot 10.2)$
 = 39.30 N./mm²

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

= $P \cdot Rm/(2t) - M1/(K1 \cdot \pi \cdot Rm^2 t)$
 = $391 \cdot 1131/(2 \cdot 10.2) - 1247/(0.2 \cdot \pi \cdot 1131^2 \cdot 10.2)$
 = 23.19 N./mm²

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

= $P \cdot Rm/(2t) + M1/(K1^* \cdot \pi \cdot Rm^2 \cdot t)$
 = $391 \cdot 1131/(2 \cdot 10.2) + 1247/(0.33 \cdot \pi \cdot 1131^2 \cdot 10.2)$
 = 20.79 N./mm²

Maximum Shear Force in the Saddle (4.15.3) [T]:

= $Q(L - 2a)/(L + (4 \cdot hm/3))$
 = $59961 (8200 - 2 \cdot 900)/(8200 + (4 \cdot 0/3))$
 = 46799.2 Kgf

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

= $K2 \cdot T / (\pi \cdot Rm \cdot t)$
 = $0.66 \cdot 46799/(1131 \cdot 10.2)$
 = 26.30 N./mm²

Decay Length (4.15.20) [x1,x2]:

= $0.78 \cdot \sqrt{(Rm \cdot t)}$
 = $0.78 \cdot \sqrt{(1131 \cdot 10.2)}$
 = 83.774 mm

Effective reinforcing plate width (4.15.24) [B1]:

= $\min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a)$
 = $\min(325 + 1.56 \cdot \sqrt{(1131 \cdot 10.2)}, 2 \cdot 900)$
 = 492.55 mm



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Wear Plate/Shell Stress ratio (4.15.26) [eta]:

$$= \min(Sr/S, 1)$$

$$= \min(106/106, 1)$$

$$= 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.64 \cdot 59961 \cdot 0.1 / (493 (10.2 + 1 \cdot 20))$$

$$= -2.54 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, L<8Rm (4.15.29) [σ_7, r^*]:

$$= -Q / (4(t + \eta \cdot tr) b1) - 12 \cdot K7 \cdot Q \cdot Rm / (L(t + \eta \cdot tr)^2)$$

$$= -59961 / (4(10.2 + 1 \cdot 20) 493) -$$

$$12 \cdot 0.016 \cdot 59961 \cdot 1131 / (8200(10.2 + 1 \cdot 20)^2)$$

$$= -26.80 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2070.1001	mm
Baseplate Thickness	Bpthk	22.0000	mm
Baseplate Width	Bpwid	350.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	14.0000	mm
Web Thickness	Webtk	14.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	262.0	N./mm ²
Height of Web at Center	Hw,c	730.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{ (R \cdot t) + \text{WearPlateWidth} }) t^3) / 12$$

$$= ((1.56 \cdot \sqrt{ (1126 \cdot 10.2) + 560 }) 10.2^3) / 12$$

$$= 6.431 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A\text{Shell}(C1 - Y)^2$$

$$= 6.43 + 74.2 (223 - 5.1)^2$$

$$= 35168.086 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	727.2	10.2	5.1	74.2	37827.3	0.352E+05
Wearplate	560.0	20.0	20.2	112.0	226240.0	0.460E+05
Web	14.0	582.0	321.2	81.5	2617137.0	0.309E+05
BasePlate	350.0	22.0	623.2	77.0	4798639.5	0.123E+06
Totals	...	...	...	344.7	7679843.5	0.236E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 7679843/345$$

$$= 222.829 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 167/2$$

$$= 96.5$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(96.5) - 0.5 \cdot \sin(96.5)^2) / (\pi - 1.68 + \sin(96.5) \cos(96.5))$$



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$$= 0.2924$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.29 \cdot 59961$$

$$= 17531.0801 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh/As) = 6.3563 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 157.1972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1760 - 1126 \cdot \sin(167/2) / 1.46$$

$$= 992.468 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 17399.3711 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 16.1425 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 174.6635 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{3(59961 + 964) \cdot 350 / (4 \cdot 2070 \cdot 175)}$$

$$= 20.827 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 167/2) (1125 + 11 + 20)$$

$$= 2297.138 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (N_{\text{ribs}} - 1)$$

$$= 2297 / (5 - 1)$$

$$= 574.285 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot B_{\text{pwid}} / 2$$

$$= 574 \cdot 350 / 2$$

$$= 1004.998 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 60926 / (2070 \cdot 350)$$

$$= 8.409 \text{ Kgf/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 1005 \cdot 8.41$$

$$= 8450.970 \text{ Kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 43.5 + 40.2$$

$$= 83.740 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / Ar$$

$$= 8451 / 83.7$$

$$= 9.897 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction



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	B	D	Y	A	AY	Io
Rib+Web	14.0	325.0	...	45.5	...	0.400E+04

Rib dimension [D]:
 = Saddle Width - Web Thickness
 = 325 - 14
 = 311.000 mm

Distance to Centroid from Datum [ytot]:
 = AY / A
 = 0/83.7
 = 0.000 mm

Distance to Centroid [C1]:
 = Saddle Width / 2
 = 325/2
 = 162.500 mm

Radius of Gyration [r]:
 $= \sqrt{\text{Total Inertia} / \text{Total Area}}$
 $= \sqrt{4005/83.7}$
 = 69.156 mm

Length of Outer Rib, step 1, [Lv]:
 = Saddle Height - cos(saddle bearing angle / 2)(radius + shlthk + wpdthk) - bpthk
 = 1760 - cos(167/2)(1125 + 11 + 20) - 22
 = 1607.137 mm

Length of Outer Rib, step 2, [Lw]:
 = 2 • sin(saddle bearing angle / 2)(radius + shlthk + wpdthk)
 = 2 • sin(167/2)(1125 + 11 + 20)
 = 2297.138 mm

Length of Outer Rib, step 3, [Bd]:
 = Baseplate length - clearance
 = 2070 - 50.8
 = 2019.300 mm

Length of Outer Rib, step 4, [Dd]:
 = (Lw - Bd)/2
 = (2297 - 2019)/2
 = 138.919 mm

Length of Outer Rib [L]:
 $= \sqrt{Lv^2 + Dd^2}$
 $= \sqrt{1607^2 + 139^2}$
 = 1613.130 mm

Intermediate Term [Cc]:
 $= \sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$
 $= \sqrt{2 \cdot \pi^2 \cdot 199943008/262}$
 = 122.739

Slenderness ratio [KL/r]:
 = KL/r
 = 1 • 1613/69.2
 = 23.326

Bending Moment [Rm]:
 = FI / (2 • Bplen) • e • L / 2
 = 5963 / (2 • 2070) • 574 • 1613/2
 = 667.182 Kg-m.



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Compressive Allowable, $KL/r < C_c$ ($23.3 < 123$) per AISC E2-1 [Sca]:
 $= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$
 $= (1 - (23.3)^2 / (2 \cdot 123^2)) 262 / (5/3 + 3 \cdot (23.3) / (8 \cdot 123) - (23.3^3) / (8 \cdot 123^3))$
 $= 148.1 \text{ N./mm}^2$

AISC Unity Check of Outside Ribs (must be ≤ 1)
 $= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$
 $= 9.9 / 148 + (667 \cdot 162 / 40049480) / 175$
 $= 0.219$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	14.0	311.0	0.0	43.5	0.0	0.400E+04
Web	574.3	14.0	0.0	80.4	0.0	13.1
Totals	...	...	...	123.9	...	0.402E+04

Distance to Centroid from Datum [ytot]:
 $= AY / A$
 $= 0 / 124$
 $= 0.000 \text{ mm}$

Distance to Centroid [C1]:
 $= \text{Saddle Width} / 2$
 $= 325 / 2$
 $= 162.500 \text{ mm}$

Length of Inner Rib [L]:
 $= \text{Saddle Height} - \text{Outside Radius} - B_{pthk}$
 $= 1760 - \cos(1156 / 2) (22 + 0 + 0) - 0$
 $= 582.000 \text{ mm}$

Radius of Gyration [r]:
 $= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$
 $= \sqrt{ (4018 / 124) }$
 $= 56.936 \text{ mm}$

Slenderness ratio [KL/r]:
 $= KL / r$
 $= 1 \cdot 582 / 56.9$
 $= 10.222$

Unit Force [Force,u]:
 $= FI / (2 \cdot \text{Baseplate Length})$
 $= 5963 / (2 \cdot 2070)$
 $= 1.440 \text{ Kg/mm}$

Moment at base of inner Rib [Mbase,c]:
 $= \text{Unit Force} \cdot e \cdot L$
 $= 1.44 \cdot 574 \cdot 582$
 $= 481.424 \text{ Kg-m.}$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:
 $= \text{Bending Moment} / \text{Section Modulus}$
 $= 481 / 247247$
 $= 19.095 \text{ N./mm}^2$

Compressive Allowable, $KL/r < C_c$ ($10.2 < 123$) per AISC E2-1 [Sca]:
 $= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$
 $= (1 - (10.2)^2 / (2 \cdot 123^2)) 262 / (5/3 + 3 \cdot (10.2) / (8 \cdot 123) - (10.2^3) / (8 \cdot 123^3))$
 $= 153.8 \text{ N./mm}^2$



AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (Mbase,c \cdot C1/I)/Sba$$

$$= 13.4/154 + (481 \cdot 162/4018)/175$$

$$= 0.196$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	8
Total Number of Bolts in Tension/Baseplate	Nbt	4
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-307 A
Bolt Allowable Stress	Stba	138.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	76.2 mm
User Defined Area in a Single Bolt	UserBoltArea	4.0400 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	3.0428 cm ²
Saddle Load Q0 (Weight)	Q0	53320.7 Kgf
Saddle Load QL (Wind/Seismic contribution)	QL	1645.1 Kgf
Maximum Transverse Force	Ft	2981.7 Kgf
Maximum Longitudinal Force	Fl	5963.4 Kgf
Saddle Bolted to Steel Foundation		No

Friction force due to thermal growth, [Ff]:

$$= 15961.7 \text{ Kgf}$$

Shear Stress in a Single Bolt, Longitudinal Direction [rb,l]:

$$= \max(Ff, Fl) / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 15962 / (3.04 \cdot 8)$$

$$= 64.3 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [rb,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 2982 / (3.04 \cdot 8)$$

$$= 12.0 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (Q0 > QL --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= Fl / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 5963 / (82.8 \cdot 8)$$

$$= 0.8829 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 1760 \cdot 2982 + 0$$

$$= 5247.85 \text{ Kg-m.}$$

Eccentricity (e):

$$= Rmom / Q0$$

$$= 5248/53321$$

$$= 98.42 \text{ mm} < Bplen/6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$

$$= \max[0, 0.88, 0]$$



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= 0.8829 cm²

Review notes about nozzle loadings and supports in the Warnings report.

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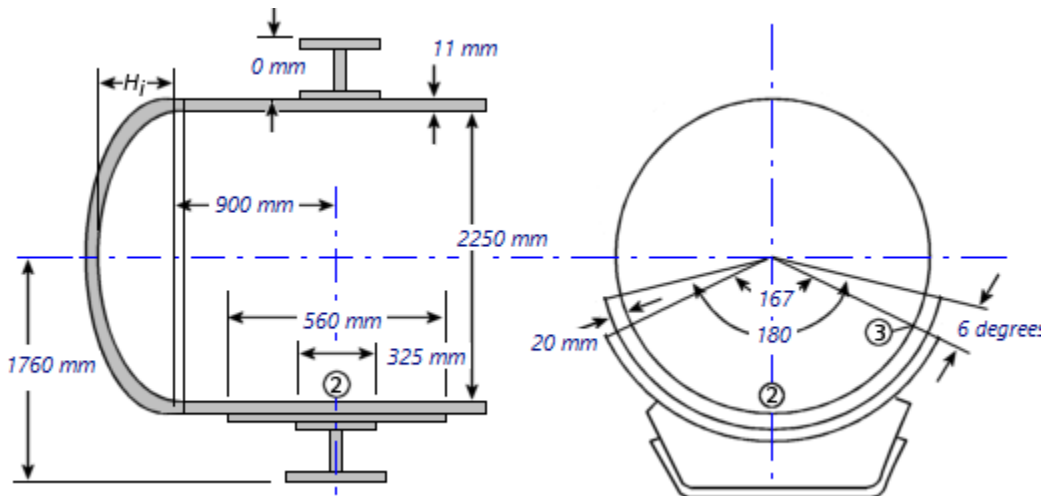
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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Test Case

Input and Calculated Values:

Vessel Mean Radius	Rm	1130.50	mm
Shell Thickness used in this Case	t	11.000	mm
Stiffened Vessel Length per 4.15.6	L	8200.00	mm
Distance from Saddle to Vessel tangent	a1 or a	900.00	mm
Saddle Width	b1 or b	325.00	mm
Saddle Bearing Angle	delta or theta	167.00	degrees
Wear Plate Width	b2 or b1	560.00	mm
Wear Plate Bearing Angle	delta2 or theta1	180.00	degrees
Wear Plate Thickness	e2 or tr	20.0	mm
Wear Plate Allowable Stress	fw or Sr	106.20	N./mm ²
Shell Allowable Stress used in Calculation		184.50	N./mm ²
Head Allowable Stress used in Calculation		184.50	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1760.00	mm
Coefficient of Friction	mu	0.30	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces	51899.01	Kgf
Pressure used in Analysis	P	505.081 KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	11.81	184.50
Long. Stress at Bottom of Midspan	40.11	184.50
Long. Stress at Top of Saddles	27.17	184.50
Long. Stress at Bottom of Saddles	25.24	184.50
Tangential Shear in Shell	21.11	109.26
Circ. Stress at Horn of Saddle	22.13	276.74
Circ. Compressive Stress in Shell	2.11	184.50



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Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$
 $= 3 (625/2 + 0) \cdot 1760/2070$
 $= 797.4 \text{ Kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / \text{Ls}$
 $= \max(251, 0) \cdot 1760/6380$
 $= 69.2 \text{ Kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$
 $= 51102 + \max(69.2, 797, 0, 0)$
 $= 51899.0 \text{ Kgf}$

Longitudinal Wind Force [FI]:
 $= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 766 (0 + \pi/4(2.27 \cdot 1.2)^2)$
 $= 4919.705 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	52863.27	Kgf
Transverse Shear Load	312.62	Kgf
Longitudinal Shear Load	250.84	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \text{Baseplate Length}$
 $= 2070.100 \text{ mm}$

The Computed K values from Table 4.15.1:

K1 = 0.1974	K2 = 0.6609	K3 = 0.3428	K4 = 0.2477
K5 = 0.6419	K6 = 0.0229	K7 = 0.0159	K8 = 0.2738
K9 = 0.1786	K10 = 0.0228	K1* = 0.3348	K6p = 0.0174
K7p = 0.0121			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$
 $= -51899 \cdot 900 [1 - (1 - 900/8200 + (1130^2 - 0^2)/(2 \cdot 900 \cdot 8200))/(1 + (4 \cdot 0)/(3 \cdot 8200))]$
 $= -1082.2 \text{ Kg-m.}$

Moment per Equation 4.15.2 [M2]:
 $= Q \cdot L/4((1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m)/3L) - 4a/L)$
 $= 51899 \cdot 8200/4((1 + 2(1130^2 - 0^2)/(8200^2))/(1 + (4 \cdot 0)/(3 \cdot 8200)) - 4 \cdot 900/8200)$
 $= 63729.5 \text{ Kg-m.}$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:
 $= P \cdot R_m/(2t) - M2/(\pi \cdot R_m^2 t)$
 $= 505 \cdot 1130/(2 \cdot 11) - 63730/(\pi \cdot 1130^2 \cdot 11)$
 $= 11.81 \text{ N./mm}^2$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:
 $= P \cdot R_m/(2t) + M2/(\pi \cdot R_m^2 \cdot t)$
 $= 505 \cdot 1130/(2 \cdot 11) + 63730/(\pi \cdot 1130^2 \cdot 11)$



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$$= 40.11 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 t)$$

$$= 505 \cdot 1130 / (2 \cdot 11) - 1082 / (0.2 \cdot \pi \cdot 1130^2 \cdot 11)$$

$$= 27.17 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot Rm / (2t) + M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 505 \cdot 1130 / (2 \cdot 11) + 1082 / (0.33 \cdot \pi \cdot 1130^2 \cdot 11)$$

$$= 25.24 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot hm/3))$$

$$= 51899 (8200 - 2 \cdot 900) / (8200 + (4 \cdot 0/3))$$

$$= 40506.5 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (Rm \cdot t)$$

$$= 0.66 \cdot 40507 / (1130 \cdot 11)$$

$$= 21.11 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(Rm \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1130 \cdot 11)}$$

$$= 86.981 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a)$$

$$= \min(325 + 1.56 \cdot \sqrt{(1130 \cdot 11)}, 2 \cdot 900)$$

$$= 498.96 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= 1.0000 \text{ Materials are the same, test case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.64 \cdot 51899 \cdot 0.1 / (499(11 + 1 \cdot 20))$$

$$= -2.11 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8Rm$ (4.15.29) [σ_7, r^*]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 12 \cdot K7 \cdot Q \cdot Rm / (L(t + \eta \cdot tr)^2)$$

$$= -51899 / (4(11 + 1 \cdot 20)499) -$$

$$12 \cdot 0.016 \cdot 51899 \cdot 1130 / (8200(11 + 1 \cdot 20)^2)$$

$$= -22.13 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2070.1001	mm
Baseplate Thickness	Bpthk	22.0000	mm
Baseplate Width	Bpwid	350.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	14.0000	mm
Web Thickness	Webtk	14.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw, c	730.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(1125 \cdot 11)} + 560)^3) / 12$$

$$= 8.136 \text{ cm}^4$$



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Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 8.14 + 80.7 (219 - 5.5)^2$$

$$= 36959.289 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	733.5	11.0	5.5	80.7	44379.1	0.370E+05
Wearplate	560.0	20.0	21.0	112.0	235200.0	0.442E+05
Web	14.0	582.0	322.0	81.5	2623656.2	0.316E+05
BasePlate	350.0	22.0	624.0	77.0	4804800.0	0.126E+06
Totals	...	...	...	351.2	7708035.5	0.239E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 7708035/351$$

$$= 219.496 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 167/2$$

$$= 96.5$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(96.5) - 0.5 \cdot \sin(96.5)^2) / (\pi - 1.68 + \sin(96.5) \cos(96.5))$$

$$= 0.2924$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.29 \cdot 51899$$

$$= 15173.8389 \text{ Kg}$$

$$\text{Tension Stress, } St = (Fh / As) = 5.5016 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1760 - 1125 \cdot \sin(167/2) / 1.46$$

$$= 993.013 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 15068.1172 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 13.5876 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(51899 + 964) \cdot 350) / (4 \cdot 2070 \cdot 173)}$$

$$= 19.475 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 167/2) (1125 + 11 + 20)$$

$$= 2297.138 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nrings} - 1)$$

$$= 2297 / (5 - 1)$$

$$= 574.285 \text{ mm}$$

Baseplate Pressure Area [Ap]:



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$$= e \cdot Bpwid / 2$$

$$= 574 \cdot 350 / 2$$

$$= 1004.998 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + SaddleWt) / (BasePlateLength \cdot BasePlateWidth)$$

$$= 52863 / (2070 \cdot 350)$$

$$= 7.296 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 1005 \cdot 7.3$$

$$= 7332.631 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= Rib \text{ Area} + Web \text{ Area}$$

$$= 43.5 + 40.2$$

$$= 83.740 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / Ar$$

$$= 7333 / 83.7$$

$$= 8.587 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	14.0	325.0	...	45.5	...	0.400E+04

Rib dimension [D]:

$$= Saddle \text{ Width} - Web \text{ Thickness}$$

$$= 325 - 14$$

$$= 311.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 83.7$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= Saddle \text{ Width} / 2$$

$$= 325 / 2$$

$$= 162.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (Total \text{ Inertia} / Total \text{ Area}) }$$

$$= \sqrt{ (4005 / 83.7) }$$

$$= 69.156 \text{ mm}$$

Length of Outer Rib, step 1, [Lv]:

$$= Saddle \text{ Height} - \cos(saddle \text{ bearing angle} / 2)(radius + shlthk + wpdthk) - bpthk$$

$$= 1760 - \cos(167 / 2)(1125 + 11 + 20) - 22$$

$$= 1607.137 \text{ mm}$$

Length of Outer Rib, step 2, [Lw]:

$$= 2 \cdot \sin(saddle \text{ bearing angle} / 2)(radius + shlthk + wpdthk)$$

$$= 2 \cdot \sin(167 / 2)(1125 + 11 + 20)$$

$$= 2297.138 \text{ mm}$$

Length of Outer Rib, step 3, [Bd]:

$$= Baseplate \text{ length} - clearance$$

$$= 2070 - 50.8$$



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= 2019.300 mm

Length of Outer Rib, step 4, [Dd]:

= (Lw - Bd)/2
 = (2297 - 2019)/2
 = 138.919 mm

Length of Outer Rib [L]:

= $\sqrt{(Lv^2 + Dd^2)}$
 = $\sqrt{(1607^2 + 139^2)}$
 = 1613.130 mm

Intermediate Term [Cc]:

= $\sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$
 = $\sqrt{(2 \cdot \pi^2 \cdot 199943008 / 260)}$
 = 123.211

Slenderness ratio [KL/r]:

= KL/r
 = 1 • 1613/69.2
 = 23.326

Bending Moment [Rm]:

= FI / (2 • Bplen) • e • L / 2
 = 251 / (2 • 2070) • 574 • 1613/2
 = 28.064 Kg-m.

Compressive Allowable, KL/r < Cc (23.3 < 123) per AISC E2-1 [Sca]:

= (1 - (KL/r)² / (2 • Cc²)) Fy / (5/3 + 3 • (KL/r) / (8 • Cc) - (KL/r)³ / (8 • Cc³))
 = (1 - (23.3)² / (2 • 123²)) 260 /
 (5/3 + 3 • (23.3) / (8 • 123) - (23.3³) / (8 • 123³))
 = 147 N./mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)

= Sc/Sca + (Rm • C1 / I) / Sba
 = 8.59/147 + (28.1 • 162/40049480) / 173
 = 0.065

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	14.0	311.0	0.0	43.5	0.0	0.400E+04
Web	574.3	14.0	0.0	80.4	0.0	13.1
Totals	...	...	...	123.9	...	0.402E+04

Distance to Centroid from Datum [ytot]:

= AY / A
 = 0/124
 = 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2
 = 325/2
 = 162.500 mm

Length of Inner Rib [L]:

= Saddle Height - Outside Radius - Bpthk
 = 1760 - cos(1156/2) (22 + 0 + 0) - 0
 = 582.000 mm

Radius of Gyration [r]:

= $\sqrt{(\text{Total Inertia} / \text{Total Area})}$
 = $\sqrt{(4018 / 124)}$



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= 56.936 mm

Slenderness ratio [KL/r]:

= KL/r
 = 1 • 582/56.9
 = 10.222

Unit Force [Force,u]:

= FI / (2 • Baseplate Length)
 = 251/(2 • 2070)
 = 0.061 Kg/mm

Moment at base of inner Rib [Mbase,c]:

= Unit Force • e • L
 = 0.061 • 574 • 582
 = 20.250 Kg-m.

Bending Stress due to Transverse Force and Weight Load [σB,base,c]:

= Bending Moment / Section Modulus
 = 20.3/247247
 = 0.803 N./mm²

Compressive Allowable, KL/r < Cc (10.2 < 123) per AISC E2-1 [Sca]:

= (1-(KL/r)²/(2•Cc²))Fy/(5/3+3•(KL/r)/(8•Cc)-(KL/r)³/(8•Cc³))
 = (1-(10.2)²/(2 • 123²))260/
 (5/3+3•(10.2)/(8• 123)-(10.2³)/(8•123³))
 = 152.6 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)

= Sc/Sca + (Mbase,c • C1/I)/Sba
 = 11.6/153 + (20.3 • 162/4018)/173
 = 0.081

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	8
Total Number of Bolts in Tension/Baseplate	Nbt	4
Bolt Material Specification		SA-307 A
Bolt Allowable Stress	Stba	48.30 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	76.2 mm
User Defined Area in a Single Bolt	UserBoltArea	4.0400 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	3.0428 cm ²
Saddle Load Q0 (Weight)	Q0	52065.9 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	69.2 Kg
Maximum Transverse Force	Ft	312.6 Kg
Maximum Longitudinal Force	F1	250.8 Kg
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τb,l]:

= FI / (Bolt Area • Number of Bolts)
 = 251/(3.04 • 8)
 = 1.0 N./mm². Must be less than 29.0 N./mm².

Shear Stress in a Single Bolt, Transverse Direction [τb,t]:

= Ft / (Bolt Area • Number of Bolts)
 = 313/(3.04 • 8)
 = 1.3 N./mm². Must be less than 29.0 N./mm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

= 0.0 (Q0 > QL --> No Uplift in Longitudinal direction)



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Bolt Area due to Shear Load [Bltarears]:
 $= F / (\text{BoltShearAllowable} \cdot N_{\text{bolts}})$
 $= 251 / (29 \cdot 8)$
 $= 0.1061 \text{ cm}^2$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
 $= B \cdot F_t + \text{Sum of X Moments}$
 $= 1760 \cdot 313 + 0$
 $= 550.22 \text{ Kg-m.}$

Eccentricity (e):
 $= R_{\text{mom}} / QO$
 $= 550 / 52066$
 $= 10.57 \text{ mm} < B_{\text{plen}} / 6 \rightarrow \text{No Uplift in Transverse direction}$

Bolt Area due to Transverse Load [Bltareart]:
 $= 0 \text{ (No Uplift)}$

Required Area of a Single Bolt [Bltarear]:
 $= \max[B_{\text{ltarearl}}, B_{\text{ltarears}}, B_{\text{ltareart}}]$
 $= \max[0, 0.11, 0]$
 $= 0.1061 \text{ cm}^2$

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	1130.50	mm
Shell Thickness used in this Case	t	11.000	mm
Stiffened Vessel Length per 4.15.6	L	8200.00	mm
Distance from Saddle to Vessel tangent	a1 or a	900.00	mm
Saddle Width	b1 or b	325.00	mm
Saddle Bearing Angle	delta or theta	167.00	degrees
Wear Plate Width	b2 or b1	560.00	mm
Wear Plate Bearing Angle	delta2 or theta1	180.00	degrees
Wear Plate Thickness	e2 or tr	20.0	mm
Wear Plate Allowable Stress	fw or Sr	106.20	N./mm ²
Shell Allowable Stress used in Calculation		184.50	N./mm ²
Head Allowable Stress used in Calculation		184.50	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1760.00	mm
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces 52014.51 Kgf
 Pressure used in Analysis P 505.081 KPa.

Horizontal Vessel Analysis Results:		Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan		11.77	184.50
Long. Stress at Bottom of Midspan		40.14	184.50
Long. Stress at Top of Saddles		27.18	184.50
Long. Stress at Bottom of Saddles		25.24	184.50
Tangential Shear in Shell		21.16	109.26



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Circ. Stress at Horn of Saddle	22.18	276.74
Circ. Compressive Stress in Shell	2.12	184.50

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= Ftr(Ft/Num\ of\ Saddles + Z\ Force\ Load) \cdot B / E$
 $= 3 (625/2 + 0) \cdot 1760/2070$
 $= 797.4\ Kg$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(Fl, wind; Sum\ of\ X\ Forces) \cdot B / Ls$
 $= \max(251, 0) \cdot 1760/6380$
 $= 69.2\ Kg$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= Saddle\ Load + \max(Fwl, Fwt, Fsl, Fst)$
 $= 51217 + \max(69.2, 797, 0, 0)$
 $= 52014.5\ Kg$

Longitudinal Wind Force [Fl]:
 $= WindScalar \cdot WindPress(Platform\ Area + (\pi/4(OD \cdot WindDiaMult)^2))$
 $= 1.1 \cdot 766 (0 + \pi/4(2.27 \cdot 1.2)^2)$
 $= 4919.705\ N$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	52978.78	Kgf
Transverse Shear Load	312.62	Kgf
Longitudinal Shear Load	250.84	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= Baseplate\ Length$
 $= 2070.100\ mm$

The Computed K values from Table 4.15.1:

K1 = 0.1974	K2 = 0.6609	K3 = 0.3428	K4 = 0.2477
K5 = 0.6419	K6 = 0.0229	K7 = 0.0159	K8 = 0.2738
K9 = 0.1786	K10 = 0.0228	K1* = 0.3348	K6p = 0.0174
K7p = 0.0121			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $Rm/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L)) / (1 + (4hm)/3L)]$
 $= -52015 \cdot 900 [1 - (1 - 900/8200 + (1130^2 - 0^2) / (2 \cdot 900 \cdot 8200)) / (1 + (4 \cdot 0) / (3 \cdot 8200))]$
 $= -1084.6\ Kg \cdot m$

Moment per Equation 4.15.2 [M2]:
 $= Q \cdot L / 4 ((1 + 2(Rm^2 - hm^2)/(L^2)) / (1 + (4hm^2)/(3L)) - 4a/L)$
 $= 52015 \cdot 8200 / 4 ((1 + 2(1130^2 - 0^2)/(8200^2)) / (1 + (4 \cdot 0) / (3 \cdot 8200)) - 4 \cdot 900/8200)$
 $= 63871.4\ Kg \cdot m$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:
 $= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 t)$
 $= 505 \cdot 1130 / (2 \cdot 11) - 63871 / (\pi \cdot 1130^2 \cdot 11)$
 $= 11.77\ N/mm^2$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:



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$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 505 \cdot 1130 / (2 \cdot 11) + 63871 / (\pi \cdot 1130^2 \cdot 11)$$

$$= 40.14 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 505 \cdot 1130 / (2 \cdot 11) - 1085 / (0.2 \cdot \pi \cdot 1130^2 \cdot 11)$$

$$= 27.18 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 505 \cdot 1130 / (2 \cdot 11) + 1085 / (0.33 \cdot \pi \cdot 1130^2 \cdot 11)$$

$$= 25.24 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 52015 (8200 - 2 \cdot 900) / (8200 + (4 \cdot 0 / 3))$$

$$= 40596.7 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 0.66 \cdot 40597 / (1130 \cdot 11)$$

$$= 21.16 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{R_m \cdot t}$$

$$= 0.78 \cdot \sqrt{1130 \cdot 11}$$

$$= 86.981 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(325 + 1.56 \cdot \sqrt{1130 \cdot 11}, 2 \cdot 900)$$

$$= 498.96 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= 1.0000 \text{ Materials are the same, test case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr))$$

$$= -0.64 \cdot 52015 \cdot 0.1 / (499(11 + 1 \cdot 20))$$

$$= -2.12 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.29) [σ_7, r^*]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 12 \cdot K_7 \cdot Q \cdot R_m / (L(t + \eta \cdot tr)^2)$$

$$= -52015 / (4(11 + 1 \cdot 20)499) -$$

$$12 \cdot 0.016 \cdot 52015 \cdot 1130 / (8200(11 + 1 \cdot 20)^2)$$

$$= -22.18 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2070.1001	mm
Baseplate Thickness	Bpthk	22.0000	mm
Baseplate Width	Bpwid	350.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	14.0000	mm
Web Thickness	Webtk	14.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	730.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{R \cdot t}) + \text{WearPlateWidth})^3 / 12$$



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$$= ((1.56 \cdot \sqrt{(1125 \cdot 11)} + 560) 11^3) / 12$$

$$= 8.136 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 8.14 + 80.7 (219 - 5.5)^2$$

$$= 36959.289 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	733.5	11.0	5.5	80.7	44379.1	0.370E+05
Wearplate	560.0	20.0	21.0	112.0	235200.0	0.442E+05
Web	14.0	582.0	322.0	81.5	2623656.2	0.316E+05
BasePlate	350.0	22.0	624.0	77.0	4804800.0	0.126E+06
Totals	...	...	...	351.2	7708035.5	0.239E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 7708035 / 351$$

$$= 219.496 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 167 / 2$$

$$= 96.5$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(96.5) - 0.5 \cdot \sin(96.5)^2) / (\pi - 1.68 + \sin(96.5) \cos(96.5))$$

$$= 0.2924$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.29 \cdot 52015$$

$$= 15207.6094 \text{ Kg}$$

Tension Stress, $St = (Fh / As) = 5.5138 \text{ N./mm}^2$
 Allowed Stress, $Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1760 - 1125 \cdot \sin(167/2) / 1.46$$

$$= 993.013 \text{ mm}$$

Bending Moment, $M = Fh \cdot d = 15101.6523 \text{ Kg-m.}$

Bending Stress, $Sb = (M \cdot C1 / I) = 13.6178 \text{ N./mm}^2$
 Allowed Stress, $Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(52015 + 964) 350) / (4 \cdot 2070 \cdot 173)}$$

$$= 19.496 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 167/2) (1125 + 11 + 20)$$

$$= 2297.138 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (N_{\text{ribs}} - 1)$$

$$= 2297 / (5 - 1)$$



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= 574.285 mm

Baseplate Pressure Area [Ap]:

= e • Bpwid / 2
 = 574 • 350/2
 = 1004.998 cm²

Bearing Pressure [Bp]:

= (Q + SaddleWt) / (BasePlateLength • BasePlateWidth)
 = 52979/(2070 • 350)
 = 7.312 Kg/cm²

Axial Load [P]:

= Ap • Bp
 = 1005 • 7.31
 = 7348.653 Kg

Area of the Rib and Web [Ar]:

= Rib Area + Web Area
 = 43.5 + 40.2
 = 83.740 cm²

Compressive Stress [Sc]:

= P/Ar
 = 7349/83.7
 = 8.606 N./mm²

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	14.0	325.0	...	45.5	...	0.400E+04

Rib dimension [D]:

= Saddle Width - Web Thickness
 = 325 - 14
 = 311.000 mm

Distance to Centroid from Datum [ytot]:

= AY / A
 = 0/83.7
 = 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2
 = 325/2
 = 162.500 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$
 = $\sqrt{4005/83.7}$
 = 69.156 mm

Length of Outer Rib, step 1, [Lv]:

= Saddle Height - cos(saddle bearing angle / 2)(radius + shlthk + wpdthk) - bpthk
 = 1760 - cos(167/2)(1125 + 11 + 20) - 22
 = 1607.137 mm

Length of Outer Rib, step 2, [Lw]:

= 2 • sin(saddle bearing angle / 2)(radius + shlthk + wpdthk)
 = 2 • sin(167/2)(1125 + 11 + 20)
 = 2297.138 mm



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Length of Outer Rib, step 3, [Bd]:
 = Baseplate length - clearance
 = 2070 - 50.8
 = 2019.300 mm

Length of Outer Rib, step 4, [Dd]:
 = (Lw - Bd)/2
 = (2297 - 2019)/2
 = 138.919 mm

Length of Outer Rib [L]:

$$= \sqrt{(Lv^2 + Dd^2)}$$

$$= \sqrt{(1607^2 + 139^2)}$$
 = 1613.130 mm

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 199943008 / 260)}$$
 = 123.211

Slenderness ratio [KL/r]:
 = KL/r
 = 1 • 1613/69.2
 = 23.326

Bending Moment [Rm]:

$$= Fl / (2 \cdot Bplen) \cdot e \cdot L / 2$$

$$= 251 / (2 \cdot 2070) \cdot 574 \cdot 1613 / 2$$
 = 28.064 Kg-m.

Compressive Allowable, KL/r < Cc (23.3 < 123) per AISC E2-1 [Sca]:

$$= (1 - (Klr)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (Klr) / (8 \cdot Cc) - (Klr)^3 / (8 \cdot Cc^3))$$

$$= (1 - (23.3)^2 / (2 \cdot 123^2)) 260 / (5/3 + 3 \cdot (23.3) / (8 \cdot 123) - (23.3)^3 / (8 \cdot 123^3))$$
 = 147 N./mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 8.61/147 + (28.1 \cdot 162/40049480) / 173$$
 = 0.065

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	14.0	311.0	0.0	43.5	0.0	0.400E+04
Web	574.3	14.0	0.0	80.4	0.0	13.1
Totals	...	...	...	123.9	...	0.402E+04

Distance to Centroid from Datum [ytot]:
 = AY / A
 = 0/124
 = 0.000 mm

Distance to Centroid [C1]:
 = Saddle Width / 2
 = 325/2
 = 162.500 mm

Length of Inner Rib [L]:
 = Saddle Height - Outside Radius - Bpthk

$$= 1760 - \cos(1156/2) (22 + 0 + 0) - 0$$
 = 582.000 mm



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Radius of Gyration [r]:
 $= \sqrt{(\text{Total Inertia} / \text{Total Area})}$
 $= \sqrt{(4018/124)}$
 $= 56.936 \text{ mm}$

Slenderness ratio [KL/r]:
 $= KL/r$
 $= 1 \cdot 582/56.9$
 $= 10.222$

Unit Force [Force,u]:
 $= FI / (2 \cdot \text{Baseplate Length})$
 $= 251/(2 \cdot 2070)$
 $= 0.061 \text{ Kg/mm}$

Moment at base of inner Rib [Mbase,c]:
 $= \text{Unit Force} \cdot e \cdot L$
 $= 0.061 \cdot 574 \cdot 582$
 $= 20.250 \text{ Kg-m.}$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:
 $= \text{Bending Moment} / \text{Section Modulus}$
 $= 20.3/247247$
 $= 0.803 \text{ N./mm}^2$

Compressive Allowable, $KL/r < C_c$ ($10.2 < 123$) per AISC E2-1 [Sca]:
 $= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$
 $= (1 - (10.2)^2 / (2 \cdot 123^2)) 260 /$
 $(5/3 + 3 \cdot (10.2) / (8 \cdot 123) - (10.2^3) / (8 \cdot 123^3))$
 $= 152.6 \text{ N./mm}^2$

AISC Unity Check of Inside Ribs (must be ≤ 1)
 $= S_c / S_{ca} + (M_{base,c} \cdot C1/I) / S_{ba}$
 $= 11.6/153 + (20.3 \cdot 162/4018) / 173$
 $= 0.081$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	8
Total Number of Bolts in Tension/Baseplate	Nbt	4
Bolt Material Specification	SA-307 A	
Bolt Allowable Stress	Stba	48.30 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	76.2 mm
User Defined Area in a Single Bolt	UserBoltArea	4.0400 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	3.0428 cm ²
Saddle Load Q0 (Weight)	Q0	52181.4 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	69.2 Kg
Maximum Transverse Force	Ft	312.6 Kg
Maximum Longitudinal Force	F1	250.8 Kg
Saddle Bolted to Steel Foundation	No	

Shear Stress in a Single Bolt, Longitudinal Direction [rb,l]:
 $= FI / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 $= 251/(3.04 \cdot 8)$
 $= 1.0 \text{ N./mm}^2$. Must be less than 29.0 N./mm².

Shear Stress in a Single Bolt, Transverse Direction [rb,t]:
 $= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 $= 313/(3.04 \cdot 8)$
 $= 1.3 \text{ N./mm}^2$. Must be less than 29.0 N./mm².

Bolt Area Calculation per Dennis R. Moss



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Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
= $F_l / (\text{BoltShearAllowable} \cdot N_{\text{bolts}})$
= $251 / (29 \cdot 8)$
= 0.1061 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
= $B \cdot F_t + \text{Sum of X Moments}$
= $1760 \cdot 313 + 0$
= 550.22 Kg-m.

Eccentricity (e):
= R_{mom} / QO
= $550 / 52181$
= 10.54 mm < $B_{\text{plen}} / 6$ --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
= $\max[\text{Bltarearl}, \text{Bltarears}, \text{Bltareart}]$
= $\max[0, 0.11, 0]$
= 0.1061 cm²

Review notes about nozzle loadings and supports in the Warnings report.

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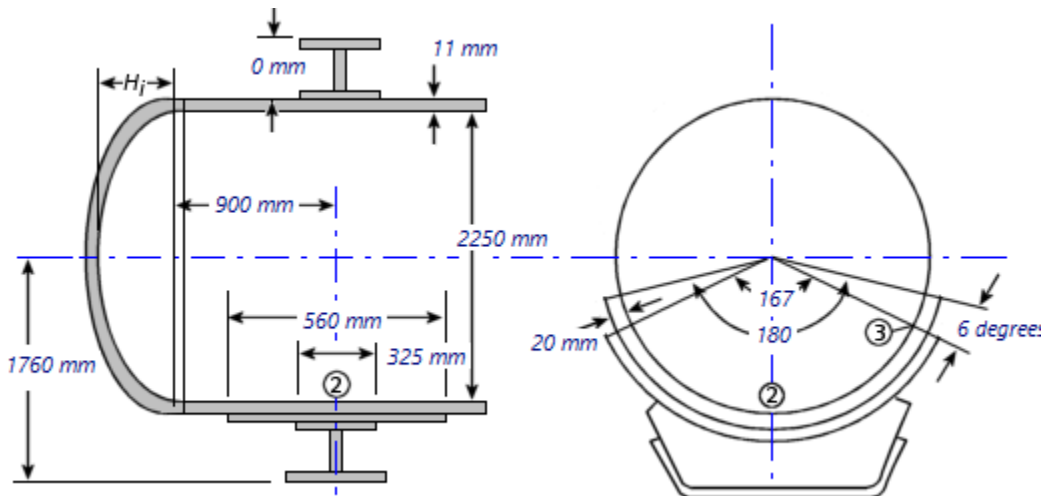
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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : External Pressure

Input and Calculated Values:

Vessel Mean Radius	Rm	1130.90	mm
Shell Thickness used in this Case	t	11.000	mm
Stiffened Vessel Length per 4.15.6	L	8200.00	mm
Distance from Saddle to Vessel tangent	a1 or a	900.00	mm
Saddle Width	b1 or b	325.00	mm
Saddle Bearing Angle	delta or theta	167.00	degrees
Wear Plate Width	b2 or b1	560.00	mm
Wear Plate Bearing Angle	delta2 or theta1	180.00	degrees
Wear Plate Thickness	e2 or tr	20.0	mm
Wear Plate Allowable Stress	fw or Sr	106.20	N./mm ²
Shell Allowable Stress used in Calculation		106.20	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1760.00	mm
Coefficient of Friction	mu	0.30	



Results for Horizontal Vessel Analysis:

Saddle Force Q, External Pressure Case		59846.61	Kgf
Pressure used in Analysis	P	-51.709	KPa.

Horizontal Vessel Analysis Results:		Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan		-20.45	-126.73
Long. Stress at Top of Midspan		20.45	106.20
Long. Stress at Bottom of Midspan		14.72	106.20
Long. Stress at Top of Saddles		-1.36	-126.73
Long. Stress at Top of Saddles		1.36	106.20
Long. Stress at Bottom of Saddles		-3.76	-126.73
Long. Stress at Bottom of Saddles		3.76	106.20



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Tangential Shear in Shell	26.25	84.96
Circ. Stress at Horn of Saddle	26.75	132.75
Circ. Compressive Stress in Shell	2.53	106.20

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= Ftr(Ft/Num\ of\ Saddles + Z\ Force\ Load) \cdot B / E$
 $= 3 (1250/2 + 0) \cdot 1760/2070$
 $= 1594.7\ Kg$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(Fl, wind; Sum\ of\ X\ Forces) \cdot B / Ls$
 $= \max(502, 0) \cdot 1760/6380$
 $= 138.4\ Kg$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(Fl, eq; Sum\ of\ X\ Forces) \cdot B / Ls$
 $= \max(5963, 0) \cdot 1760/6380$
 $= 1645.1\ Kg$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= Ftr(Ft/Num\ of\ Saddles + Z\ Force\ Load) \cdot B / E$
 $= 3 (5963/2 + 0) \cdot 1760/2070$
 $= 7605.1\ Kg$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= Saddle\ Load + \max(Fwl, Fwt, Fsl, Fst)$
 $= 52242 + \max(138, 1595, 1645, 7605)$
 $= 59846.6\ Kg$

Longitudinal Wind Force [Fl]:
 $= WindScalar \cdot WindPress(Platform\ Area + (\pi/4(OD \cdot WindDiaMult)^2))$
 $= 1.1 \cdot 766 (0 + \pi/4(2.27 \cdot 1.2)^2)$
 $= 4919.705\ N$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	60810.88	Kgf
Transverse Shear Load	2981.68	Kgf
Longitudinal Shear Load	5963.35	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= Baseplate\ Length$
 $= 2070.100\ mm$

The Computed K values from Table 4.15.1:

K1 = 0.1974	K2 = 0.6609	K3 = 0.3428	K4 = 0.2477
K5 = 0.6419	K6 = 0.0229	K7 = 0.0159	K8 = 0.2738
K9 = 0.1786	K10 = 0.0228	K1* = 0.3348	K6p = 0.0174
K7p = 0.0121			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $Rm/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L)) / (1 + (4hm)/3L)]$
 $= -59847 \cdot 900 [1 - (1 - 900/8200 + (1131^2 - 0^2) / (2 \cdot 900 \cdot 8200)) / (1 + (4 \cdot 0) / (3 \cdot 8200))]$
 $= -1244.6\ Kg \cdot m$

Moment per Equation 4.15.2 [M2]:



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$$\begin{aligned}
 &= Q \cdot L / 4 \cdot ((1 + 2(Rm^2 - hm^2) / (L^2)) / (1 + (4hm^2) / (3L)) - 4a / L) \\
 &= 59847 \cdot 8200 / 4 \cdot ((1 + 2(1131^2 - 0^2) / (8200^2)) / (1 + (4 \cdot 0) / (3 \cdot 8200)) - 4 \cdot 900 / 8200) \\
 &= 73492.1 \text{ Kg-m.}
 \end{aligned}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$\begin{aligned}
 &= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 \cdot t) \\
 &= -51.7 \cdot 1131 / (2 \cdot 10.2) - 73492 / (\pi \cdot 1131^2 \cdot 10.2) \\
 &= -20.45 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$\begin{aligned}
 &= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t) \\
 &= -51.7 \cdot 1131 / (2 \cdot 10.2) + 73492 / (\pi \cdot 1131^2 \cdot 10.2) \\
 &= 14.72 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$\begin{aligned}
 &= P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t) \\
 &= -51.7 \cdot 1131 / (2 \cdot 10.2) - 1245 / (0.2 \cdot \pi \cdot 1131^2 \cdot 10.2) \\
 &= -1.36 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$\begin{aligned}
 &= P \cdot Rm / (2t) + M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t) \\
 &= -51.7 \cdot 1131 / (2 \cdot 10.2) + 1245 / (0.33 \cdot \pi \cdot 1131^2 \cdot 10.2) \\
 &= -3.76 \text{ N./mm}^2
 \end{aligned}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$\begin{aligned}
 &= Q(L - 2a) / (L + (4 \cdot hm / 3)) \\
 &= 59847 \cdot (8200 - 2 \cdot 900) / (8200 + (4 \cdot 0 / 3)) \\
 &= 46709.6 \text{ Kgf}
 \end{aligned}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$\begin{aligned}
 &= K2 \cdot T / (Rm \cdot t) \\
 &= 0.66 \cdot 46710 / (1131 \cdot 10.2) \\
 &= 26.25 \text{ N./mm}^2
 \end{aligned}$$

Decay Length (4.15.20) [x_1, x_2]:

$$\begin{aligned}
 &= 0.78 \cdot \sqrt{(Rm \cdot t)} \\
 &= 0.78 \cdot \sqrt{(1131 \cdot 10.2)} \\
 &= 83.774 \text{ mm}
 \end{aligned}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$\begin{aligned}
 &= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a) \\
 &= \min(325 + 1.56 \cdot \sqrt{(1131 \cdot 10.2)}, 2 \cdot 900) \\
 &= 492.55 \text{ mm}
 \end{aligned}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$\begin{aligned}
 &= \min(Sr / S, 1) \\
 &= \min(106 / 106, 1) \\
 &= 1.0000
 \end{aligned}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$\begin{aligned}
 &= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr)) \\
 &= -0.64 \cdot 59847 \cdot 0.1 / (493(10.2 + 1 \cdot 20)) \\
 &= -2.53 \text{ N./mm}^2
 \end{aligned}$$

Circ. Comp. Stress at Horn of Saddle, $L < 8Rm$ (4.15.29) [σ_7, r_{*}]:

$$\begin{aligned}
 &= -Q / (4(t + \eta \cdot tr)b1) - 12 \cdot K7 \cdot Q \cdot Rm / (L(t + \eta \cdot tr)^2) \\
 &= -59847 / (4(10.2 + 1 \cdot 20)493) - 12 \cdot 0.016 \cdot 59847 \cdot 1131 / (8200(10.2 + 1 \cdot 20)^2) \\
 &= -26.75 \text{ N./mm}^2
 \end{aligned}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2070.1001	mm
Baseplate Thickness	Bpthk	22.0000	mm
Baseplate Width	Bpwid	350.0000	mm



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Number of Ribs (inc. outside ribs)	Nribs	5
Rib Thickness	Ribtk	14.0000 mm
Web Thickness	Webtk	14.0000 mm
Web Location	Webloc	Center
Saddle Design Temperature		21.1 °C
Saddle Yield Stress	Sy	262.0 N./mm ²
Height of Web at Center	Hw,c	730.0 mm
Friction Coefficient	mu	0.300

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:
 $= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3)/12$
 $= ((1.56 \cdot \sqrt{(1126 \cdot 10.2)} + 560)10.2^3)/12$
 $= 6.431 \text{ cm}^4$

Resolved Inertia for the Shell [I,shell]:
 $= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$
 $= 6.43 + 74.2 (223 - 5.1)^2$
 $= 35168.086 \text{ cm}^4$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	727.2	10.2	5.1	74.2	37827.3	0.352E+05
Wearplate	560.0	20.0	20.2	112.0	226240.0	0.460E+05
Web	14.0	582.0	321.2	81.5	2617137.0	0.309E+05
BasePlate	350.0	22.0	623.2	77.0	4798639.5	0.123E+06
Totals	...	...	...	344.7	7679843.5	0.236E+06

Distance to Centroid [C1]:
 $= AY / A$
 $= 7679843/345$
 $= 222.829 \text{ mm}$

Angle [β]:
 $= 180 - \text{Saddle Angle}/2$
 $= 180 - 167/2$
 $= 96.5$

Saddle Splitting Coefficient [K1]:
 $= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta)\cos(\beta))$
 $= (1 + \cos(96.5) - 0.5 \cdot \sin(96.5)^2) / (\pi - 1.68 + \sin(96.5)\cos(96.5))$
 $= 0.2924$

Saddle Splitting Force [Fh]:
 $= K1 \cdot Q$
 $= 0.29 \cdot 59847$
 $= 17497.5000 \text{ Kg}$

Tension Stress, St = (Fh/As) = 6.3441 N./mm²
 Allowed Stress, Sa = 0.6 * Yield Str = 157.1972 N./mm²

Saddle Splitting Dimension [d]:
 $= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$
 $= 1760 - 1126 \cdot \sin(167/2)/1.46$
 $= 992.468 \text{ mm}$

Bending Moment, M = Fh * d = 17366.0449 Kg-m.

Bending Stress, Sb = (M * C1 / I) = 16.1116 N./mm²
 Allowed Stress, Sa = 2/3 * Yield Str = 174.6635 N./mm²

Minimum Thickness of Baseplate per Moss:

$= \sqrt{3(Q + \text{SaddleWt})\text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$



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$$= \sqrt{3(59847 + 964)350 / (4 \cdot 2070 \cdot 175)}$$

$$= 20.808 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 167/2) (1125 + 11 + 20)$$

$$= 2297.138 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 2297 / (5 - 1)$$

$$= 574.285 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 574 \cdot 350 / 2$$

$$= 1004.998 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 60811 / (2070 \cdot 350)$$

$$= 8.393 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= \text{Ap} \cdot \text{Bp}$$

$$= 1005 \cdot 8.39$$

$$= 8435.038 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 43.5 + 40.2$$

$$= 83.740 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / \text{Ar}$$

$$= 8435 / 83.7$$

$$= 9.878 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	14.0	325.0	...	45.5	...	0.400E+04

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 325 - 14$$

$$= 311.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= \text{AY} / \text{A}$$

$$= 0 / 83.7$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 325 / 2$$

$$= 162.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$



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$$= \sqrt{4005/83.7}$$

$$= 69.156 \text{ mm}$$

Length of Outer Rib, step 1, [Lv]:

$$= \text{Saddle Height} - \cos(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1760 - \cos(167/2) (1125 + 11 + 20) - 22$$

$$= 1607.137 \text{ mm}$$

Length of Outer Rib, step 2, [Lw]:

$$= 2 \cdot \sin(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk})$$

$$= 2 \cdot \sin(167/2) (1125 + 11 + 20)$$

$$= 2297.138 \text{ mm}$$

Length of Outer Rib, step 3, [Bd]:

$$= \text{Baseplate length} - \text{clearance}$$

$$= 2070 - 50.8$$

$$= 2019.300 \text{ mm}$$

Length of Outer Rib, step 4, [Dd]:

$$= (Lw - Bd) / 2$$

$$= (2297 - 2019) / 2$$

$$= 138.919 \text{ mm}$$

Length of Outer Rib [L]:

$$= \sqrt{Lv^2 + Dd^2}$$

$$= \sqrt{1607^2 + 139^2}$$

$$= 1613.130 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 199943008 / 262)}$$

$$= 122.739$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1613 / 69.2$$

$$= 23.326$$

Bending Moment [Rm]:

$$= Fl / (2 \cdot Bplen) \cdot e \cdot L / 2$$

$$= 5963 / (2 \cdot 2070) \cdot 574 \cdot 1613 / 2$$

$$= 667.182 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < Cc$ ($23.3 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (23.3)^2 / (2 \cdot 123^2)) 262 / (5/3 + 3 \cdot (23.3) / (8 \cdot 123) - (23.3^3) / (8 \cdot 123^3))$$

$$= 148.1 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 9.88 / 148 + (667 \cdot 162 / 40049480) / 175$$

$$= 0.219$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	14.0	311.0	0.0	43.5	0.0	0.400E+04
Web	574.3	14.0	0.0	80.4	0.0	13.1
Totals	...	...	...	123.9	...	0.402E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$



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= 0/124
 = 0.000 mm

Distance to Centroid [C1]:
 = Saddle Width / 2
 = 325/2
 = 162.500 mm

Length of Inner Rib [L]:
 = Saddle Height - Outside Radius - Bpthk
 = 1760 - cos(1156/2)(22 + 0 + 0) - 0
 = 582.000 mm

Radius of Gyration [r]:
 = $\sqrt{(\text{Total Inertia} / \text{Total Area})}$
 = $\sqrt{(4018/124)}$
 = 56.936 mm

Slenderness ratio [KL/r]:
 = KL/r
 = 1 • 582/56.9
 = 10.222

Unit Force [Force,u]:
 = FI / (2 • Baseplate Length)
 = 5963/(2 • 2070)
 = 1.440 Kg/mm

Moment at base of inner Rib [Mbase,c]:
 = Unit Force • e • L
 = 1.44 • 574 • 582
 = 481.424 Kg-m.

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:
 = Bending Moment / Section Modulus
 = 481/247247
 = 19.095 N./mm²

Compressive Allowable, KL/r < Cc (10.2 < 123) per AISC E2-1 [Sca]:
 = $(1 - (KL/r)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
 = $(1 - (10.2)^2 / (2 \cdot 123^2)) 262 /$
 $(5/3 + 3 \cdot (10.2) / (8 \cdot 123) - (10.2^3) / (8 \cdot 123^3))$
 = 153.8 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)
 = Sc/Sca + (Mbase,c • C1/I)/Sba
 = 13.4/154 + (481 • 162/4018)/175
 = 0.196

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	1130.90	mm
Shell Thickness used in this Case	t	11.000	mm
Stiffened Vessel Length per 4.15.6	L	8200.00	mm
Distance from Saddle to Vessel tangent	a1 or a	900.00	mm
Saddle Width	b1 or b	325.00	mm
Saddle Bearing Angle	delta or theta	167.00	degrees
Wear Plate Width	b2 or b1	560.00	mm
Wear Plate Bearing Angle	delta2 or theta1	180.00	degrees
Wear Plate Thickness	e2 or tr	20.0	mm
Wear Plate Allowable Stress	fw or Sr	106.20	N./mm ²



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Shell Allowable Stress used in Calculation	106.20	N./mm ²
Head Allowable Stress used in Calculation	138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle	1.00	
Circumferential Efficiency at Mid-Span	1.00	
Distance from Saddle Base to Centerline	B	1760.00 mm
Coefficient of Friction	mu	0.00

Results for Horizontal Vessel Analysis:

Saddle Force Q, External Pressure Case	59961.47	Kgf
Pressure used in Analysis	P	-51.709 KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	-20.49	-126.73
Long. Stress at Top of Midspan	20.49	106.20
Long. Stress at Bottom of Midspan	14.75	106.20
Long. Stress at Top of Saddles	-1.36	-126.73
Long. Stress at Top of Saddles	1.36	106.20
Long. Stress at Bottom of Saddles	-3.76	-126.73
Long. Stress at Bottom of Saddles	3.76	106.20
Tangential Shear in Shell	26.30	84.96
Circ. Stress at Horn of Saddle	26.80	132.75
Circ. Compressive Stress in Shell	2.54	106.20

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3 (1250/2 + 0) \cdot 1760/2070$
 $= 1594.7 \text{ Kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l,wind}; \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(502, 0) \cdot 1760/6380$
 $= 138.4 \text{ Kgf}$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(F_{l,eq}; \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(5963, 0) \cdot 1760/6380$
 $= 1645.1 \text{ Kgf}$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3 (5963/2 + 0) \cdot 1760/2070$
 $= 7605.1 \text{ Kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$
 $= 52356 + \max(138, 1595, 1645, 7605)$
 $= 59961.5 \text{ Kgf}$

Longitudinal Wind Force [Fl]:
 $= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (OD \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 766 (0 + \pi/4 (2.27 \cdot 1.2)^2)$
 $= 4919.705 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	60925.73	Kgf
Transverse Shear Load	2981.68	Kgf



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Longitudinal Shear Load 5963.35 Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 = Baseplate Length
 = 2070.100 mm

The Computed K values from Table 4.15.1:

K1 = 0.1974	K2 = 0.6609	K3 = 0.3428	K4 = 0.2477
K5 = 0.6419	K6 = 0.0229	K7 = 0.0159	K8 = 0.2738
K9 = 0.1786	K10 = 0.0228	K1* = 0.3348	K6p = 0.0174
K7p = 0.0121			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a \left[1 - \left(1 - \frac{a}{L} + \frac{(R_m^2 - h_m^2)}{(2a \cdot L)} \right) / \left(1 + \frac{(4h_m)}{3L} \right) \right]$$

$$= -59961 \cdot 900 \left[1 - \left(1 - \frac{900}{8200} + \frac{(1131^2 - 0^2)}{(2 \cdot 900 \cdot 8200)} \right) / \left(1 + \frac{(4 \cdot 0)}{(3 \cdot 8200)} \right) \right]$$

$$= -1247.0 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 \left(\frac{(1 + 2(R_m^2 - h_m^2)/(L^2))}{(1 + (4h_m^2)/(3L))} - \frac{4a}{L} \right)$$

$$= 59961 \cdot 8200 / 4 \left(\frac{(1 + 2(1131^2 - 0^2)/(8200^2))}{(1 + (4 \cdot 0)/(3 \cdot 8200))} - \frac{4 \cdot 900}{8200} \right)$$

$$= 73633.2 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 t)$$

$$= -51.7 \cdot 1131 / (2 \cdot 10.2) - 73633 / (\pi \cdot 1131^2 \cdot 10.2)$$

$$= -20.49 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= -51.7 \cdot 1131 / (2 \cdot 10.2) + 73633 / (\pi \cdot 1131^2 \cdot 10.2)$$

$$= 14.75 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 t)$$

$$= -51.7 \cdot 1131 / (2 \cdot 10.2) - 1247 / (0.2 \cdot \pi \cdot 1131^2 \cdot 10.2)$$

$$= -1.36 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= -51.7 \cdot 1131 / (2 \cdot 10.2) + 1247 / (0.33 \cdot \pi \cdot 1131^2 \cdot 10.2)$$

$$= -3.76 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot h_m / 3))$$

$$= 59961 (8200 - 2 \cdot 900) / (8200 + (4 \cdot 0 / 3))$$

$$= 46799.2 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 0.66 \cdot 46799 / (1131 \cdot 10.2)$$

$$= 26.30 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1131 \cdot 10.2)}$$

$$= 83.774 \text{ mm}$$



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Effective reinforcing plate width (4.15.24) [B1]:
 $= \min(b + 1.56 \cdot \sqrt{ Rm \cdot t }, 2a)$
 $= \min(325 + 1.56 \cdot \sqrt{ 1131 \cdot 10.2 }, 2 \cdot 900)$
 $= 492.55 \text{ mm}$

Wear Plate/Shell Stress ratio (4.15.26) [eta]:
 $= \min(Sr/S, 1)$
 $= \min(106/106, 1)$
 $= 1.0000$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:
 $= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$
 $= -0.64 \cdot 59961 \cdot 0.1 / (493 (10.2 + 1 \cdot 20))$
 $= -2.54 \text{ N./mm}^2$

Circ. Comp. Stress at Horn of Saddle, L<8Rm (4.15.29) [σ_7, r^*]:
 $= -Q / (4(t + \eta \cdot tr) b1) - 12 \cdot K7 \cdot Q \cdot Rm / (L(t + \eta \cdot tr)^2)$
 $= -59961 / (4(10.2 + 1 \cdot 20) 493) -$
 $12 \cdot 0.016 \cdot 59961 \cdot 1131 / (8200(10.2 + 1 \cdot 20)^2)$
 $= -26.80 \text{ N./mm}^2$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2070.1001	mm
Baseplate Thickness	Bpthk	22.0000	mm
Baseplate Width	Bpwid	350.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	14.0000	mm
Web Thickness	Webtk	14.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	262.0	N./mm ²
Height of Web at Center	Hw, c	730.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:
 $= ((1.56 \cdot \sqrt{ R \cdot t } + \text{WearPlateWidth}) t^3) / 12$
 $= ((1.56 \cdot \sqrt{ 1126 \cdot 10.2 } + 560) 10.2^3) / 12$
 $= 6.431 \text{ cm}^4$

Resolved Inertia for the Shell [I,shell]:
 $= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$
 $= 6.43 + 74.2 (223 - 5.1)^2$
 $= 35168.086 \text{ cm}^4$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	727.2	10.2	5.1	74.2	37827.3	0.352E+05
Wearplate	560.0	20.0	20.2	112.0	226240.0	0.460E+05
Web	14.0	582.0	321.2	81.5	2617137.0	0.309E+05
BasePlate	350.0	22.0	623.2	77.0	4798639.5	0.123E+06
Totals	...	...	...	344.7	7679843.5	0.236E+06

Distance to Centroid [C1]:
 $= AY / A$
 $= 7679843/345$
 $= 222.829 \text{ mm}$

Angle [β]:
 $= 180 - \text{Saddle Angle}/2$
 $= 180 - 167/2$
 $= 96.5$



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Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(96.5) - 0.5 \cdot \sin(96.5)^2) / (\pi - 1.68 + \sin(96.5) \cos(96.5))$$

$$= 0.2924$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.29 \cdot 59961$$

$$= 17531.0801 \text{ Kg}$$

$$\text{Tension Stress, } St = (Fh / As) = 6.3563 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 157.1972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1760 - 1126 \cdot \sin(167/2) / 1.46$$

$$= 992.468 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 17399.3711 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 16.1425 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 174.6635 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(59961 + 964) \cdot 350) / (4 \cdot 2070 \cdot 175)}$$

$$= 20.827 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) \cdot \text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk}$$

$$= 2 \cdot \cos(90 - 167/2) \cdot (1125 + 11 + 20)$$

$$= 2297.138 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 2297 / (5 - 1)$$

$$= 574.285 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 574 \cdot 350 / 2$$

$$= 1004.998 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 60926 / (2070 \cdot 350)$$

$$= 8.409 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 1005 \cdot 8.41$$

$$= 8450.970 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 43.5 + 40.2$$

$$= 83.740 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / Ar$$

$$= 8451 / 83.7$$

$$= 9.897 \text{ N./mm}^2$$



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Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	14.0	325.0	...	45.5	...	0.400E+04

Rib dimension [D]:
 = Saddle Width - Web Thickness
 = 325 - 14
 = 311.000 mm

Distance to Centroid from Datum [ytot]:
 = AY / A
 = 0/83.7
 = 0.000 mm

Distance to Centroid [C1]:
 = Saddle Width / 2
 = 325/2
 = 162.500 mm

Radius of Gyration [r]:
 = $\sqrt{\text{Total Inertia} / \text{Total Area}}$
 = $\sqrt{4005/83.7}$
 = 69.156 mm

Length of Outer Rib, step 1, [Lv]:
 = Saddle Height - $\cos(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$
 = 1760 - $\cos(167/2) (1125 + 11 + 20) - 22$
 = 1607.137 mm

Length of Outer Rib, step 2, [Lw]:
 = $2 \cdot \sin(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk})$
 = $2 \cdot \sin(167/2) (1125 + 11 + 20)$
 = 2297.138 mm

Length of Outer Rib, step 3, [Bd]:
 = Baseplate length - clearance
 = 2070 - 50.8
 = 2019.300 mm

Length of Outer Rib, step 4, [Dd]:
 = $(Lw - Bd) / 2$
 = $(2297 - 2019) / 2$
 = 138.919 mm

Length of Outer Rib [L]:
 = $\sqrt{Lv^2 + Dd^2}$
 = $\sqrt{1607^2 + 139^2}$
 = 1613.130 mm

Intermediate Term [Cc]:
 = $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$
 = $\sqrt{2 \cdot \pi^2 \cdot 199943008 / 262}$
 = 122.739

Slenderness ratio [KL/r]:
 = KL/r
 = $1 \cdot 1613 / 69.2$
 = 23.326

Bending Moment [Rm]:



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$$= F_l / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 5963 / (2 \cdot 2070) \cdot 574 \cdot 1613 / 2$$

$$= 667.182 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ ($23.3 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (Kl/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (Kl/r) / (8 \cdot C_c) - (Kl/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (23.3)^2 / (2 \cdot 123^2)) 262 /$$

$$(5/3 + 3 \cdot (23.3) / (8 \cdot 123) - (23.3^3) / (8 \cdot 123^3))$$

$$= 148.1 \text{ N/mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 9.9 / 148 + (667 \cdot 162 / 40049480) / 175$$

$$= 0.219$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	14.0	311.0	0.0	43.5	0.0	0.400E+04
Web	574.3	14.0	0.0	80.4	0.0	13.1
Totals	...	...	...	123.9	...	0.402E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 124$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 325 / 2$$

$$= 162.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - B_{pthk}$$

$$= 1760 - \cos(1156/2) (22 + 0 + 0) - 0$$

$$= 582.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(4018 / 124)}$$

$$= 56.936 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 582 / 56.9$$

$$= 10.222$$

Unit Force [Force,u]:

$$= F_l / (2 \cdot \text{Baseplate Length})$$

$$= 5963 / (2 \cdot 2070)$$

$$= 1.440 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 1.44 \cdot 574 \cdot 582$$

$$= 481.424 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 481 / 247247$$

$$= 19.095 \text{ N/mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($10.2 < 123$) per AISC E2-1 [Sca]:



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$$\begin{aligned}
 &= (1 - (Klr)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (Klr) / (8 \cdot Cc) - (Klr^3) / (8 \cdot Cc^3)) \\
 &= (1 - (10.2)^2 / (2 \cdot 123^2)) 262 / \\
 &\quad (5/3 + 3 \cdot (10.2) / (8 \cdot 123) - (10.2^3) / (8 \cdot 123^3)) \\
 &= 153.8 \text{ N./mm}^2
 \end{aligned}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$\begin{aligned}
 &= Sc/Sca + (Mbase,c \cdot C1/I) / Sba \\
 &= 13.4/154 + (481 \cdot 162/4018) / 175 \\
 &= 0.196
 \end{aligned}$$

Review notes about nozzle loadings and supports in the Warnings report.

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 Nozzle Calcs.: N5-Tube Inlet Nozl: 9 6:24pm Sep 8,2026

Input, Nozzle Desc: N5-Tube Inlet

From: 30

Pressure for Reinforcement Calculations	P	1081.470	KPa.
Temperature for Internal Pressure	Temp	82	°C
Design External Pressure	Pext	103.42	KPa.
Temperature for External Pressure	Tempex	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2250.00	mm
Design Length of Section	L	999.9695	mm
Shell Finished (Minimum) Thickness	t	20.0000	mm
Shell Internal Corrosion Allowance	c	3.2000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		934.70	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

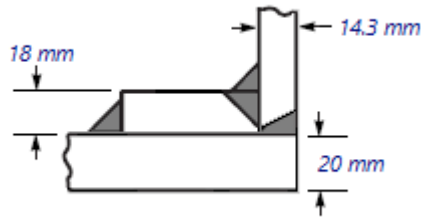
Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		18.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		40	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	0.0000	mm
Pad Material		SA-312 TP304	
Pad Allowable Stress at Temperature	Sp	137.51	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	657.2000	mm
Thickness of Pad	te	18.0000	mm
Weld leg size between Pad and Shell	Wp	14.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	18.0000	mm
Reinforcing Pad Width		100.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



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 Nozzle Calcs.: N5-Tube Inlet Nozl: 9 6:24pm Sep 8,2026



Reinforcement CALCULATION, Description: N5-Tube Inlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation 18.000 in.
 Actual Thickness Used in Calculation 0.562 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1081 \cdot 1128 / (138 \cdot 1 - 0.6 \cdot 1081)$
 $= 8.8839 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 1081 \cdot 229 / (118 \cdot 1 + 0.4 \cdot 1081)$
 $= 2.0876 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.0391 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	861.3008	mm
Parallel to Vessel Wall, opening length	d	430.6504	mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		42.0000	mm

Weld Strength Reduction Factor [fr1]:
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118 / 138)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138 / 138)$
 $= 0.996$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.86, 1)$
 $= 0.855$

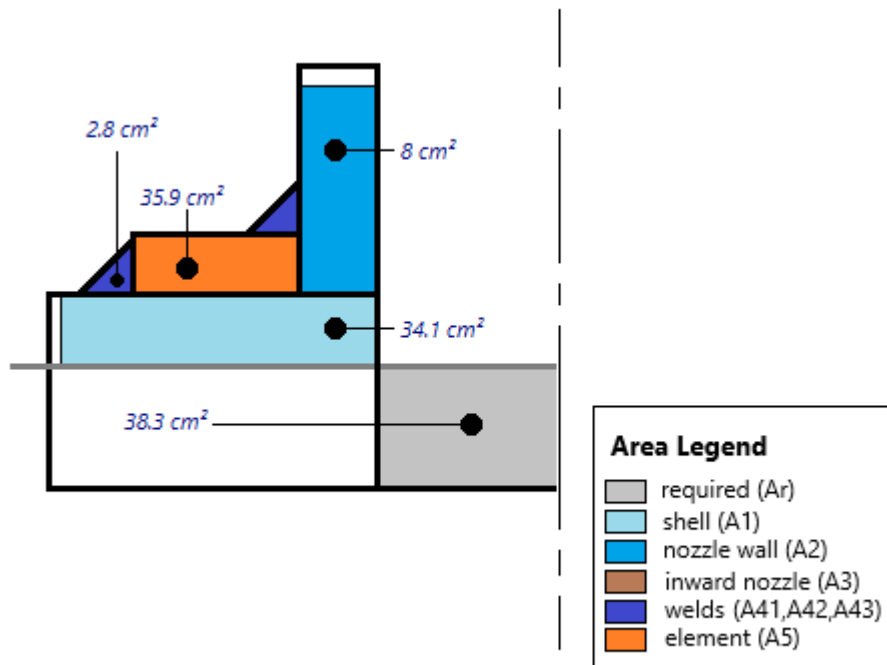


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	38.258	11.164	NA
Area in Shell	A1	34.091	50.020	NA
Area in Nozzle Wall	A2	8.035	8.788	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	2.808	2.808	NA
Area in Element	A5	35.873	35.873	NA
TOTAL AREA AVAILABLE	Atot	80.808	97.490	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.
 The area available with the given pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (431 \cdot 8.88 \cdot 1 + 2 \cdot 13.3 \cdot 8.88 \cdot 1 \cdot (1 - 1))$$

$$= 38.258 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 431(1 \cdot 16.8 - 1 \cdot 8.88) - 2 \cdot 13.3$$

$$(1 \cdot 16.8 - 1 \cdot 8.88) \cdot (1 - 1)$$

$$= 34.091 \text{ cm}^2$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$$

$$= (2 \cdot 42) \cdot (13.3 - 2.09) \cdot 0.86$$



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$$= 8.035 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:
 $= W_o^2 \cdot fr_3 + (W_i - can/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$
 $= 10^2 \cdot 0.86 + (0)^2 \cdot 0.86 + 14^2 \cdot 1$
 $= 2.808 \text{ cm}^2$

Area Available in Element [A5]:
 $= (\min(D_p, D_L) - (\text{Nozzle OD})) \cdot (\min(t_p, t_{lp}, t_e)) \cdot fr_4$
 $= (657 - 457) \cdot 18 \cdot 1$
 $= 35.873 \text{ cm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 3.0876 \text{ mm}$
Wall Thickness per UG16(b),	$tr_{16b} = 4.7000 \text{ mm}$
Wall Thickness, shell/head, internal pressure	$tr_{b1} = 12.0839 \text{ mm}$
Wall Thickness	$tb_1 = \max(tr_{b1}, tr_{16b}) = 12.0839 \text{ mm}$
Wall Thickness, shell/head, external pressure	$tr_{b2} = 4.0459 \text{ mm}$
Wall Thickness	$tb_2 = \max(tr_{b2}, tr_{16b}) = 4.7000 \text{ mm}$
Wall Thickness per table UG-45	$tb_3 = 9.3312 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb_3, \max(tb_1, tb_2)]$
 $= \min[9.33, \max(12.1, 4.7)]$
 $= 9.3312 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(t_a, t_b)$
 $= \max(3.09, 9.33)$
 $= 9.3312 \text{ mm}$

Available Nozzle Neck Thickness = $0.875 \cdot 14.275 = 12.490 \text{ mm} \rightarrow \text{OK}$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 26.8,	Allowable	: 118.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 268.2 N./mm ²	Passed
Occasional	: 8.5,	Allowable	: 156.9 N./mm ²	Passed
Shear	: 14.0,	Allowable	: 82.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 12.5$, $t_r = 2.09$, $c = 1 \text{ mm}$, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.18$, Temp. Reduction = 78°C

Min Metal Temp. w/o impact per UCS-66, Curve A	-1 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, $t_g = 12.5$, $t_r = 2.09$, $c = 1 \text{ mm}$, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.18$, Temp. Reduction = 78°C

Min Metal Temp. w/o impact per UCS-66, Curve B	-22 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C



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Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 12.5, tr = 2.09, c = 1 mm, E* = 1
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.18$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-22 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-44 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $1081.47/1960.01 = 0.552$

Weld Size Calculations, Description: N5-Tube Inlet

Intermediate Calc. for nozzle/shell Welds	Tmin	13.2748 mm
Intermediate Calc. for pad/shell Welds	TminPad	16.8000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$
Pad Weld	$8.4000 = 0.5 \cdot T_{\text{minPad}}$	$9.8980 = 0.7 \cdot W_p \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= (Ar - A1) \cdot Sv$
 $= (38.3 - 34.1) \cdot 138$
 $= 5864.31 \text{ Kg}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A5 + A4 - (Wi - Can / .707)^2 \cdot fr2) \cdot Sv$
 $= (8.04 + 35.9 + 2.81 - 0 \cdot 0.86) \cdot 138$
 $= 65737.92 \text{ Kg}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 457 \cdot 10 \cdot 0.49 \cdot 118$
 $= 42342. \text{ Kg}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$
 $= (3.14/2.0) \cdot 657 \cdot 14 \cdot 0.49 \cdot 138$
 $= 99301. \text{ Kg}$

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot Seg$
 $= (3.14/2) \cdot 457 \cdot 18 \cdot 0.74 \cdot 138$
 $= 134135. \text{ Kg}$

Strength of Failure Paths:

Failure Path [Path 1-1]:



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$$= (\text{SPEW} + \text{SNGW}) = (99301 + 0) = 99301 \text{ Kg}$$

Summary of Failure Path Calculations:

Path 1-1 = 99301 Kg, must exceed W = 5864 Kg or W1 = 65737 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1094 KPa.

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 103 KPa.

The Drop for this Nozzle is : 23.4706 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 243.4706 mm

Input Echo, WRC297 Item 1, Description: N5-Tube Inlet :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.2000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	2250.000	mm
Shell Thickness	20.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	1.0000	mm
Nozzle Diameter	457.200	mm
Nozzle Thickness	14.2748	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc 118.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 118.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	657.200	mm
Thickness of Reinforcing Pad	18.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSTAINED loads.

Design Internal Pressure	Dp	1081.47	KPa.
Radial Load	P	-3441.77	Kgf
Circumferential Shear	Vc	2294.50	Kgf
Longitudinal Shear	Vl	2294.50	Kgf
Circumferential Moment	Mc	2889.98	Kg-m.
Longitudinal Moment	Ml	2889.98	Kg-m.
Torsional Moment	Mt	4128.54	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:



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Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 457.2 + 2 \cdot 1.65 \cdot \sqrt{1145.6(20.0 - 3.2)}$$

$$= 915.010 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 2291.200 mm
 Nozzle Outside Diameter (d) = 457.200 mm
 Vessel Thickness used (T) = 34.800 mm
 Nozzle Thickness used (t) = 13.275 mm
 T / t = 2.622
 d / t = 34.441

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.619$$

Nr/P = 0.076
 Mr/P = 0.086
 M0/P = 0.065
 N0/P = 0.188
 MrD/Mc = 0.172
 NrDL/Mc = 0.107
 M0d/Mc = 0.155
 N0DL/Mc = 0.192
 MrD/Ml = 0.111
 NrDL/Ml = 0.077
 M0D/Ml = 0.093
 N0DL/Ml = 0.329

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	10	-10	10	-10
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-16	-16	16	16
Outplane Bending (ML)	-28	28	28	-28
Normal Pressure Stress	34	35	34	35
Outplane Stress Summary	5	41	96	18

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	14	-14	14	-14
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-3	-3	3	3
Inplane Bending (ML)	-33	33	33	-33
Inplane Pressure Stress	17	17	17	17
Inplane Stress Summary	-4	35	71	-24

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0



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Outplane Shear (V1)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3
Shear Stress Summary	4	4	2	2

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Two * Max Shear Stress	12	43	96	43

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	D1 Right Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	10	-10	10	-10
Outplane Membrane (Mc)	-9	-9	9	9
Outplane Bending (Mc)	-47	47	47	-47
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	17	17	17	17
Outplane Stress Summary	-24	49	90	-26

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	D1 Right Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	14	-14	14	-14
Inplane Membrane (Mc)	-5	-5	5	5
Inplane Bending (Mc)	-52	52	52	-52
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	34	35	34	35
Inplane Stress Summary	-7	70	109	-23

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	D1 Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (V1)	0	0	0	0
Torsional Shear (Mt)	3	3	3	3
Shear Stress Summary	2	2	4	4

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	D1 Right Inside
Two * Max Shear Stress	24	71	110	29

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
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Hoop Membrane (P)	5	5	5	5
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-16	-16	16	16
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	20	21	20	21
Hoop Stress Summary	8	10	42	43

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	54	-54	54	-54
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-14	-14	14	14
Axial Bending (ML)	-152	152	152	-152
Axial Pressure Stress	10	10	10	10
Axial Stress Summary	-99	95	233	-180

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	9	9	9	9
Shear Stress Summary	11	11	6	6

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	110	96	233	225

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	5	5	5	5
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-9	-9	9	9
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	20	21	20	21
Hoop Stress Summary	15	17	35	36

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	54	-54	54	-54
Axial Membrane (Mc)	-14	-14	14	14
Axial Bending (Mc)	-250	250	250	-250



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Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	10	10	10	10

Axial Stress Summary	-197	193	331	-278
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-2	-2	2	2
Shear due to Torsion	9	9	9	9
Shear Stress Summary	6	6	11	11

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	213	193	331	316

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$
 $= 457.2 + 2 \cdot 1.65 \cdot \sqrt{1136.6(20.0 - 3.2)}$
 $= 913.208 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2273.200 mm
Nozzle Outside Diameter	(d) =	657.200 mm
Vessel Thickness used	(T) =	16.800 mm
Nozzle Thickness used	(t) =	113.275 mm
T / t	=	0.148
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 3.363$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.056
Mr/P	=	0.097
M0/P	=	0.029
N0/P	=	0.048
MrD/Mc	=	0.262
NrDL/Mc	=	0.099
M0d/Mc	=	0.078
N0DL/Mc	=	0.073
MrD/Ml	=	0.074
NrDL/Ml	=	0.066
M0D/Ml	=	0.022
N0DL/Ml	=	0.057

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au	Al	Bu	B1
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	20	-20	20	-20
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0



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Outplane Membrane (ML)	-8	-8	8	8
Outplane Bending (ML)	-19	19	19	-19
Normal Pressure Stress	72	73	72	73

Outplane Stress Summary	69	69	126	47
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Inplane Membrane (P)	6	6	6	6
Inplane Bending (P)	69	-69	69	-69
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-10	-10	10	10
Inplane Bending (ML)	-67	67	67	-67
Inplane Pressure Stress	36	36	36	36

Inplane Stress Summary	34	30	190	-84
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3

Shear Stress Summary	4	4	2	2
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Two * Max Shear Stress	70	70	190	131
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	20	-20	20	-20
Outplane Membrane (Mc)	-11	-11	11	11
Outplane Bending (Mc)	-71	71	71	-71
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	36	36	36	36

Outplane Stress Summary	-20	82	145	-39
-------------------------	-----	----	-----	-----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	6	6	6	6
Inplane Bending (P)	69	-69	69	-69
Inplane Membrane (Mc)	-15	-15	15	15
Inplane Bending (Mc)	-240	240	240	-240
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0



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 Nozzle Calcs.: N5-Tube Inlet Nozl: 9 6:24pm Sep 8,2026

Inplane Pressure Stress	72	73	72	73
Inplane Stress Summary	-106	235	403	-214

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)				
	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	3	3	3	3
Shear Stress Summary	2	2	4	4

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)				
	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	106	235	403	214

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	34	35	34	35	34	35	34	35
Circ. Pl (SUS)	-11	-11	21	21	-3	-3	7	7
Circ. Q (SUS)	-18	18	38	-38	-38	38	66	-66
Long. Pm (SUS)	17	17	17	17	17	17	17	17
Long. Pl (SUS)	-1	-1	5	5	-4	-4	14	14
Long. Q (SUS)	-19	19	47	-47	-37	37	57	-57
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	34.0	35.0	34.0	35.0	34.0	35.0	34.0	35.0
Pm+Pl (SUS)	23.0	24.0	55.0	56.0	31.0	32.0	41.0	42.0
Pm+Pl+Q (Total)	10.0	43.1	93.4	43.4	24.5	70.4	107.5	28.2

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	35.00	138.00	Passed
Pm+Pl (SUS)	56.00	207.00	Passed
Pm+Pl+Q (TOTAL)	107.46	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of	Stress Values at
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Stress Int. Location	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	72	73	72	73	72	73	72	73
Circ. Pl (SUS)	-3	-3	13	13	-9	-9	21	21
Circ. Q (SUS)	1	-1	39	-39	-171	171	309	-309
Long. Pm (SUS)	36	36	36	36	36	36	36	36
Long. Pl (SUS)	-4	-4	16	16	-6	-6	16	16
Long. Q (SUS)	2	-2	136	-136	-51	51	91	-91
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	72.0	73.0	72.0	73.0	72.0	73.0	72.0	73.0
Pm+Pl (SUS)	69.0	70.0	85.0	86.0	63.0	64.0	93.0	94.0
Pm+Pl+Q (Total)	70.4	69.4	188.1	131.1	108.0	235.0	402.1	215.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	73.00	138.00	Passed
Pm+Pl (SUS)	94.02	207.00	Passed
Pm+Pl+Q (TOTAL)	402.06	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	20	21	20	21	20	21	20	21
Circ. Pl (SUS)	-11	-11	21	21	-4	-4	14	14
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	10	10	10	10	10	10	10	10
Long. Pl (SUS)	-13	-13	15	15	-13	-13	15	15
Long. Q (SUS)	-98	98	206	-206	-196	196	304	-304
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	9	9	9	9	9	9	9	9
Pm (SUS)	20.0	21.0	20.0	21.0	20.0	21.0	20.0	21.0
Pm+Pl (SUS)	12.6	13.6	41.2	42.2	19.4	20.4	34.4	35.4
Pm+Pl+Q (Total)	112.2	96.4	231.3	223.4	215.5	193.3	329.4	314.8

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	21.00	118.00	Passed



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Pm+Pl (SUS)	42.23	177.00	Passed
Pm+Pl+Q (TOTAL)	329.41	353.99	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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 Nozzle Calcs.: N1-Shell inlet Nozl: 10 6:24pm Sep 8,2026

Input, Nozzle Desc: N1-Shell inlet

From: 50

Pressure for Reinforcement Calculations	P	791.335	KPa.
Temperature for Internal Pressure	Temp	130	°C
Design External Pressure	Pext	51.71	KPa.
Temperature for External Pressure	Tempex	130	°C
Parent Material		SA-240 304	
Parent Allowable Stress at Temperature	Sv	106.20	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2250.00	mm
Design Length of Section	L	8200.0000	mm
Shell Finished (Minimum) Thickness	t	24.0000	mm
Shell Internal Corrosion Allowance	c	0.8000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		5598.70	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

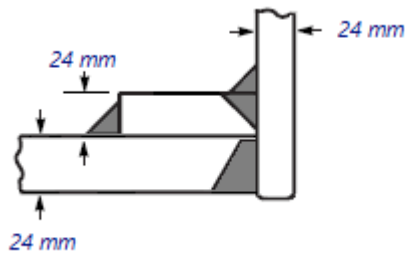
Type of Element Connected to the Parent : Nozzle

Material		SA-312 TP304	
Material UNS Number		S30400	
Material Specification/Type	Smls. & wld. pipe		
Allowable Stress at Temperature	Sn	133.20	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		36.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	24.0000	mm
Flange Material		SA-182 F304	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	0.8000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	20.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	22.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	1614.4000	mm
Thickness of Pad	te	24.0000	mm
Weld leg size between Pad and Shell	Wp	20.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	24.0000	mm
Reinforcing Pad Width		350.0000	mm
Flange Class		150	
Flange Grade		GR 2.1	
Flange Series		B	

The Pressure Design option was MAWP + static head (to the nozzle).



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 Nozzle Calcs.: N1-Shell inlet Nozl: 10 6:24pm Sep 8,2026



Reinforcement CALCULATION, Description: N1-Shell inlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation 36.000 in.
 Actual Thickness Used in Calculation 0.945 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 791 \cdot 1126 / (106 \cdot 1 - 0.6 \cdot 791)$
 $= 8.4269 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 791 \cdot 457 / (133 \cdot 1 + 0.4 \cdot 791)$
 $= 2.7099 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.4428 mm

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 1736.0000 mm
 Parallel to Vessel Wall, opening length d 868.0000 mm
 Normal to Vessel Wall (Thickness Limit), pad side Tlwp 58.0000 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 133 / 106)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 133 / 106)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138 / 106)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$

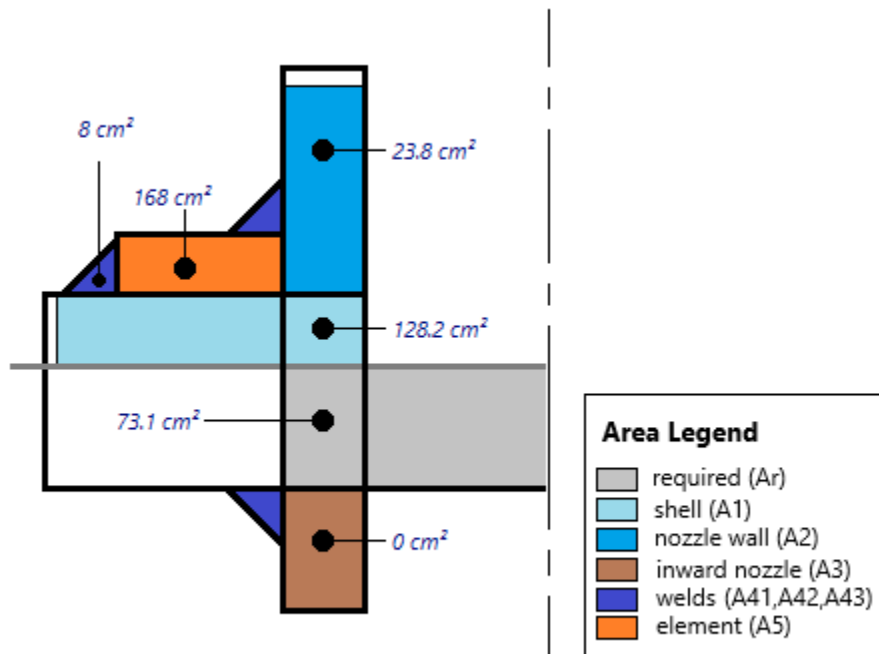


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	73.146	42.448	NA
Area in Shell	A1	128.230	116.480	NA
Area in Nozzle Wall	A2	23.768	25.238	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	8.000	8.000	NA
Area in Element	A5	168.000	168.000	NA
TOTAL AREA AVAILABLE	Atot	327.999	317.719	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.
 The area available with the given pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (868 \cdot 8.43 \cdot 1 + 2 \cdot 23.2 \cdot 8.43 \cdot 1 \cdot (1 - 1))$
 $= 73.146 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 868(1 \cdot 23.2 - 1 \cdot 8.43) - 2 \cdot 23.2$
 $(1 \cdot 23.2 - 1 \cdot 8.43) \cdot (1 - 1)$
 $= 128.230 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 58) \cdot (23.2 - 2.71) \cdot 1$



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= 23.768 cm²

Area Available in Welds [A41 + A42 + A43]:
 = $W_o^2 \cdot fr3 + (W_i - can/0.707)^2 \cdot fr2 + W_p^2 \cdot fr4$
 = $20^2 \cdot 1 + (0)^2 \cdot 1 + 20^2 \cdot 1$
 = 8.000 cm²

Area Available in Element [A5]:
 = $(\min(D_p, D_L) - (\text{Nozzle OD})) \cdot (\min(t_p, t_{lp}, t_e)) \cdot fr4$
 = $(1614 - 914) \cdot 24 \cdot 1$
 = 168.000 cm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.5099 mm
Wall Thickness per UG16(b),	tr16b = 2.3000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 9.2269 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 9.2269 mm
Wall Thickness, shell/head, external pressure	trb2 = 1.3484 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 2.3000 mm
Wall Thickness per table UG-45	tb3 = 9.1312 mm

Determine Nozzle Thickness candidate [tb]:
 = min[tb3, max(tb1, tb2)]
 = min[9.13, max(9.23, 2.3)]
 = 9.1312 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 = max(ta, tb)
 = max(3.51, 9.13)
 = 9.1312 mm

Available Nozzle Neck Thickness = 24.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	17.5,	Allowable	:	133.2 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	321.5 N./mm ²	Passed
Occasional	:	7.2,	Allowable	:	177.2 N./mm ²	Passed
Shear	:	7.9,	Allowable	:	93.2 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Pad Weld for Nozzle skipped, as Nozzle is not a Carbon Steel material.

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 24, tr = 2.62, c = 0.8 mm, E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.11$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-2 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 24, tr = 6.48, c = 0.8 mm, E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.28$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-2 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C



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Nozzle-Shell Weld for Nozzle skipped as Nozzle is not a Carbon Steel material.

Gov. MDMT of the Reinforcement Pad : -104 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 1302.0000 mm
 Parallel to Vessel Wall, opening length d 651.0000 mm
 Normal to Vessel Wall (Thickness Limit), pad side Tlwp 58.0000 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Reinforcement Areas for Large Nozzle (Per App 1-7(a)): (cm²)

AREA AVAILABLE, A1 to A6		No Pad	With Pad
Area Required [Internal Pressure] AR		48.766	48.766
Area Available in Shell A1		64.115	64.115
Area Available in Nozzle Wall A2		23.768	23.768
Area Available in Inward Nozzle A3		0.000	0.000
Area Available in Welds A4		4.000	4.000
Area Available in Pad A5		0.000	93.024
Area Available in Hub A6		0.000	0.000
TOTAL AREA AVAILABLE Atot		91.884	184.908

The area available without a pad is Sufficient.
 The area available with the given pad is Sufficient.

The Nozzle does not fall within the scope of the diameter requirements of App. 1-7(b). Therefore, the Membrane and Bending stress checks are not required.

Weld Size Calculations, Description: N1-Shell inlet

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm
 Intermediate Calc. for pad/shell Welds TminPad 19.0000 mm

Results Per UW-16.1:

Required Thickness Actual Thickness
 Nozzle Weld 6.0000 = Min per Code 14.1400 = 0.7 * Wo mm
 Pad Weld 9.5000 = 0.5*TminPad 14.1400 = 0.7 * Wp mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_r1 \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (73.1 - 128 + 2 \cdot 23.2 \cdot 1 \cdot (1 \cdot 23.2 - 8.43)) \cdot 106)$
 $= \max(0, -52229.30) \text{ Kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i \cdot \text{Can}/.707)^2 \cdot f_r2) \cdot S_v$
 $= (23.8 + 168 + 8 - 0 \cdot 1) \cdot 106$
 $= 216332.95 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_r1)) \cdot S_v$
 $= (23.8 + 0 + 4 + (10.8)) \cdot 106$
 $= 41728.40 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_r1)) \cdot S$



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$$= (23.8 + 0 + 8 + 168 + (10.8)) \cdot 106$$

$$= 227990.38 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$$

$$= (3.14/2.0) \cdot 914 \cdot 20 \cdot 0.49 \cdot 133$$

$$= 191186. \text{ Kgf}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$$

$$= (3.14/2.0) \cdot 1614 \cdot 20 \cdot 0.49 \cdot 106$$

$$= 269124. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.14 \cdot 446) \cdot (24 - 0.8) \cdot 0.7 \cdot 133$$

$$= 308784. \text{ Kgf}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.14/2) \cdot 914 \cdot 24 \cdot 0.74 \cdot 138$$

$$= 358956. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$$

$$= (3.14/2.0) \cdot 914 \cdot (22 - 0) \cdot 0.74 \cdot 133$$

$$= 317603. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (269124 + 308784) = 577908 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (191186 + 358956 + 317603 + 0) = 867745 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (Spew + Tngw + Sinw)$$

$$= (269124 + 317603 + 0) = 586727 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 577908 Kgf, must exceed W = 0 Kgf	or W1 = 216332 Kgf
Path 2-2 = 867745 Kgf, must exceed W = 0 Kgf	or W2 = 41728 Kgf
Path 3-3 = 586727 Kgf, must exceed W = 0 Kgf	or W3 = 227990 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1815 KPa.

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 51.7 KPa.

The Drop for this Nozzle is : 97.0928 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 421.0928 mm

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 868.00 mm

Warning:



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Review Appendix 1-7 (b)(3) for notes regarding externally applied loads to large nozzles and U-2(g).

Input Echo, WRC297 Item 1, Description: N1-Shell inlet:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	0.8000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	2250.000	mm
Shell Thickness	24.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-240 304	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 106.20	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	0.8000	mm
Nozzle Diameter	914.400	mm
Nozzle Thickness	24.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-312 TP304	
Nozzle Cold S.I. Allowable	SNmc 138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 133.20	N./mm ²

Diameter of Reinforcing Pad	1614.400	mm
Thickness of Reinforcing Pad	24.0000	mm

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.*

Design Internal Pressure	Dp	791.33	KPa.
Radial Load	P	-6883.56	Kgf
Circumferential Shear	Vc	4589.03	Kgf
Longitudinal Shear	Vl	4589.03	Kgf
Circumferential Moment	Mc	11559.91	Kg-m.
Longitudinal Moment	Ml	11559.91	Kg-m.
Torsional Moment	Mt	16514.19	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$$

$$= 914.4 + 2 \cdot 1.65 \cdot \sqrt{1149.4(24.0 - 0.8)}$$

$$= 1453.282 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2298.800	mm
Nozzle Outside Diameter	(d) =	914.400	mm
Vessel Thickness used	(T) =	47.200	mm
Nozzle Thickness used	(t) =	23.200	mm
T / t	=	2.034	
d / t	=	39.414	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.776$$



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 Nozzle Calcs.: N1-Shell inlet Nozl: 10 6:24pm Sep 8,2026

Nr/P = 0.065
 Mr/P = 0.073
 MØ/P = 0.042
 NØ/P = 0.150
 MrD/Mc = 0.172
 NrDL/Mc = 0.096
 MØd/Mc = 0.113
 NØDL/Mc = 0.199
 MrD/Ml = 0.086
 NrDL/Ml = 0.053
 MØD/Ml = 0.048
 NØDL/Ml = 0.268

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	7	-7	7	-7
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-14	-14	14	14
Outplane Bending (ML)	-15	15	15	-15
Normal Pressure Stress	18	19	18	19
Outplane Stress Summary	0	17	61	15

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	13	-13	13	-13
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-2	-2	2	2
Inplane Bending (ML)	-28	28	28	-28
Inplane Pressure Stress	9	9	9	9
Inplane Stress Summary	-7	23	56	-27

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2
Shear Stress Summary	3	3	1	1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	9	25	62	43

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	7	-7	7	-7
Outplane Membrane (Mc)	-11	-11	11	11
Outplane Bending (Mc)	-37	37	37	-37
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	9	9	9	9

Outplane Stress Summary	-27	32	70	-20
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	13	-13	13	-13
Inplane Membrane (Mc)	-5	-5	5	5
Inplane Bending (Mc)	-57	57	57	-57
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	18	19	18	19

Inplane Stress Summary	-29	60	96	-44
------------------------	-----	----	----	-----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	2	2	2	2

Shear Stress Summary	1	1	3	3
----------------------	---	---	---	---

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stress Intensities)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	30	60	97	44

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stresses in the hoop direction)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Hoop Membrane (P)	4	4	4	4
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-14	-14	14	14
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	17	18	17	18

Hoop Stress Summary	6	7	36	37
---------------------	---	---	----	----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stresses Normal to pipe cross-section)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Axial Membrane (P)	1	1	1	1



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Axial Bending (P)	30	-30	30	-30
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-8	-8	8	8
Axial Bending (ML)	-82	82	82	-82
Axial Pressure Stress	8	8	8	8
Axial Stress Summary	-49	53	130	-95

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside
Shear due to (Vc)	1	1	-1	-1
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	5	5	5	5
Shear Stress Summary	6	6	3	3

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Outside	Inside	Outside	Inside
Two * Max Shear Stress	58	54	130	133

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside
Hoop Membrane (P)	4	4	4	4
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-11	-11	11	11
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	17	18	17	18
Hoop Stress Summary	10	11	32	33

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	30	-30	30	-30
Axial Membrane (Mc)	-8	-8	8	8
Axial Bending (Mc)	-171	171	171	-171
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	8	8	8	8
Axial Stress Summary	-139	142	220	-184

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-1	-1	1	1
Shear due to Torsion	5	5	5	5
Shear Stress Summary	3	3	6	6



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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	150	142	220	218

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 914.4 + 2 \cdot 1.65 \cdot \sqrt{1137.4(24.0 - 0.8)}$$

$$= 1450.461 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2274.800 mm
Nozzle Outside Diameter	(d) =	1614.400 mm
Vessel Thickness used	(T) =	23.200 mm
Nozzle Thickness used	(t) =	373.200 mm
T / t	=	0.062
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 7.027$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.045
Mr/P	=	0.071
MØ/P	=	0.019
NØ/P	=	0.024
MrD/Mc	=	0.238
NrDL/Mc	=	0.091
MØd/Mc	=	0.066
NØDL/Mc	=	0.044
MrD/Ml	=	0.036
NrDL/Ml	=	0.035
MØD/Ml	=	0.012
NØDL/Ml	=	0.020

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	14	-14	14	-14
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-2	-2	2	2
Outplane Bending (ML)	-9	9	9	-9
Normal Pressure Stress	38	38	38	38
Outplane Stress Summary	43	34	67	20

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	5	5	5	5
Inplane Bending (P)	53	-53	53	-53
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-4	-4	4	4



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 Nozzle Calcs.: N1-Shell inlet Nozl: 10 6:24pm Sep 8,2026

Inplane Bending (ML)	-28	28	28	-28
Inplane Pressure Stress	19	19	19	19
Inplane Stress Summary	45	-5	110	-52

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE				
(Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1
Shear Stress Summary	2	2	0	0

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	47	39	110	73

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	14	-14	14	-14
Outplane Membrane (Mc)	-5	-5	5	5
Outplane Bending (Mc)	-51	51	51	-51
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	19	19	19	19
Outplane Stress Summary	-21	53	93	-38

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	5	5	5	5
Inplane Bending (P)	53	-53	53	-53
Inplane Membrane (Mc)	-11	-11	11	11
Inplane Bending (Mc)	-186	186	186	-186
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	38	38	38	38
Inplane Stress Summary	-101	165	295	-183

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	1	1	1	1
Shear Stress Summary	0	0	2	2

Vessel Stresses (N./mm²):

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CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	101	165	295	183

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	18	19	18	19	18	19	18	19
Circ. Pl (SUS)	-10	-10	18	18	-4	-4	6	6
Circ. Q (SUS)	-8	8	22	-22	-44	44	70	-70
Long. Pm (SUS)	9	9	9	9	9	9	9	9
Long. Pl (SUS)	-1	-1	3	3	-7	-7	15	15
Long. Q (SUS)	-15	15	41	-41	-30	30	44	-44
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	18.0	19.0	18.0	19.0	18.0	19.0	18.0	19.0
Pm+Pl (SUS)	8.0	9.0	36.0	37.0	14.0	15.0	24.0	25.0
Pm+Pl+Q (Total)	8.1	23.6	58.7	44.2	31.2	59.1	94.2	45.2

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	19.00	106.20	Passed
Pm+P1 (SUS)	37.00	159.30	Passed
Pm+P1+Q (TOTAL)	94.15	318.60	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was $3 * S_{mh}$.*

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

[illegible]



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Pm+Pl (SUS)	39.0	39.0	43.0	43.0	32.0	32.0	54.0	54.0
Pm+Pl+Q (Total)	45.6	39.1	109.0	73.0	101.0	165.0	293.0	185.0

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	38.00	106.20	Passed
Pm+Pl (SUS)	54.00	159.30	Passed
Pm+Pl+Q (TOTAL)	293.00	318.60	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	17	18	17	18	17	18	17	18
Circ. Pl (SUS)	-10	-10	18	18	-7	-7	15	15
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	8	8	8	8	8	8	8	8
Long. Pl (SUS)	-7	-7	9	9	-7	-7	9	9
Long. Q (SUS)	-52	52	112	-112	-141	141	201	-201
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	5	5	5	5	5	5	5	5
Pm (SUS)	17.0	18.0	17.0	18.0	17.0	18.0	17.0	18.0
Pm+Pl (SUS)	7.2	8.1	35.1	36.1	10.1	11.1	32.1	33.1
Pm+Pl+Q (Total)	59.2	53.8	129.2	131.2	150.2	142.1	218.2	217.3

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	18.00	133.20	Passed
Pm+Pl (SUS)	36.05	199.80	Passed
Pm+Pl+Q (TOTAL)	218.19	399.60	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



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 Nozzle Calcs.: N4-Shell outlet Nozl: 11 6:24pm Sep 8,2026

Input, Nozzle Desc: N4-Shell outlet

From: 50

Pressure for Reinforcement Calculations	P	813.389	KPa.
Temperature for Internal Pressure	Temp	130	°C
Design External Pressure	Pext	51.71	KPa.
Temperature for External Pressure	Tempex	130	°C
Parent Material		SA-240 304	
Parent Allowable Stress at Temperature	Sv	106.20	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2250.00	mm
Design Length of Section	L	8200.0000	mm
Shell Finished (Minimum) Thickness	t	24.0000	mm
Shell Internal Corrosion Allowance	c	0.8000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		5598.70	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

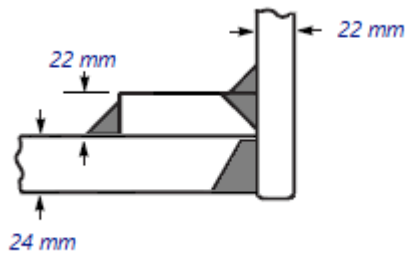
Type of Element Connected to the Parent : Nozzle

Material		SA-312 TP304	
Material UNS Number		S30400	
Material Specification/Type	Smls. & wld. pipe		
Allowable Stress at Temperature	Sn	133.20	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		180.00	deg
Diameter		36.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	22.0000	mm
Flange Material		SA-182 F304	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	0.8000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	20.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	22.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	1514.4000	mm
Thickness of Pad	te	22.0000	mm
Weld leg size between Pad and Shell	Wp	20.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	22.0000	mm
Reinforcing Pad Width		300.0000	mm
Flange Class		150	
Flange Grade		GR 2.1	
Flange Series		B	

The Pressure Design option was MAWP + static head (to the nozzle).



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 Nozzle Calcs.: N4-Shell outlet Nozl: 11 6:24pm Sep 8,2026



Reinforcement CALCULATION, Description: N4-Shell outlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	36.000 in.
Actual Thickness Used in Calculation	0.866 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 813 \cdot 1126 / (106 \cdot 1 - 0.6 \cdot 813)$
 $= 8.6629 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 813 \cdot 457 / (133 \cdot 1 + 0.4 \cdot 813)$
 $= 2.7853 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.4428 mm

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1743.9999 mm
Parallel to Vessel Wall, opening length	d	871.9999 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		58.0000 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 133 / 106)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 133 / 106)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138 / 106)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$



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 Nozzle Calcs.: N4-Shell outlet Nozl: 11 6:24pm Sep 8,2026

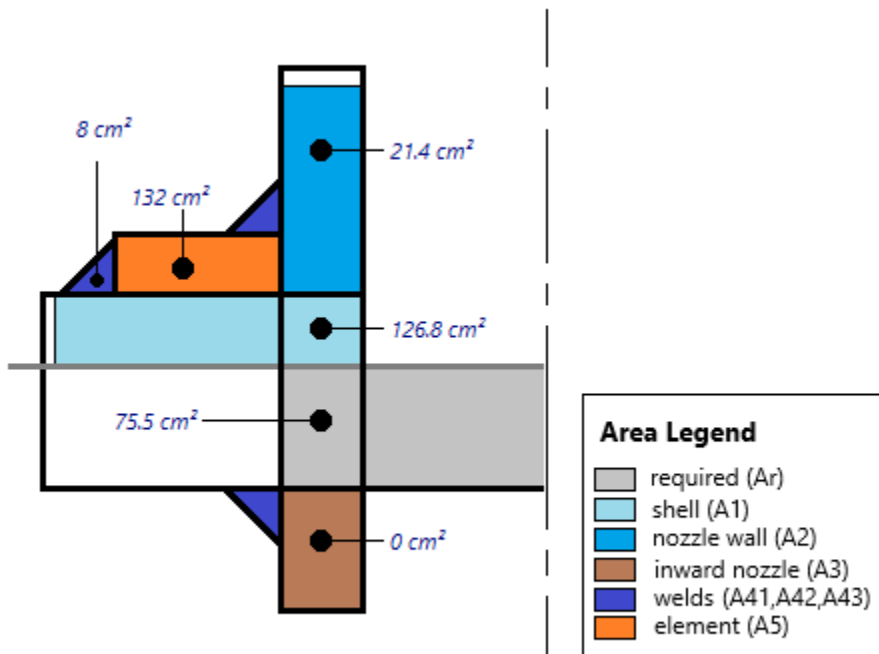


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	75.540	42.643	NA
Area in Shell	A1	126.764	117.017	NA
Area in Nozzle Wall	A2	21.361	22.918	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	8.000	8.000	NA
Area in Element	A5	132.000	132.000	NA
TOTAL AREA AVAILABLE	Atot	288.125	279.936	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.
 The area available with the given pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (872 \cdot 8.66 \cdot 1 + 2 \cdot 21.2 \cdot 8.66 \cdot 1 \cdot (1 - 1))$
 $= 75.540 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 872(1 \cdot 23.2 - 1 \cdot 8.66) - 2 \cdot 21.2$
 $(1 \cdot 23.2 - 1 \cdot 8.66) \cdot (1 - 1)$
 $= 126.764 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 58) \cdot (21.2 - 2.79) \cdot 1$



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= 21.361 cm²

Area Available in Welds [A41 + A42 + A43]:
 = $W_o^2 \cdot fr_3 + (W_i - can/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$
 = $20^2 \cdot 1 + (0)^2 \cdot 1 + 20^2 \cdot 1$
 = 8.000 cm²

Area Available in Element [A5]:
 = $(\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr_4$
 = $(1514 - 914) \cdot 22 \cdot 1$
 = 132.000 cm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.5853 mm
Wall Thickness per UG16(b),	tr16b = 2.3000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 9.4629 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 9.4629 mm
Wall Thickness, shell/head, external pressure	trb2 = 1.3484 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 2.3000 mm
Wall Thickness per table UG-45	tb3 = 9.1312 mm

Determine Nozzle Thickness candidate [tb]:
 = min[tb3, max(tb1, tb2)]
 = min[9.13, max(9.46, 2.3)]
 = 9.1312 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 = max(ta, tb)
 = max(3.59, 9.13)
 = 9.1312 mm

Available Nozzle Neck Thickness = 22.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	19.4,	Allowable	:	133.2 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	319.6 N./mm ²	Passed
Occasional	:	8.2,	Allowable	:	177.2 N./mm ²	Passed
Shear	:	8.6,	Allowable	:	93.2 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle to Flange Weld skipped as Nozzle is not a Carbon Steel material.

Nozzle-Pad Weld for Nozzle skipped, as Nozzle is not a Carbon Steel material.

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 22, tr = 2.69, c = 0.8 mm, E* = 1
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.13$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-5 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 22, c = 0.8 mm, E* = 1
 Thickness Ratio = $tr \cdot E^*/(tg - c) = -1$, Temp. Reduction = 0 °C
 Attachment governing, Attachment stress = Shell stress per Code



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Min Metal Temp. w/o impact per UCS-66, Curve B -5 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle-Shell Weld for Nozzle skipped as Nozzle is not a Carbon Steel material.

Gov. MDMT of the Reinforcement Pad : -104 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 1308.0000 mm
 Parallel to Vessel Wall, opening length d 654.0000 mm
 Normal to Vessel Wall (Thickness Limit), pad side Tlwp 58.0000 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Reinforcement Areas for Large Nozzle (Per App 1-7(a)): (cm²)

AREA AVAILABLE, A1 to A6		No Pad	With Pad
Area Required [Internal Pressure] AR		50.363	50.363
Area Available in Shell A1		63.382	63.382
Area Available in Nozzle Wall A2		19.520	21.361
Area Available in Inward Nozzle A3		0.000	0.000
Area Available in Welds A4		4.000	4.000
Area Available in Pad A5		0.000	86.592
Area Available in Hub A6		0.000	0.000
TOTAL AREA AVAILABLE Atot		86.902	175.335

The area available without a pad is Sufficient.
 The area available with the given pad is Sufficient.

The Nozzle does not fall within the scope of the diameter requirements of App. 1-7(b). Therefore, the Membrane and Bending stress checks are not required.

Weld Size Calculations, Description: N4-Shell outlet

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm
 Intermediate Calc. for pad/shell Welds TminPad 19.0000 mm

Results Per UW-16.1:

Required Thickness Actual Thickness
 Nozzle Weld 6.0000 = Min per Code 14.1400 = 0.7 * Wo mm
 Pad Weld 9.5000 = 0.5*TminPad 14.1400 = 0.7 * Wp mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_r1 \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (75.5 - 127 + 2 \cdot 21.2 \cdot 1 \cdot (1 \cdot 23.2 - 8.66)) \cdot 106)$
 $= \max(0, -48796.41) \text{ Kg}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4 - (W_i - C_{an}/.707)^2 \cdot f_r2) \cdot S_v$
 $= (21.4 + 132 + 8 - 0 \cdot 1) \cdot 106$
 $= 174740.88 \text{ Kg}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_r1)) \cdot S_v$
 $= (21.4 + 0 + 4 + (9.84)) \cdot 106$
 $= 38116.44 \text{ Kg}$



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Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$$

$$= (21.4 + 0 + 8 + 132 + (9.84)) \cdot 106$$

$$= 185393.33 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.14/2.0) \cdot 914 \cdot 20 \cdot 0.49 \cdot 133$$

$$= 191186. \text{ Kgf}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$$

$$= (3.14/2.0) \cdot 1514 \cdot 20 \cdot 0.49 \cdot 106$$

$$= 252454. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$$

$$= (3.14 \cdot 447) \cdot (22 - 0.8) \cdot 0.7 \cdot 133$$

$$= 282798. \text{ Kgf}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$$

$$= (3.14/2) \cdot 914 \cdot 22 \cdot 0.74 \cdot 138$$

$$= 329043. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gvi} \cdot 0.74 \cdot S_{ng}$$

$$= (3.14/2.0) \cdot 914 \cdot (22 - 0) \cdot 0.74 \cdot 133$$

$$= 317603. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (S_{PEW} + S_{NW}) = (252454 + 282798) = 535252 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (S_{onw} + T_{pgw} + T_{ngw} + S_{inw})$$

$$= (191186 + 329043 + 317603 + 0) = 837832 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (S_{pew} + T_{ngw} + S_{inw})$$

$$= (252454 + 317603 + 0) = 570057 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 535252 Kgf, must exceed W = 0 Kgf	or W1 = 174740 Kgf
Path 2-2 = 837832 Kgf, must exceed W = 0 Kgf	or W2 = 38116 Kgf
Path 3-3 = 570056 Kgf, must exceed W = 0 Kgf	or W3 = 185393 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1815 KPa.

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 51.7 KPa.

The Drop for this Nozzle is : 97.0928 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 421.0928 mm

This nozzle or its diameter limit intersects weld seam # [2] on this element. The distance between the nozzle and seam is, 552.93 mm , while the distance from the center



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of the nozzle to the edge of the diameter limit is: 872.00 mm

Warning:
 Review Appendix 1-7 (b)(3) for notes regarding externally applied loads
 to large nozzles and U-2(g).

Input Echo, WRC297 Item 1, Description: N4-Shell outle:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	0.8000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	2250.000	mm
Shell Thickness	24.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-240 304	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 106.20	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	0.8000	mm
Nozzle Diameter	914.400	mm
Nozzle Thickness	22.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-312 TP304	
Nozzle Cold S.I. Allowable	SNmc 138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 133.20	N./mm ²
Diameter of Reinforcing Pad	1514.400	mm
Thickness of Reinforcing Pad	22.0000	mm

Note:
 External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	813.39	KPa.
Radial Load	P	-6883.56	Kgf
Circumferential Shear	Vc	4589.03	Kgf
Longitudinal Shear	Vl	4589.03	Kgf
Circumferential Moment	Mc	11559.91	Kg-m.
Longitudinal Moment	Ml	11559.91	Kg-m.
Torsional Moment	Mt	16514.19	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$
 $= 914.4 + 2 \cdot 1.65 \cdot \sqrt{1148.4(24.0 - 0.8)}$
 $= 1453.047 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2296.800	mm
Nozzle Outside Diameter	(d) =	914.400	mm
Vessel Thickness used	(T) =	45.200	mm
Nozzle Thickness used	(t) =	21.200	mm
T / t	=	2.132	
d / t	=	43.132	



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$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.838$$

Nr/P = 0.054
 Mr/P = 0.068
 M0/P = 0.041
 N0/P = 0.155
 MrD/Mc = 0.165
 NrDL/Mc = 0.091
 M0d/Mc = 0.112
 N0DL/Mc = 0.210
 MrD/Ml = 0.078
 NrDL/Ml = 0.048
 M0D/Ml = 0.046
 N0DL/Ml = 0.271

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	8	-8	8	-8
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-16	-16	16	16
Outplane Bending (ML)	-16	16	16	-16
Normal Pressure Stress	19	20	19	20
Outplane Stress Summary	0	17	66	17

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	13	-13	13	-13
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-2	-2	2	2
Inplane Bending (ML)	-28	28	28	-28
Inplane Pressure Stress	9	9	9	9
Inplane Stress Summary	-6	23	56	-27

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2
Shear Stress Summary	3	3	2	2

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	9	25	66	45

Vessel Stresses (N./mm²):



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CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	8	-8	8	-8
Outplane Membrane (Mc)	-12	-12	12	12
Outplane Bending (Mc)	-40	40	40	-40
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	9	9	9	9
Outplane Stress Summary	-30	35	76	-21

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	13	-13	13	-13
Inplane Membrane (Mc)	-5	-5	5	5
Inplane Bending (Mc)	-60	60	60	-60
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	19	20	19	20
Inplane Stress Summary	-30	63	100	-45

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	2	2	2	2
Shear Stress Summary	2	2	3	3

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	32	63	101	46

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	5	5	5	5
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-16	-16	16	16
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	19	20	19	20
Hoop Stress Summary	8	9	41	42

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to	Au Top	Al Top	Bu Bottom	Bl Bottom
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pipe cross-section)	Outside	Inside	Outside	Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	37	-37	37	-37
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-8	-8	8	8
Axial Bending (ML)	-90	90	90	-90
Axial Pressure Stress	9	9	9	9
Axial Stress Summary	-50	55	147	-107

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE				
(Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	1	1	-1	-1
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	5	5	5	5
Shear Stress Summary	7	7	4	4

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stress Intensities	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	60	56	147	150

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses in the	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
hoop direction)				
Hoop Membrane (P)	5	5	5	5
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-12	-12	12	12
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	19	20	19	20
Hoop Stress Summary	11	12	37	38

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses Normal to	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
pipe cross-section)				
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	37	-37	37	-37
Axial Membrane (Mc)	-8	-8	8	8
Axial Bending (Mc)	-198	198	198	-198
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	9	9	9	9
Axial Stress Summary	-159	163	255	-216

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-1	-1	1	1
Shear due to Torsion	5	5	5	5



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Shear Stress Summary	4	4	7	7
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	171	163	255	254

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$
 $= 914.4 + 2 \cdot 1.65 \cdot \sqrt{1137.4(24.0 - 0.8)}$
 $= 1450.461 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2274.800 mm
Nozzle Outside Diameter	(d) =	1514.400 mm
Vessel Thickness used	(T) =	23.200 mm
Nozzle Thickness used	(t) =	321.200 mm
T / t	=	0.072
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 6.592$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.045
Mr/P	=	0.071
M0/P	=	0.019
N0/P	=	0.026
MrD/Mc	=	0.239
NrDL/Mc	=	0.092
M0d/Mc	=	0.067
N0DL/Mc	=	0.048
MrD/Ml	=	0.039
NrDL/Ml	=	0.036
M0D/Ml	=	0.012
N0DL/Ml	=	0.021

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	14	-14	14	-14
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	-2	-2	2	2
Outplane Bending (Ml)	-10	10	10	-10
Normal Pressure Stress	39	39	39	39
Outplane Stress Summary	43	35	69	21

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Inplane Membrane (P)	5	5	5	5
Inplane Bending (P)	53	-53	53	-53



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Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-5	-5	5	5
Inplane Bending (ML)	-32	32	32	-32
Inplane Pressure Stress	19	19	19	19
Inplane Stress Summary	41	0	115	-55

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE		Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)		Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	0	0	0	0	0
Outplane Shear (Vl)	0	0	0	0	0
Outplane Shear (Mt)	1	1	1	1	1
Shear Stress Summary	2	2	1	1	1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE		Au	Al	Bu	Bl
(Stress Intensities)		Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress	45	36	115	76	

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)		Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	3	3	3	3	3
Outplane Bending (P)	14	-14	14	-14	
Outplane Membrane (Mc)	-6	-6	6	6	
Outplane Bending (Mc)	-56	56	56	-56	
Outplane Membrane (ML)	0	0	0	0	
Outplane Bending (ML)	0	0	0	0	
Normal Pressure Stress	19	19	19	19	
Outplane Stress Summary	-25	57	100	-41	

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)		Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	5	5	5	5	
Inplane Bending (P)	53	-53	53	-53	
Inplane Membrane (Mc)	-12	-12	12	12	
Inplane Bending (Mc)	-199	199	199	-199	
Inplane Membrane (ML)	0	0	0	0	
Inplane Bending (ML)	0	0	0	0	
Inplane Pressure Stress	39	39	39	39	
Inplane Stress Summary	-114	178	309	-194	

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)		Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0	0
Outplane Shear (Vl)	0	0	0	0	0
Torsional Shear (Mt)	1	1	1	1	1



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 Nozzle Calcs.: N4-Shell outlet Nozl: 11 6:24pm Sep 8,2026

Shear Stress Summary 1 1 2 2

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	114	178	309	194

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	19	20	19	20	19	20	19	20
Circ. Pl (SUS)	-11	-11	21	21	-4	-4	6	6
Circ. Q (SUS)	-8	8	24	-24	-47	47	73	-73
Long. Pm (SUS)	9	9	9	9	9	9	9	9
Long. Pl (SUS)	-1	-1	3	3	-7	-7	17	17
Long. Q (SUS)	-15	15	41	-41	-32	32	48	-48
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	19.0	20.0	19.0	20.0	19.0	20.0	19.0	20.0
Pm+Pl (SUS)	8.0	9.0	40.0	41.0	15.0	16.0	26.0	26.0
Pm+Pl+Q (Total)	8.1	23.6	64.4	46.2	33.2	63.1	98.2	47.2

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	20.00	106.20	Passed
Pm+Pl (SUS)	41.00	159.30	Passed
Pm+Pl+Q (TOTAL)	98.17	318.60	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	39	39	39	39	39	39	39	39
Circ. Pl (SUS)	1	1	5	5	-7	-7	17	17
Circ. Q (SUS)	4	-4	24	-24	-146	146	252	-252
Long. Pm (SUS)	19	19	19	19	19	19	19	19
Long. Pl (SUS)	0	0	10	10	-3	-3	9	9
Long. Q (SUS)	21	-21	85	-85	-42	42	70	-70
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0



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Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	39.0	39.0	39.0	39.0	39.0	39.0	39.0	39.0
Pm+Pl (SUS)	40.0	40.0	44.0	44.0	32.0	32.0	56.0	56.0
Pm+Pl+Q (Total)	44.2	38.1	114.0	76.0	114.0	178.0	308.0	196.0

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	39.00	106.20	Passed
Pm+Pl (SUS)	56.00	159.30	Passed
Pm+Pl+Q (TOTAL)	308.00	318.60	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	19	20	19	20	19	20	19	20
Circ. Pl (SUS)	-11	-11	21	21	-7	-7	17	17
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	9	9	9	9	9	9	9	9
Long. Pl (SUS)	-7	-7	9	9	-7	-7	9	9
Long. Q (SUS)	-53	53	127	-127	-161	161	235	-235
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	5	5	5	5	5	5	5	5
Pm (SUS)	19.0	20.0	19.0	20.0	19.0	20.0	19.0	20.0
Pm+Pl (SUS)	8.2	9.1	40.0	41.0	12.1	13.1	36.1	37.1
Pm+Pl+Q (Total)	60.2	55.8	145.2	150.2	171.2	163.1	253.2	254.3

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	20.00	133.20	Passed
Pm+Pl (SUS)	41.04	199.80	Passed
Pm+Pl+Q (TOTAL)	254.28	399.60	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



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 Nozzle Calcs.: N6-Tube Outlet Nozl: 12 6:24pm Sep 8,2026

Input, Nozzle Desc: N6-Tube Outlet

From: 70

Pressure for Reinforcement Calculations	P	1103.525	KPa.
Temperature for Internal Pressure	Temp	82	°C
Design External Pressure	Pext	103.42	KPa.
Temperature for External Pressure	Tempex	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2250.00	mm
Design Length of Section	L	1000.0294	mm
Shell Finished (Minimum) Thickness	t	20.0000	mm
Shell Internal Corrosion Allowance	c	3.2000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		10234.70	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

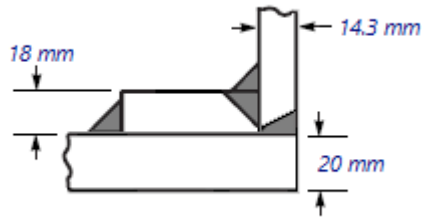
Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		180.00	deg
Diameter		18.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		40	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	0.0000	mm
Pad Material		SA-312 TP304	
Pad Allowable Stress at Temperature	Sp	137.51	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	657.2000	mm
Thickness of Pad	te	18.0000	mm
Weld leg size between Pad and Shell	Wp	14.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	18.0000	mm
Reinforcing Pad Width		100.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



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Reinforcement CALCULATION, Description: N6-Tube Outlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation 18.000 in.
 Actual Thickness Used in Calculation 0.562 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1104 \cdot 1128 / (138 \cdot 1 - 0.6 \cdot 1104)$
 $= 9.0659 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 1104 \cdot 229 / (118 \cdot 1 + 0.4 \cdot 1104)$
 $= 2.1300 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.0391 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	861.3008	mm
Parallel to Vessel Wall, opening length	d	430.6504	mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		42.0000	mm

Weld Strength Reduction Factor [fr1]:
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118 / 138)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138 / 138)$
 $= 0.996$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.86, 1)$
 $= 0.855$

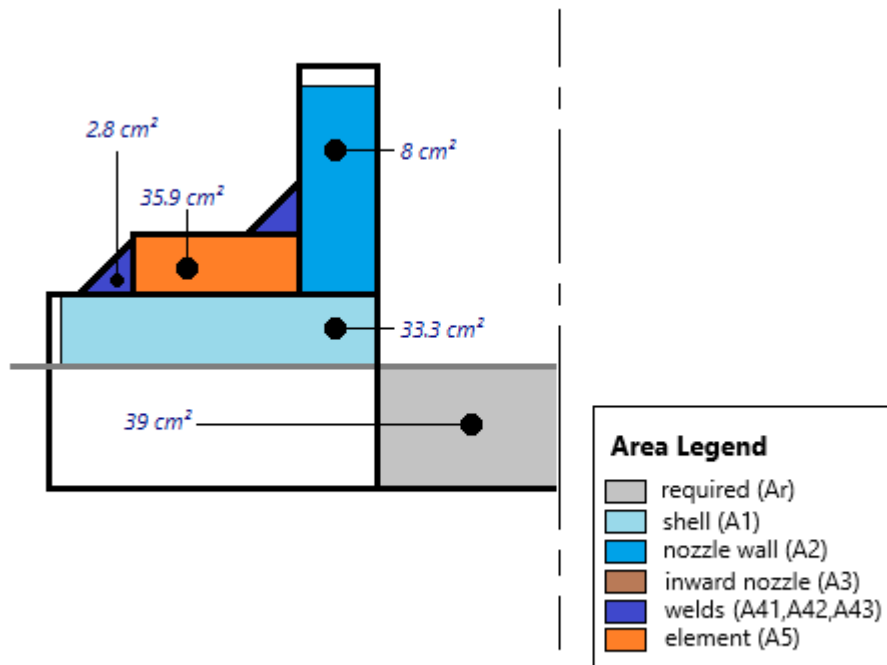


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	39.042	11.176	NA
Area in Shell	A1	33.307	49.997	NA
Area in Nozzle Wall	A2	8.005	8.788	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	2.808	2.808	NA
Area in Element	A5	35.873	35.873	NA
TOTAL AREA AVAILABLE	Atot	79.993	97.467	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.
 The area available with the given pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (431 \cdot 9.07 \cdot 1 + 2 \cdot 13.3 \cdot 9.07 \cdot 1 \cdot (1 - 1))$
 $= 39.042 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 431(1 \cdot 16.8 - 1 \cdot 9.07) - 2 \cdot 13.3$
 $(1 \cdot 16.8 - 1 \cdot 9.07) \cdot (1 - 1)$
 $= 33.307 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 42) \cdot (13.3 - 2.13) \cdot 0.86$



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= 8.005 cm²

Area Available in Welds [A41 + A42 + A43]:
 = $W_o^2 \cdot fr_3 + (W_i - can/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$
 = $10^2 \cdot 0.86 + (0)^2 \cdot 0.86 + 14^2 \cdot 1$
 = 2.808 cm²

Area Available in Element [A5]:
 = $(\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr_4$
 = $(657 - 457) \cdot 18 \cdot 1$
 = 35.873 cm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.1300 mm
Wall Thickness per UG16(b),	tr16b = 4.7000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 12.2659 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 12.2659 mm
Wall Thickness, shell/head, external pressure	trb2 = 4.0459 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.7000 mm
Wall Thickness per table UG-45	tb3 = 9.3312 mm

Determine Nozzle Thickness candidate [tb]:
 = min[tb3, max(tb1, tb2)]
 = min[9.33, max(12.3, 4.7)]
 = 9.3312 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 = max(ta, tb)
 = max(3.13, 9.33)
 = 9.3312 mm

Available Nozzle Neck Thickness = 0.875 * 14.275 = 12.490 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	26.9,	Allowable	:	118.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	268.1 N./mm ²	Passed
Occasional	:	8.7,	Allowable	:	156.9 N./mm ²	Passed
Shear	:	14.0,	Allowable	:	82.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 12.5, tr = 2.13, c = 1 mm, E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.19$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-1 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 12.5, tr = 2.13, c = 1 mm, E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.19$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-22 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C



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Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 12.5, tr = 2.13, c = 1 mm, E* = 1
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.19$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-22 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-43 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $1103.52/1960.01 = 0.563$

Weld Size Calculations, Description: N6-Tube Outlet

Intermediate Calc. for nozzle/shell Welds	Tmin	13.2748 mm
Intermediate Calc. for pad/shell Welds	TminPad	16.8000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$
Pad Weld	$8.4000 = 0.5 \cdot T_{minPad}$	$9.8980 = 0.7 \cdot W_p \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= (Ar - A1) \cdot Sv$
 $= (39 - 33.3) \cdot 138$
 $= 8070.65 \text{ Kg}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A5 + A4 - (Wi - Can / .707)^2 \cdot fr2) \cdot Sv$
 $= (8 + 35.9 + 2.81 - 0 \cdot 0.86) \cdot 138$
 $= 65695.05 \text{ Kg}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 457 \cdot 10 \cdot 0.49 \cdot 118$
 $= 42342. \text{ Kg}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$
 $= (3.14/2.0) \cdot 657 \cdot 14 \cdot 0.49 \cdot 138$
 $= 99301. \text{ Kg}$

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot Seg$
 $= (3.14/2) \cdot 457 \cdot 18 \cdot 0.74 \cdot 138$
 $= 134135. \text{ Kg}$

Strength of Failure Paths:

Failure Path [Path 1-1]:



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$$= (\text{SPEW} + \text{SNGW}) = (99301 + 0) = 99301 \text{ Kg}$$

Summary of Failure Path Calculations:

Path 1-1 = 99301 Kg, must exceed W = 8070 Kg or W1 = 65695 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1300 KPa.

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 103 KPa.

The Drop for this Nozzle is : 23.4706 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 243.4706 mm

Input Echo, WRC297 Item 1, Description: N6-Tube Outlet:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.2000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	2250.000	mm
Shell Thickness	20.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	1.0000	mm
Nozzle Diameter	457.200	mm
Nozzle Thickness	14.2748	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc 118.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 118.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	657.200	mm
Thickness of Reinforcing Pad	18.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	1103.52	KPa.
Radial Load	P	-3441.77	Kgf
Circumferential Shear	Vc	2294.50	Kgf
Longitudinal Shear	Vl	2294.50	Kgf
Circumferential Moment	Mc	2889.98	Kg-m.
Longitudinal Moment	Ml	2889.98	Kg-m.
Torsional Moment	Mt	4128.54	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:



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Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 457.2 + 2 \cdot 1.65 \cdot \sqrt{1145.6(20.0 - 3.2)}$$

$$= 915.010 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 2291.200 mm
 Nozzle Outside Diameter (d) = 457.200 mm
 Vessel Thickness used (T) = 34.800 mm
 Nozzle Thickness used (t) = 13.275 mm
 T / t = 2.622
 d / t = 34.441

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.619$$

Nr/P = 0.076
 Mr/P = 0.086
 M0/P = 0.065
 N0/P = 0.188
 MrD/Mc = 0.172
 NrDL/Mc = 0.107
 M0d/Mc = 0.155
 N0DL/Mc = 0.192
 MrD/Ml = 0.111
 NrDL/Ml = 0.077
 M0D/Ml = 0.093
 N0DL/Ml = 0.329

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	10	-10	10	-10
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-16	-16	16	16
Outplane Bending (ML)	-28	28	28	-28
Normal Pressure Stress	35	36	35	36
Outplane Stress Summary	5	42	96	18

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	14	-14	14	-14
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-3	-3	3	3
Inplane Bending (ML)	-33	33	33	-33
Inplane Pressure Stress	17	17	17	17
Inplane Stress Summary	-3	35	71	-24

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0



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Outplane Shear (V1)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3
Shear Stress Summary	4	4	2	2

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Two * Max Shear Stress	13	44	96	43

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	D1 Right Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	10	-10	10	-10
Outplane Membrane (Mc)	-9	-9	9	9
Outplane Bending (Mc)	-47	47	47	-47
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	17	17	17	17
Outplane Stress Summary	-23	49	91	-25

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	D1 Right Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	14	-14	14	-14
Inplane Membrane (Mc)	-5	-5	5	5
Inplane Bending (Mc)	-52	52	52	-52
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	35	36	35	36
Inplane Stress Summary	-6	71	110	-23

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	D1 Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (V1)	0	0	0	0
Torsional Shear (Mt)	3	3	3	3
Shear Stress Summary	2	2	4	4

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	D1 Right Inside
Two * Max Shear Stress	24	71	111	29

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
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Hoop Membrane (P)	5	5	5	5
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-16	-16	16	16
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	20	22	20	22
Hoop Stress Summary	9	10	42	44

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	54	-54	54	-54
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-14	-14	14	14
Axial Bending (ML)	-152	152	152	-152
Axial Pressure Stress	10	10	10	10
Axial Stress Summary	-99	95	233	-180

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	9	9	9	9
Shear Stress Summary	11	11	6	6

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	110	96	233	225

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	5	5	5	5
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-9	-9	9	9
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	20	22	20	22
Hoop Stress Summary	16	17	35	37

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	54	-54	54	-54
Axial Membrane (Mc)	-14	-14	14	14
Axial Bending (Mc)	-250	250	250	-250



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Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	10	10	10	10

Axial Stress Summary	-197	193	331	-278
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
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Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-2	-2	2	2
Shear due to Torsion	9	9	9	9

Shear Stress Summary	6	6	11	11
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
---	---------------	--------------	---------------	--------------

Two * Max Shear Stress	213	193	331	316
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Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 457.2 + 2 \cdot 1.65 \cdot \sqrt{1136.6(20.0 - 3.2)}$$

$$= 913.208 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D)	=	2273.200 mm
Nozzle Outside Diameter (d)	=	657.200 mm
Vessel Thickness used (T)	=	16.800 mm
Nozzle Thickness used (t)	=	113.275 mm
T / t	=	0.148
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 3.363$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.056
Mr/P	=	0.097
M0/P	=	0.029
N0/P	=	0.048
MrD/Mc	=	0.262
NrDL/Mc	=	0.099
M0d/Mc	=	0.078
N0DL/Mc	=	0.073
MrD/Ml	=	0.074
NrDL/Ml	=	0.066
M0D/Ml	=	0.022
N0DL/Ml	=	0.057

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	20	-20	20	-20
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0



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Outplane Membrane (ML)	-8	-8	8	8
Outplane Bending (ML)	-19	19	19	-19
Normal Pressure Stress	73	74	73	74

Outplane Stress Summary	71	71	128	48
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Inplane Membrane (P)	6	6	6	6
Inplane Bending (P)	69	-69	69	-69
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-10	-10	10	10
Inplane Bending (ML)	-67	67	67	-67
Inplane Pressure Stress	36	36	36	36

Inplane Stress Summary	35	31	190	-83
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3

Shear Stress Summary	4	4	2	2
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Two * Max Shear Stress	71	71	190	132
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	20	-20	20	-20
Outplane Membrane (Mc)	-11	-11	11	11
Outplane Bending (Mc)	-71	71	71	-71
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	36	36	36	36

Outplane Stress Summary	-20	82	146	-38
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	6	6	6	6
Inplane Bending (P)	69	-69	69	-69
Inplane Membrane (Mc)	-15	-15	15	15
Inplane Bending (Mc)	-240	240	240	-240
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0



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Inplane Pressure Stress	73	74	73	74
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Inplane Stress Summary	-105	236	405	-213
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)		Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	--	-----------------------	----------------------	------------------------	-----------------------

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	3	3	3	3

Shear Stress Summary	2	2	4	4
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)		Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Two * Max Shear Stress	105	236	405	213
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WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	35	36	35	36	35	36	35	36
Circ. Pl (SUS)	-11	-11	21	21	-3	-3	7	7
Circ. Q (SUS)	-18	18	38	-38	-38	38	66	-66
Long. Pm (SUS)	17	17	17	17	17	17	17	17
Long. Pl (SUS)	-1	-1	5	5	-4	-4	14	14
Long. Q (SUS)	-19	19	47	-47	-37	37	57	-57
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	35.0	36.0	35.0	36.0	35.0	36.0	35.0	36.0
Pm+Pl (SUS)	24.0	25.0	56.0	57.0	32.0	33.0	42.0	43.0
Pm+Pl+Q (Total)	10.8	44.0	94.4	44.4	24.5	71.4	108.4	27.9

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	36.00	138.00	Passed
Pm+Pl (SUS)	57.00	207.00	Passed
Pm+Pl+Q (TOTAL)	108.44	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of	Stress Values at
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Stress Int. Location	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	73	74	73	74	73	74	73	74
Circ. Pl (SUS)	-3	-3	13	13	-9	-9	21	21
Circ. Q (SUS)	1	-1	39	-39	-171	171	309	-309
Long. Pm (SUS)	36	36	36	36	36	36	36	36
Long. Pl (SUS)	-4	-4	16	16	-6	-6	16	16
Long. Q (SUS)	2	-2	136	-136	-51	51	91	-91
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	73.0	74.0	73.0	74.0	73.0	74.0	73.0	74.0
Pm+Pl (SUS)	70.0	71.0	86.0	87.0	64.0	65.0	94.0	95.0
Pm+Pl+Q (Total)	71.4	70.4	188.1	132.1	107.0	236.0	403.1	214.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	74.00	138.00	Passed
Pm+Pl (SUS)	95.02	207.00	Passed
Pm+Pl+Q (TOTAL)	403.06	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	20	22	20	22	20	22	20	22
Circ. Pl (SUS)	-11	-11	21	21	-4	-4	14	14
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	10	10	10	10	10	10	10	10
Long. Pl (SUS)	-13	-13	15	15	-13	-13	15	15
Long. Q (SUS)	-98	98	206	-206	-196	196	304	-304
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	9	9	9	9	9	9	9	9
Pm (SUS)	20.0	22.0	20.0	22.0	20.0	22.0	20.0	22.0
Pm+Pl (SUS)	12.6	14.6	41.2	43.2	19.4	21.4	34.4	36.4
Pm+Pl+Q (Total)	112.2	96.4	231.3	224.4	215.5	193.3	329.4	315.8

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	22.00	118.00	Passed



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Pm+Pl (SUS)	43.22	177.00	Passed
Pm+Pl+Q (TOTAL)	329.41	353.99	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Input Echo, Tubesheet Number 1, Description: Fixed Tubesheet

Shell Data:

Main Shell Description: main Shell 1

Shell Maximum Design Pressure	Psd,max	380.00 KPa.
Shell Maximum Operating Pressure	Psox,max	380.00 KPa.
Shell Minimum Operating Pressure	Psox,min	0.00 KPa.
Shell Thickness	ts	12.0000 mm
Shell Internal Corrosion Allowance	cas	0.8000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	2250.000 mm
Shell Circumferential Joint Efficiency	Esw	0.850
Shell Temperature for Internal Pressure	Ts	130.00 °C
Shell Material		SA-240 304

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		S30400
Shell Allowable Stress at Temperature	Ss	106.20 N./mm ²
Shell Allowable Stress at Ambient		138.00 N./mm ²

Channel Data:

Channel Description: LHS CH Shell

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	1000.00 KPa.
Channel Maximum Operating Pressure	Ptox,max	1000.00 KPa.
Channel Minimum Operating Pressure	Ptox,min	0.00 KPa.
Channel Thickness	tc	14.0000 mm
Channel Internal Corrosion Allowance	cac	3.2000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	2250.000 mm
Channel Design Temperature	TEMPC	82.00 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	138.00 N./mm ²
Channel Allowable Stress at Ambient		138.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	4500
Tube Wall Thickness	et	1.6500 mm
Tube Outside Diameter	D	19.0500 mm
Total Straight Tube Length	Lt	8300.00 mm
Straight Tube Length (bet. inner tubsht faces) L		8200.00 mm
Design Temperature of the Tubes		82.00 °C
Tube Material		SA-213 TP316L
Tube Material UNS Number		S31603
Is this a Welded Tube		No
Tube Material Specification used		Smls. tube
Tube Allowable Stress at Temperature		115.00 N./mm ²
Tube Allowable Stress At Ambient		115.00 N./mm ²
Tube Yield Stress At design Temperature	Syt	151.17 N./mm ²
Tube Pitch (Center to Center Spacing)	P	23.2100 mm
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	1.6500 mm
Groove Weld Leg	ag	1.6500 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	1107.050 mm



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Largest Center-to-Center Tube Distance	U1	159.7280 mm
Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	0.0000 mm
Area of Tube Layout	Ap	20443.4 cm ²
Perimeter of Tube Layout	Cp	6127.4 mm

Tubesheet Data:

Tubesheet TYPE: Fixed Tubesheet Exchanger, Conf A

Tubesheet Design Metal Temperature	T	82.00 °C
Tubesheet Material		SA-266 2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number		K03506
Tubesheet Allowable Stress at Temperature	S	138.00 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	138.00 N./mm ²
Thickness of Tubesheet	h	50.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	3.2000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	3.2000 mm
Tubesheet Outside Diameter	A	2278.000 mm

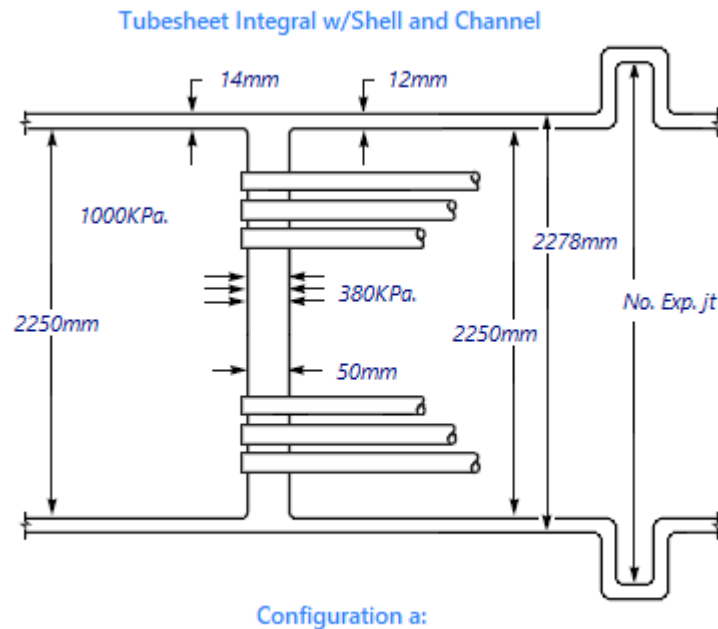
Area of the Untubed Lanes	AL	0.0 cm ²
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Additional Data for Fixed/Floating Tubesheet Exchangers:

Unsupported Tube Span under consideration	l	700.000 mm
Tube End condition corresponding to Span (l)	k	1.00
Ignore Radial Thermal Exp. effects (4.18.8/4.18.9)		Yes
Perform Differential Pressure Design		No
Run Multiple Load Cases		Yes
Shell Side Min. Design Pressure	Ps _{d,min}	-51.7095 KPa.
Channel Side Min. Design Pressure	P _{t,d,min}	-103.4190 KPa.
Mean Shell Metal Temp. along Shell len.	T _{sm}	0.00 °C
Mean Tube Metal Temp. along Tube length	T _{tm}	0.00 °C
Junction Stress Reduction option		Perform Plastic Calculation
Tubesheet Integral with		Both
Tubesheet Extended as Flange		No
Is Exchanger in Creep range (skip EP, Use 3S for Sps)		No



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ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties:

Shell - TE-1 Austenitic Group 3 Steels Group 1
 Shell - TM-1 Material Group G

Th. Exp. Coeff. Metal Temp. along Len	0.0 °C	0.0000152988 /°C
Elastic Mod. at Design Temperature	130.0 °C	0.18733E+09 KPa.
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000152988 /°C
Elastic Mod. at Metal Temp. along Len	0.0 °C	0.19650E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.19512E+09 KPa.

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
 Channel - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Design Temperature	82.0 °C	0.19921E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tubes - TE-1 Austenitic Group 3 Steels Group 1
 Tubes - TM-1 Material Group G

Th. Exp. Coeff. Metal Temp. along Len	0.0 °C	0.0000152988 /°C
Elastic Mod. at Design Temperature	82.0 °C	0.19047E+09 KPa.
Elastic Mod. at Metal Temp. along Len	0.0 °C	0.19650E+09 KPa.
Elastic Mod. at Tubsht. Design Temp.	82.0 °C	0.19047E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.19512E+09 KPa.

TubeSheet - TE-1 Carbon & Low Alloy Steels, Group 1
 TubeSheet - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	82.0 °C	0.19921E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Note:



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The Elasticity and Alpha values are taken from Tables in ASME II D.
 Please insure these properties are consistent with the
 type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:
 $= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= (1052 \cdot (19/2 - 0)) / (115 \cdot 1 + 0.4 \cdot 1052)$
 $= 0.0868 + 0.0000 = 0.0868 \text{ mm}$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart HA-4 at 82.00 °C
 Elastic Modulus for Material 188659120.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
1.6500	19.05	8200.00	11.55	50.0000	0.0082522	85.16

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 85.2) / (3 \cdot 11.5) = 9834 \text{ KPa.}$

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.2891	19.05	8200.00	65.90	50.0000	0.0002533	23.89

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 23.9) / (3 \cdot 65.9) = 483 \text{ KPa.}$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.2891 mm
Allowable Internal Pressure at Corroded thickness	21402.77 KPa.
Required Internal Design Pressure	1051.71 KPa.
Allowable External Pressure at Corroded thickness	9834.12 KPa.
Required External Design Pressure	483.42 KPa.
Required Thickness due to Shell Side pressure	0.2891 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Results for ASME Fixed Tubesheet Calculations for Configuration a:

Results for Tubesheet Calculations Original Thickness:

ASME Section VIII-2 Part 4.18.8.4 Step 1:

Tube expansion depth ratio $[\rho]$:
 $= l_{tx} / h$ (modified for corrosion if present)
 $= 0/50$
 $= 0.0000$ (must be $0 \leq \rho \leq 1$)

Effective tube hole diameter $[d^*]$:
 $= \max(dt - 2 \cdot tt(EtT/E) \cdot (StT/S) \cdot (\rho); dt - 2 \cdot tt)$
 $= \max(19 - 2 \cdot 1.65 (190467648/199214080) \cdot (115/138) \cdot (0); 19 - 2 \cdot 1.65)$
 $= 19.0500 \text{ mm}$

Equivalent outer tube limit circle diameter $[Do]$:

$= 2 \cdot ro + dt$
 $= 2 \cdot 1107 + 19$
 $= 2233.150 \text{ mm}$

Determine the basic ligament efficiency for shear $[\mu]$:

$= (p - dt) / p$
 $= (23.2 - 19) / 23.2$
 $= 0.179$



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Equivalent outer tube limit radius [ao]:

$$= Do / 2$$

$$= 2233/2$$

$$= 1116.5751 \text{ mm}$$

Effective tube pitch [p*]:

$$= p / \sqrt{(1 - 4 \cdot \min(AL \cdot CNVfactor; 4 \cdot Do \cdot p) / (\pi \cdot Do^2))}$$

$$= 23.2 / \sqrt{(1 - 4 \cdot \min(0 \cdot 100 ; 4 \cdot 2233 \cdot 23.2) / (\pi \cdot 2233^2))}$$

$$= 23.2100 \text{ mm}$$

Effective ligament efficiency for bending [μ^*]:

$$= (p^* - d^*) / p^*$$

$$= (23.2 - 19) / 23.2$$

$$= 0.1792$$

Ratio [ps]:

$$= as / ao$$

$$= 1125/1117$$

$$= 1.0075$$

Ratio [pc]:

$$= ac / ao$$

$$= 1125/1117$$

$$= 1.0075$$

Compute parameter [xt]:

$$= 1 - Nt / ((dt - 2 \cdot tt) / (2 \cdot ao))^2$$

$$= 1 - 4500 / ((19 - 2 \cdot 1.65) / (2 \cdot 1117))^2$$

$$= 0.7762$$

Determine parameter [xs]:

$$= 1 - Nt / (dt / (2 \cdot ao))^2$$

$$= 1 - 4500 / (19 / (2 \cdot 1117))^2$$

$$= 0.6725$$

Determine the value [h'g]:

$$= \max((hg - CATC); 0) \text{ (For pressure only cases)}$$

$$= \max((0 - 0); 0)$$

$$= 0.000 \text{ mm}$$

ASME Section VIII-2 Part 4.18.8.4 Step 2:

Axial Shell Stiffness [Ks]:

$$= \pi \cdot ts (Ds + ts) Es / L$$

$$= \pi \cdot 12 (2250 + 12) 187327008 / 8200$$

$$= 198658.3438 \text{ Kg/mm}$$

Axial Tube Stiffness [Kt]:

$$= \pi \cdot tt (Dt - tt) Et / L$$

$$= \pi \cdot 1.65 (19 - 1.65) 190467648 / 8200$$

$$= 213.6422 \text{ Kg/mm}$$

Stiffness Factor [Ks,t]:

$$= Ks / (Nt \cdot Kt)$$

$$= 198658 / (4500 \cdot 214)$$

$$= 0.20664$$

Rigidity Ratio [J]:

$$= 1 / (1 + Ks / Kj)$$

$$= 1 / (1 + 198658 / 0)$$

$$= 1.0000 (= 1 \text{ if No Exp. Jt.})$$

Shell coefficient [β_s]:

$$= ((12(1 - \nu_s^2))^{1/4}) / ((Ds + ts) ts)$$

$$= ((12(1 - 0.31^2))^{1/4}) / ((2250 + 12) 12)$$



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= 0.0110 1/mm

Shell coefficient [ks]:

$$= \beta_s \cdot E_s \cdot t_s^3 / (6(1 - \nu_s^2))$$

$$= 0.011 \cdot 187327008 \cdot 12^3 / (6(1 - 0.31^2))$$

$$= 657447360.0000 \text{ KPa} \cdot \text{mm}^2$$

Shell coefficient [λs]:

$$= 6(D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$

$$= 6(2250 + 12) \cdot 657447360 / 50^3 \cdot (1 + 50 \cdot 0.011 + (50 \cdot 0.011)^2 / 2)$$

$$= 121523904.0000 \text{ KPa}$$

Shell coefficient [δs]:

$$= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot \nu_s / (2(D_s + t_s)))$$

$$= 2250^2 / (4 \cdot 187327008 \cdot 12) \cdot (1 - 2250 \cdot 0.31 / (2(2250 + 12)))$$

$$= 0.47619\text{E}+00 \text{ mm/N} \cdot \text{mm}^2$$

Channel coefficient [βc]:

$$= ((12(1 - \nu_c^2))^{1/4}) / \sqrt{(D_c + t_c)t_c}$$

$$= ((12(1 - 0.3^2))^{1/4}) / \sqrt{(2250 + 14)14}$$

$$= 0.0102 \text{ 1/mm}$$

Channel coefficient [kc]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - \nu_c^2))$$

$$= 0.01 \cdot 199214080 \cdot 14^3 / (6(1 - 0.3^2))$$

$$= 1022268480.0000 \text{ KPa} \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λc]:

$$= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (Cm\Delta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot Cq\Delta)$$

$$= 6(2250 + 14 / \cos(0)) \cdot 1022268480 / 50^3 \cdot (1 + 50 \cdot 0.01 \cdot (1 + 1) / 2 + (50 \cdot 0.01)^2 / 2 \cdot 1)$$

$$= 182285728.0 \text{ KPa}$$

Channel coefficient [δc]:

$$= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot \nu_c / (2(D_c + t_c)))$$

$$= 2250^2 / (4 \cdot 199214080 \cdot 14) \cdot (1 - 2250 \cdot 0.3 / (2(2250 + 14)))$$

$$= 0.38612\text{E}+00 \text{ mm/N} \cdot \text{mm}^2$$

Channel coefficient [zetac]:

$$zetac = 0$$

ASME Section VIII-2 Part 4.18.8.4 Step 3:

$h/p = 2.154244$; $\mu^* = 0.179233$; $E^* = 26176462 \text{ KPa}$
 $E^*/E = 0.131399$; $\nu^* = 0.491147$

Note: As h/p (2.154) is > 2, data values for $h/p = 2$ were used.

Tube Bundle Stiffness Factor [Xa]:

$$= ((24(1 - \nu^*)Nt \cdot Et \cdot t_t(dt - t_t)ao^2) / (E^* \cdot L \cdot H^3))^{1/4}$$

$$= ((24(1 - 0.49^2)4500 \cdot 190467648 \cdot 1.65(19 - 1.65)1117^2) / (26176462 \cdot 8200 \cdot 50^3))^{1/4}$$

$$= 12.0125$$

Values from Table 4.18.3

$$Z_d = 0.000838 ; Z_v = 0.006899 ; Z_m = 0.117549$$

$$Z_a = 0.214350\text{E}+06 ; Z_w = 0.006899$$

ASME Section VIII-2 Part 4.18.8.4 Step 4:

Diameter ratio [K]:



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$$= A / Do$$

$$= 2278/2233$$

$$= 1.0201$$

Compute coefficient [F]:

$$= (1 - v^*) / E^* \cdot (\lambda s + \lambda c + E \cdot \ln(K))$$

$$= (1 - 0.49) / 26176462 \cdot (121523904 + 182285728 + 199214080 \cdot \ln(1.02))$$

$$= 5.9829$$

Compute parameter [ϕ]:

$$= (1 + v^*) F$$

$$= (1 + 0.49) 5.98$$

$$= 8.9213$$

Compute parameter [Q1]:

$$= (\psi s - 1 - \phi \cdot Zv) / (1 + \phi \cdot Zm)$$

$$= (1.01 - 1 - 8.92 \cdot 0.0069) / (1 + 8.92 \cdot 0.12)$$

$$= -0.026360$$

Compute Parameter [Qz1]:

$$= (Zd + Q1 \cdot Zw) / 2 \cdot Xa^4$$

$$= (0.00084 + -0.026 \cdot 0.0069) / 2 \cdot 12^4$$

$$= 6.8303$$

Compute parameter [Qz2]:

$$= (Zv + Q1 \cdot Zm) / 2 \cdot Xa^4$$

$$= (0.0069 + -0.026 \cdot 0.12) / 2 \cdot 12^4$$

$$= 39.5683$$

Compute parameter [U]:

$$= (Zw + (\psi s - 1) Zm) Xa^4 / (1 + \phi \cdot Zm)$$

$$= (0.0069 + (1.01 - 1) 0.12) 12^4 / (1 + 8.92 \cdot 0.12)$$

$$= 79.1365$$

ASME Section VIII-2 Part 4.18.8.4 Step 5:

Determine factor [γ_b]:

$$= 0 \text{ (config a)}$$

Compute Parameter [γ]:

$$= 0.000 \text{ mm (For Pressure only cases)}$$

Calculate parameter [ωs]:

$$= \psi s \cdot k_s \cdot \beta s \cdot \delta S (1 + h \cdot \beta s)$$

$$= 1.01 \cdot 657447360 \cdot 0.011 \cdot 0.47618675 (1 + 50 \cdot 0.011)$$

$$= 5388.3989 \text{ mm}^2$$

Calculate parameter [$\omega^* s$]:

$$= a o^2 (\psi s^2 - 1) \cdot (\psi s - 1) / 4 - \omega s$$

$$= 1117^2 (1.01^2 - 1) \cdot (1.01 - 1) / 4 - 5388$$

$$= -5352.7754 \text{ mm}^2$$

Parameter [ωc]:

$$= \psi c \cdot k_c [\delta C (CmTheta \cdot \beta C + Cq\Delta \cdot h \cdot \beta C^2) - \zeta c (CmTheta + h \cdot CqTheta \cdot h \cdot \beta C / 2)]$$

$$= 1.01 \cdot 1022268480 [0.38612181 (1 + 0.01 \cdot 1 \cdot 50) - 0.000 (1/50 + 1 \cdot 0.01/2)]$$

$$= 6134.2666 \text{ mm}^2$$

Parameter [$\omega^* c$]:

$$= a o^2 [(\psi c^2 + 1) \cdot (\psi c - 1) / 4 - (\psi s - 1) / 2] - \omega c$$

$$= 1117^2 [(1.01^2 + 1) \cdot (1.01 - 1) / 4 - (1.01 - 1) / 2] - 6134$$

$$= -6098.6431 \text{ mm}^2$$



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Pressure [P*S]:
 = 0 For Pressure only cases or Configurations d,e,f,A,B,C,D

Pressure [P*C]:
 = 0 For Pressure only cases or Configurations b,c,d,B,C,D

ASME Section VIII-2 Part 4.18.8.4 Step 6:

Pressure [P's]:

$$= P_s \{ x_s + 2(1 - x_s) v_t + [2/K_{st} \cdot (D_s/D_o)^2] v_s - [(p_s^2 - 1)/(J \cdot K_{st})] - [(1 - J)/(2J \cdot K_{st})][(D_j^2 - (D_s)^2)/D_o^2] \}$$

$$= 380 \{ 0.67 + 2(1 - 0.67)0.31 + [2/0.21 \cdot (2250/2233)^2]0.31 - [(1.01^2 - 1)/(1 \cdot 0.21)] - [(1 - 1)/(2 \cdot 1 \cdot 0.21)][(0^2 - (2250)^2)/2233^2] \}$$

 = 1462.2941 KPa.

Pressure [P't]:

$$= [x_t + 2(1 - x_t) v_t + 1/(J \cdot K_{st})] P_t$$

$$= [0.78 + 2(1 - 0.78)0.31 + 1/(1 \cdot 0.21)] 1000$$

 = 5754.3535 KPa.

Pressure [P_γ]:

$$= N_t \cdot K_t \cdot \gamma / (\pi \cdot a_o^2)$$

$$= 4500 \cdot 214 \cdot 0 / (3.14 \cdot 1117^2)$$

 = 0.0000 KPa.

Pressure [P_w]:

$$= -\gamma_b \cdot U \cdot W^* / (2 \cdot \pi \cdot a_o^2)$$

$$= -0 \cdot 79.1 \cdot 0 / (2 \cdot \pi \cdot 1117^2)$$

 = -0.0000 KPa.

Calculate the pressure [Prim]:

$$= -(U/a_o^2) \cdot (\omega_s \cdot P_s - \omega_c \cdot P_t)$$

$$= -(79.1/1117^2) \cdot (-5353 \cdot 380 - 6099 \cdot 1000)$$

 = -257.9988 KPa.

Calculate the pressure [P_ω]:

$$= (U/a_o^2) \cdot (\omega_s \cdot P_s - \omega_c \cdot P_c)$$

$$= (79.1/1117^2) \cdot (5388 \cdot 0 - 6134 \cdot 0)$$

 = 0.0000 KPa.

Determine the effective pressure [Pe]:

$$= J \cdot K_{st} / (1 + J \cdot K_{st} (Q_{z1} + (p_s - 1) Q_{z2})) \cdot (P_s - P_t + P_\gamma + P_\omega + P_w + P_{\text{Prim}})$$

$$= 0.1000E+01 \cdot 0.21 / (1 + 1 \cdot 0.21 (6.83 + (1.01 - 1)39.6)) \cdot (1462 - 5754 + 0 + 0 + -0 + -258)$$

 = -380.1765 KPa.

ASME Section VIII-2 Part 4.18.8.4 Step 7:

Determine factor [Q2]:

$$= [((\omega_s \cdot P_s - \omega_c \cdot P_t) - (\omega_s \cdot P_s - \omega_c \cdot P_c)) + W^* \cdot \gamma_b / (2 \cdot \pi)] / (1 + \phi \cdot Z_m)$$

$$= [((-5353 \cdot 380 - 6099 \cdot 1000) - (5388 \cdot 0 - 6134 \cdot 0)) \cdot 0.0001 + 0 \cdot 0 / (2 \cdot \pi)] / (1 + 8.92 \cdot 0.12)$$

 = 202.3187 Kg_f

Calculate factor [Q3]:

$$= Q_1 + 2 \cdot Q_2 / (P_e \cdot a_o^2)$$

$$= -0.026 + 2 \cdot 202 / (-380 \cdot 1117^2)$$

 = -0.03473

Fm Value from Table 4.18.3 = 0.017366



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Tubesheet Bending Stress - original thickness [σ]:

$$= (1.5 \cdot F_m / \mu \cdot) \cdot (2 \cdot a_o / (H - h'g))^2 \cdot P_e$$

$$= (1.5 \cdot 0.017 / 0.18) \cdot (2 \cdot 1117 / (50 - 0))^2 \cdot -380$$

$$= -110.2250 \text{ N./mm}^2$$

Allowable Tubesheet Bending Stress [σ allowed]:

$$= 1.5 \cdot S = 1.5 \cdot 138 = 207 \text{ N./mm}^2$$

Tubesheet Bending Stress - final thickness [σ_f]:

$$= (1.5 \cdot F_m / \mu \cdot) \cdot (2 \cdot a_o / (h - h'g))^2 \cdot P_e$$

$$= (1.5 \cdot 0.017 / 0.18) \cdot (2 \cdot 1117 / (34 - 0))^2 \cdot -347$$

$$= -206.9945 \text{ N./mm}^2$$

Reqd Tubesheet Thickness, for Bending Stress (including CA) [HReqB]:

$$= h + C_{ats} + C_{atc}$$

$$= 34 + 0 + 0$$

$$= 34.0049 \text{ mm}$$

ASME Section VIII-2 Part 4.18.8.4 Step 8:

Shear Stress check [τ_{limit}]:

$$= (2 \cdot \mu \cdot h / a_o) \cdot \min(0.8S; 0.533S_y)$$

$$= (2 \cdot 0.18 \cdot 50 / 1117) \cdot 110$$

$$= 1.7721 \text{ N./mm}^2$$

Shear Stress is not required to be computed; [P_e] <= τ_{limit}

Could not converge on a minimum required tubesheet thk. to satisfy the Shear Stress criteria.



Use the best Estimated Required Tubesheet Thickness value as design guidance & Compare the Actual Stresses versus Allowable Stresses.

Tubesheet thickness = 6.4107 mm
 Tubesheet Shear Stress = 53.3976 N./mm²

Required Tubesheet Thickness for Given Loadings (Including CA) [H reqd]:

$$= \max(H_{reqB}, H_{reqS})$$

$$= \max(34, 6.41)$$

$$= 34.0049 \text{ mm}$$

ASME Section VIII-2 Part 4.18.8.4 Step 9:

The F_{tmin} and F_{tmax} Coefficients from Table 4.18.4:

$$F_{tmin} = -0.38, F_{tmax} = 6.23$$

First Extreme Tube Axial Stress from among all the tubes [σ_{t1}]:

$$= ((P_s \cdot x_s - P_t \cdot x_t) - P_e \cdot F_{tmin}) / (X_t - X_s)$$

$$= ((380 \cdot 0.67 - 1000 \cdot 0.78) - (-380) \cdot -0.38) / (0.78 - 0.67)$$

$$= -6.4147 \text{ N./mm}^2$$

Second Extreme value of Tube Axial Stress from among all the tubes [σ_{t2}]:

$$= ((P_s \cdot x_s - P_t \cdot x_t) - P_e \cdot F_{tmax}) / (X_t - X_s)$$

$$= ((380 \cdot 0.67 - 1000 \cdot 0.78) - (-380) \cdot 6.23) / (0.78 - 0.67)$$

$$= 17.8297 \text{ N./mm}^2$$

Maximum Tube Axial Stress [$\sigma_{t,max}$]:

$$= \max(|\sigma_{t,1}|; |\sigma_{t,2}|) = 17.8 \text{ N./mm}^2$$



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Allowable Tube Stress [σ_t, A]:
 = Sot = 114.9979 N./mm²

Check for Buckling as some of the tubes are in compression.

Determine the Factor of Safety [Fs]:
 = max((3.25 - 0.25(Zd + Q3 • Zw)Xa⁴); 1.25)
 = max((3.25 - 0.25(0.00084 + -0.035 • 0.0069)12⁴); 1.25)
 = 1.2500 (Should be ≤ 2)

Determine the factor [rt]:

$$= \left(\sqrt{\frac{dt^2}{4} + \left(\frac{dt}{2} - \frac{tt}{2} \right)^2} \right) / 4$$

$$= \left(\sqrt{\frac{19^2}{4} + \left(\frac{19}{2} - \frac{1.65}{2} \right)^2} \right) / 4$$

 = 6.1794 mm

Determine the factor [Ct]:

$$= \sqrt{\frac{2 \cdot \pi^2 \cdot Et / Syt}{}}$$

$$= \sqrt{\frac{2 \cdot \pi^2 \cdot 190467648 / 151}{}}$$

 = 157.7078

Determine the factor [Ft]:
 = k • L/r
 = 1 • 700/6.18
 = 113.2791

Buckling Allowable Stress [Stb]:
 = (Sy,t/Fs) • (1 - Ft/(2 • Ct))
 = (151/1.25) • (1 - 113/(2 • 158))
 = 77.503 N./mm² (Never greater than Sot)

Note: The Axial Compressive stress in tubes is within limits.

Largest tube-to-tubesheet Joint Load [Wt]:
 = $\sigma_{t,max} \cdot \text{Tube Area}$
 = 17.8 • 0.9
 = 163.98 Kgf

Tube Metal Area [At]:

$$= \pi (d_o - t) t$$

$$= \pi (19 - 1.65) 1.65$$

 = 0.902 cm²

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$= \pi \cdot t (d_o - t) S$$

$$= \pi \cdot 1.65 (19 - 1.65) 115$$

 = 1057.656 Kgf

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t)$$

$$= 0.85 \cdot \pi \cdot 1.65 (19 + 0.67 \cdot 1.65) 115$$

 = 1041.3762 Kgf

Weld Strength Factor [fw]:
 = S / S_w where S_w = min(S and S_t)
 = 115/115
 = 1.000

Design Strength Ratio [fd]:
 = 1 for full strength welds

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t)$$



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$$= 0.55 \cdot \pi \cdot 1.65 (19 + 0.67 \cdot 1.65) 115$$

$$= 673.8317 \text{ Kgf}$$

Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{(0.75 \cdot do)^2 + 1.07 \cdot t \cdot (do - t) \cdot fw \cdot fd} - 0.75 \cdot do \right)$$

$$= 2 \left(\sqrt{(0.75 \cdot 19)^2 + 1.07 \cdot 1.65 \cdot (19 - 1.65) 1 \cdot 1} - 0.75 \cdot 19 \right)$$

$$= 2.075 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.0374 mm

Minimum Required Groove Weld Leg [agr]: 1.0374 mm

Tube-Tubesheet Jt allowable, 1058 is >= tube strength 1058 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

ASME Section VIII-2 Part 4.18.8.4 Step 10:

Shell Axial Membrane Allowable Stress:

$$= S_s \cdot E_{sw} = 106 \cdot 0.85 = 90.3 \text{ N./mm}^2$$

Axial Membrane Stress in Shell [σ_{s,m}]:

$$= a_o^2 / ((D_s + t_s) t_s) \cdot [P_e + (p_s^2 - 1)(P_s - P_t)] + a_s^2 \cdot P_t / ((D_s + t_s) t_s)$$

$$= 1117^2 / ((2250 + 12) 12) \cdot [-380 + (1.01^2 - 1)(380 - 1000)] + 1125^2 \cdot 1000 / ((2250 + 12) 12)$$

$$= 28.7349 \text{ N./mm}^2$$

ASME Section VIII-2 Part 4.18.8.4 Step 11:

Note:

For a given Shell thickness of 12 mm, the minimum Shell length adjacent to the tubesheet should be 295.770 mm

Shell Membrane Stress due to Joint Interaction [σ_{s,m}]:

$$= a_o^2 / ((D_s + t_s) t_s) [P_e + (p_s^2 - 1)(P_s - P_t)] + a_s^2 \cdot P_t / ((D_s + t_s) t_s)$$

$$= 1117^2 / ((2250 + 12) 12) [-380 + (1.01^2 - 1)(380 - 1000)] + 1125^2 \cdot 1000 / ((2250 + 12) 12)$$

$$= 28.7349 \text{ N./mm}^2$$

Shell Bending Stress due to Joint Interaction [σ_{s,b}]:

$$= 6 \cdot k_s / t_s^2 \{ \beta_s [\delta_s \cdot P_s + a_s^2 \cdot P \cdot s / (E_s \cdot t_s)] + 6(1 - \nu^2) / (E \cdot h) (a_o / h)^3 [1 + h \cdot \beta_s / 2] [P_e (Z_v + Z_m \cdot Q_1) + 2 / a_o^2 \cdot Z_m \cdot Q_2] \}$$

$$= 6 \cdot 657447360 / 12^2 \{ 0.011 [0.47618675 \cdot 380 + 1125^2 \cdot 0 / (2.25E+09)] + 6(1 - 0.49^2) / (26176462) (1117 / 50)^3 [1 + 50 \cdot 0.011 / 2] [-380 (0.0069 + 0.12 \cdot -0.026) + 2 / 1117^2 \cdot 0.12 \cdot 202] \}$$

$$= -17.8548 \text{ N./mm}^2$$

Shell Stress Summation vs. Allowable

$$\therefore |\sigma_{s,m}| + |\sigma_{s,b}| \leq 1.5 \cdot S_s$$

$$\therefore |28.7| + |-17.9| \leq 159 \text{ N./mm}^2$$

$$\therefore 46.6, \text{ must be } \leq 159 \text{ N./mm}^2$$

Note:

For a given Channel thickness of 14 mm, the minimum Channel length adjacent to the tubesheet should be 319.468 mm

Channel Membrane Stress due to Joint Interaction [σ_{c,m}]:

$$= a_c^2 \cdot P_t / ((D_c + t_c) t_c \cdot \cos(\theta))$$

$$= 1125^2 \cdot 1000 / ((2250 + 14) 14 \cdot \cos(0))$$

$$= 39.932 \text{ N./mm}^2$$

Channel Bending Stress due to Joint Interaction [σ_{c,b}]:



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$$\begin{aligned}
 &= 6 \cdot kc/tc^2 \cdot \{ (\delta_c \cdot Cm\Delta \cdot \beta_c - \zeta_c \cdot Cm\Theta) \cdot Pt \\
 &\quad - 6((1 - \nu^2)/E^*) \cdot (ao/h)^3 \cdot (Cm\Theta + \\
 &\quad h \cdot \beta_c/2 \cdot Cm\Delta) \cdot [Pe(Zv + Zm \cdot Q1) + 2/ao^2 \cdot Zm \cdot Q2] \} \\
 &= 6 \cdot 1022268480/14^2 \cdot \{ (0.38612181 \cdot 1 \cdot 0.01 - 0.000 \cdot 1) \cdot 1000 \\
 &\quad - 6((1 - 0.49^2)/26176462) \cdot \\
 &\quad (1117/50)^3 \cdot (1 + 50 \cdot 0.01/2 \cdot 1) \cdot [-380 (0.0069 + 0.12 \\
 &\quad \cdot -0.026) + 2/1117^2 \cdot 0.12 \cdot 202] \} \\
 &= 204.8660 \text{ N./mm}^2
 \end{aligned}$$

Channel Stress Summation vs. Allowable

:: | $\sigma_{c,m}$ | + | $\sigma_{c,b}$ | $\leq$ 1.5 $\cdot$ Sc
 :: |39.9| + |205| $\leq$ 207 N./mm²
 :: 245 must be $\leq$ 207 N./mm² (See Elast-Plast iteration below)

Stress/Force Summary for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet Bend. Stress	110.2 <=	207.0 N./mm ²	Pass
Tubesheet Shear Stress	14.2 <=	110.4 N./mm ²	Pass
Maximum Tube Stress	17.8 <=	115.0 N./mm ²	Pass
Minimum Tube Stress (Buckling)	-6.4 <=	-77.5 N./mm ²	Pass
Maximum Force on any one Tube	164.0 <=	1057.7 Kgf	Pass
Axial Membrane Stress in Shell	28.7 <=	90.3 N./mm ²	Pass
Shell Stress (jt. inter.)	46.6 <=	159.3 N./mm ²	Pass
Channel Stress (jt. inter.)	244.8 <=	207.0 N./mm ²	See below

Additional Results for the 2nd Elastic-Plastic Iteration.



These calculations are not intended to be used in the creep range. Check Part 4.18 in Section VIII Division 2.

Compute Allowable [Sstars]:
 = min(Sys, SPS,s/2)
 = min(160, 319/2) = 160 N./mm²

Compute Allowable [S*c]:
 = min(Syc, SPS,c/2)
 = min(243, 485/2) = 243 N./mm²

Compute Factor [facts]:
 = min([1.4 - 0.4 $\cdot$ abs(σ_s ,b) / Sstars], 1)
 = min([1.4 - 0.4 $\cdot$ abs(-17.9) / 160], 1) = 1

Compute Factor [factc]:
 = min([1.4 - 0.4 $\cdot$ abs(σ_c ,b) / Sstarc], 1)
 = min([1.4 - 0.4 $\cdot$ abs(205) / 243], 1) = 1

Note: Since facts & factc are both $\geq$ 1, the design Passes.

Stress/Force Summary for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Shell Stress (jt. inter.)	46.6 <=	159.3 N./mm ²	Pass
Channel Stress (jt. inter.)	244.8 <=	485.2 N./mm ²	Pass

Thickness Results for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
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Tubesheet Thickness :	34.005	50.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.037	1.650	Ok
Tube-Tubesheet Groove Weld Leg :	1.037	1.650	Ok

Fixed Tubesheet results per ASME Div. 2 Part 4.18.8 2025

Results for 16 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Bend	Stresses Allwd	Shear	Allwd	Case Type	Pass/ Fail
D1uc	42.550	...	159	207	20	110	Ps+Pt-Th	D1 Ok
D2uc	20.836	...	58	207	7	110	Ps+Pt-Th	D2 Ok
D3uc	34.005	...	110	207	14	110	Ps+Pt-Th	D3 Ok
D4uc	6.534	...	10	207	1	110	Ps+Pt-Th	D4 Ok
O1uc	24.314	...	153	460	19	110	Ps+Pt+Th	O1 Ok
O2uc	6.360	...	43	460	5	110	Ps+Pt+Th	O2 Ok
O3uc	18.893	...	110	460	14	110	Ps+Pt+Th	O3 Ok
O4uc	38.094	...	...	460	...	110	Ps+Pt+Th	O4 Ok
D1c	42.128	...	142	207	21	110	Ps+Pt-Th-c	D1 Ok
D2c	25.866	...	60	207	8	110	Ps+Pt-Th-c	D2 Ok
D3c	34.312	...	92	207	15	110	Ps+Pt-Th-c	D3 Ok
D4c	11.928	...	8	207	1	110	Ps+Pt-Th-c	D4 Ok
O1c	27.511	...	140	460	20	110	Ps+Pt+Th-c	O1 Ok
O2c	12.943	...	46	460	5	110	Ps+Pt+Th-c	O2 Ok
O3c	22.650	...	94	460	15	110	Ps+Pt+Th-c	O3 Ok
O4c	38.094	...	...	460	...	110	Ps+Pt+Th-c	O4 Ok
Max:	42.5503	... mm		0.766		0.191	(Str. Ratio)	

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures
 derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
 Load Case Tables

[(+)-Th]:

With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:

Operating Load Cases using the Maximum and Minimum Operating Pressures and
 Operating Temperatures

Shell Axial Membrane Stress Summary:

Case#	----	Shell Stresses	----	:	----	Shell Band Stress	----	:	Pass
	Ten	Allwd	Cmp	Allwd	:	Ten	Allwd	Cmp	Allwd : Fail
D1uc	21	90	...	...	:	...	...	...	Ok
D2uc	4	90	...	...	:	...	...	...	Ok
D3uc	29	90	...	...	:	...	...	...	Ok
D4uc	3	90	-3	-60	:	...	...	...	Ok
O1uc	22	319	...	...	:	...	...	...	Ok



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02uc	6	319	...	...	...	...	...	...	Ok
03uc	29	319	...	...	...	...	...	...	Ok
04uc	...	319	...	-60	...	...	...	...	Ok
D1c	25	90	...	...	...	...	...	...	Ok
D2c	4	90	...	...	...	...	...	...	Ok
D3c	32	90	...	...	...	...	...	...	Ok
D4c	4	90	-4	-59	...	...	...	...	Ok
01c	26	319	...	...	...	...	...	...	Ok
02c	7	319	...	...	...	...	...	...	Ok
03c	32	319	...	...	...	...	...	...	Ok
04c	...	319	...	-59	...	...	...	...	Ok

Max RATIO 0.358 0.060 ...

Tube, Shell and Channel Stress Summary:

Case#	---- Tube Stresses				Tube Loads		Shell Stress		Channel Stress		Pass
	Ten	Allwd	Cmp	Allwd	Ld	Allwd	Stress	Allwd	Stress	Allwd	
D1uc	24	114	-10	-77	218	1057	126	159	272	485	Ok
D2uc	8	114	-8	-77	70	1057	90	159	52	206	Ok
D3uc	18	114	-6	-77	164	1057	47	159	245	485	Ok
D4uc	2	114	-2	-77	15	1057	4	159	25	206	Ok
01uc	23	229	-9	-77	212	2115	116	319	269	485	Ok
02uc	5	229	-5	-77	48	2115	83	319	24	485	Ok
03uc	18	229	-6	-77	164	2115	47	319	245	485	Ok
04uc	...	229	...	-48	...	2115	...	319	...	485	Ok
D1c	27	114	-10	-77	245	1057	164	319	349	485	Ok
D2c	8	114	-8	-77	77	1057	102	159	63	206	Ok
D3c	20	114	-7	-77	185	1057	76	319	319	485	Ok
D4c	2	114	-2	-77	17	1057	5	159	32	206	Ok
01c	26	229	-10	-77	238	2115	153	319	346	485	Ok
02c	6	229	-6	-77	53	2115	91	319	27	485	Ok
03c	20	229	-7	-77	185	2115	76	319	319	485	Ok
04c	...	229	...	-48	...	2115	...	319	...	485	Ok

Max RATIO 0.232 0.128 0.232 0.790 0.720

Summary of Thickness Comparisons for 16 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	42.550	50.000	Ok
Tube thickness :	0.289	1.650	Ok
Tube-tubesheet fillet weld leg :	1.037	1.650	Ok
Tube-tubesheet groove weld leg :	1.037	1.650	Ok

Min Shell length of thk, (12.000) adj. to tubesheet: 295.770 mm

Min Channel length of thk, (14.000) adj. to tubesheet: 319.468 mm

Note: This is a full strength Tube to Tubesheet Joint.

Summary of Axial Differential Expansion between Shell and Tubes :

Due to Thermal Expansion Shell Extends by : +.000 mm
 Due to Pressure Shell Extends by : +.028 mm
 Due to Pressure + Thermal Shell Extends by : +.028 mm

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	0 shellside Stress Rat.	Shellside MAWP	0 tubeside Stress Rat.
Tubesheet Bending Stress	1395.57	0.999	1713.98	1.000
Tubesheet Shear Stress	5431.27	1.000	7758.69	1.000



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Tube Tensile Stress	4441.44	1.000	7613.84	1.000
Tube Compressive Stress	3011.93	0.363	5131.39	1.000
Tube-Tubesheet Joint load	4441.44	1.000	7613.84	1.000
Shell Stress (Axial, Junction)	1384.21	0.992	1332.83	1.000
Tubesheet-Channel Junction	1384.21	0.992	1332.83	0.195
Tube Pressure Stress	21402.76	1.000	9834.11	1.000
Minimum MAWP	1384.21		1332.83	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
Tubesheet Bending Stress	1384.21	0.998	1903.70	0.979
Tubesheet Shear Stress	5816.75	1.000	8519.07	1.000
Tube Tensile Stress	5040.37	1.000	8482.32	1.000
Tube Compressive Stress	9102.61	1.000	6256.66	1.000
Tube-Tubesheet Joint load	5040.37	1.000	8482.32	1.000
Shell Stress (Axial, Junction)	1384.21	0.998	1903.70	1.000
Tubesheet-Channel Junction	1384.21	0.998	1903.70	0.979
Tube Pressure Stress	21402.76	1.000	10774.75	1.000
Minimum MAPnc	1384.21		1903.70	

(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

Summary of Exchanger Pressure and Thermal Displacements: (mm)

Load Case#	Due to Pressure		Due to Thermal Exp.		Computed Delta Ms	
	UnCorroded	Corroded	UnCorroded	Corroded	UnCorroded	Corroded
1	0.0281	0.0281	...	...	...	...
2	-0.0259	-0.0260	...	...	...	...
3	0.0015	0.0015	...	...	...	...
4	0.0006	0.0006	...	...	...	...
5	0.0249	0.0249	...	...	...	...
6	-0.0234	-0.0234	...	...	...	...
7	0.0015	0.0015	...	...	...	...
8	...	...	...	...	...	...

Tubesheet MDMT Calculations:

Shell Side MDMT calculation:

Governing thickness on the shell side per figure UCS-66.3 (e):

= tubesheet thickness/4
 = 50/4
 = 12.500 mm

MDMT reduction ratio per UCS 66 (b)(1)(b):

= min(ps/Shellside MAPnc, 1)
 = min(380/1904, 1)
 = 0.200

Thickness Ratio = 0.2, Temperature Reduction per Fig. UCS 66.1 = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -22 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Channel Side MDMT calculation:

Governing Channel side thickness per figure UCS-66.3 (e):



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= max(tubesheet thk/4, min(tubesheet thk, channel thickness))
= max(50/4, min(50, 14))
= 14.000 mm

MDMT reduction ratio per UCS 66 (b)(1)(b):
= min(pt/Tubeside MAPnc, 1)
= min(1000/1384, 1)
= 0.722

Thickness Ratio = 0.72, Temperature Reduction per Fig. UCS 66.1 = 15 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-18 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-34 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

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PV Elite® Pressure Vessel Design Notes

Project Information	
Project Number	
Client	
Date	
Designer	
Design Code	
Additional Standards	

Design Conditions		
Internal Design Pressure:		
External Design Pressure:		
Design Int. Temperature:		
Design Ext. Temperature:		
Corrosion Allowance:		

Vessel Geometry		
Vessel Type		
Diameter		
Tangent-to-Tangent Length		
Head Type		
Number of Segments		

Material Specifications	
Shell:	
Head:	
Nozzle:	

Nozzle/Opening Details	ID/OD	Size	Rating	Matl.	Reinf.

Miscellaneous		
Support Type		
Number of Supports		
Support Material		
Seismic Code		
Wind Code		
Insulation		
Pressure Test		
Radiography		
Fatigue Analysis		

Analysis Notes: